



SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (AUTONOMOUS)

(Affiliated to Andhra University, Visakhapatnam), (Recognised by AICTE, New Delhi)

Accredited by NAAC with 'A' Grade

Recognised as Scientific and Industrial Research Organisation

CHINNA AMIRAM (P.O):: BHIMAVARAM :: W.G.Dt., A.P., INDIA :: PIN: 534 204

COMMON SCHEME OF INSTRUCTION & EXAMINATION

(Regulation R16)

I/IV B.TECH

(With effect from **2016-2017** Admitted Batch onwards)

Under Choice Based Credit System

GROUP – A (CIVIL, CSE, IT) Branches

I-SEMESTER

Code No.	Course	Credits	Lecture Hrs	Tutorial Hrs	Lab Hrs	Total Contact Hrs/Week	Sessional Marks	Exam Marks	Total Marks
B16 ENG 1101	English *	4	3	1	--	4	30	70	100
B16 ENG 1102	Mathematics-I *	4	3	1	--	4	30	70	100
B16 ENG 1103	Mathematics-II *	4	3	1	--	4	30	70	100
B16 ENG 1104	Chemistry	4	3	1	--	4	30	70	100
B16 ENG 1106	Computer Programming Using C & Numerical Methods	4	3	1	--	4	30	70	100
B16 ENG 1108	History of Science and Technology	2	2	--	--	2	30	70	100
B16 ENG 1110	Chemistry Lab	2	--	--	3	3	50	50	100
B16 ENG 1112	Computer programming Using C & Numerical Methods Lab	2	--	--	3	3	50	50	100
B16 ENG 1114	Sports (Audit)	--	--	--	--	3	--	--	--
Total		26	17	5	6	31	280	520	800

Note: Add-On Course: Technical English *

*** Common to both Group - A and Group - B**

Code: B16 ENG 1101

ENGLISH
(Common to All Branches)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

COURSE OBJECTIVES:

Reading Skills

1. Addressing explicit and implicit meanings of a text on current topics.
2. Understanding the context.
3. Learning new words and phrases.
4. Using words and phrases in different contexts.

Writing Skills

1. Using the basic structure of a sentence.
2. Applying relevant writing formats to create paragraphs, essays, letters, emails, reports and presentations.
3. Retaining a logical flow while writing.
4. Planning and executing an assignment creatively.

Interactive Skills

1. Analyzing a topic of discussion and relating to it.
2. Participating in discussions and influencing them.
3. Communicating ideas effectively.
4. Presenting ideas coherently within a stipulated time.

Life Skills and Core Skills

1. Examining self-attributes and identifying areas that require improvement: self-diagnosis and self-motivation.
2. Adapting to a given situation and developing a functional approach to finding solutions: adaptability and problem solving.
3. Understanding the importance of helping others: community services and enthusiasm.

COURSE OUTCOMES:

1. The overall performance of the students will be enhanced after the course; they will be in a position to make presentations on topics of current interests – politics, famous personalities, science and technology, tourism, work and business environment, with increased public speaking skills.
2. Students will be able to read, listen, speak and write effectively in both academic and non-academic environment.
3. The students will be updated with certain real life situations, which they can handle when come face to face.

SYLLABUS

Listening Skills

Conversations: Life in a Hostel – Eating Away those Blues – Meeting Carl Jung – A Documentary on the Big Cat – A Consultant Interviewing Employees – A Conversation about a Business Idea.

Speaking Skills

Your Favorite Holiday Destination – Describe Yourself – Why we need to save our Tiger – A Dialogue – Your First Interview – Pair Work: Setting up a New Business.

Reading Skills

Reading Comprehension: Famous People – What is Personality, Personality based on Blood Groups – News Report, Magazine Article, Mobile Towers and Health – An Excerpt from a Short Story, An Excerpt from a Biography – Open Letter to Prime Minister, Business Dilemmas: An Email Exchange – A Review of IPL: The Inside Story, Mark Zuckerberg: World's Youngest Billionaire.

Writing Skills

Letter Writing, Essay Writing, Email Writing, Report Writing, Paragraph Writing, Drafting a Pamphlet, Argument Writing, Dialogue Writing.

Grammar

Types of Sentences, Articles, Prepositions, Gerunds and Infinitives, Conjunctions, Tense, Quantifiers, Punctuations, Correction of Sentences, Fill-in the Blanks.

Vocabulary

Synonyms, Antonyms, Idioms, One Word Substitution.

Life Skills and Core Skills

Self Awareness and Self Motivation – Communication, Adaptability – Motivation, Problem Solving – Personal Presentation Skills, Stress Management – Professionalism – Ethics.

TEXT BOOK:

1. Life Through Language: A Holistic Approach to Language Learning. Board of Editors, Pearson Publishers, India 2013.

REFERENCE BOOKS:

1. Basic Vocabulary. Edgar Thorpe, Showick Thorpe. Pearson P. 2008
2. Quick Solutions to Common Errors in English, Angela Bunt. MacMillan P. 2008
3. Know Your English (Volume 1 & 2), by Dr. S. Upendra, Universities Press, India 2012
4. Business Communication Strategies. Maathukutty Monipally. Tata Mc Grahill P. 2009.

MATHEMATICS - I
(Common to All Branches)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

COURSE OBJECTIVES:

Students learn

1. How to find partial derivatives, Jacobians, how to change from one set of variables to another set of variables
2. Know Geometrical interpretation of partial derivatives
3. Learn applications to errors and approximations, maxima and minima
4. How to form ordinary differential equations (ODE) and solve equations of first order and first degree.
5. Applications of first order ODE to orthogonal trajectories, simple electrical circuits, Newton's law of cooling, Law of natural growth and decay
6. How to solve higher order linear differential equations with constant coefficients, Cauchy's linear equation, Legendre's linear equation and simultaneous first order ODEs
7. How to obtain Fourier Series for a periodic function of period 2π and Period $2C$ (C is a constant)
8. How to obtain Half range Cosine & Sine series and also Parseval's formula

COURSE OUTCOMES:

Students will be able to

1. Find partial derivatives, expand a function of more than one variable in a Taylor series and utilize them for errors and approximations, maxima and minima.
2. Solve a first order ODE and also find orthogonal trajectories and solve problems related to simple applications.
3. Solve a given higher order ODE, an equation with constant coefficients, a Cauchy's equation or a Legendre's equation.
4. Utilize knowledge of Fourier series for solving partial differential equations and also in understanding courses like Signals & Systems

SYLLABUS

Partial Differentiation

Functions of two or more variables – Partial derivatives – Homogeneous Functions – Euler's Theorem – Total Derivative – Change of Variables – Jacobians – Geometrical Interpretation: Tangent Plane and Normal to a Surface.

Applications of Partial Differentiation

Taylor's Theorem for functions of two variables – Errors and Approximations – Total Differential – Maxima and Minima of functions of two variables – Lagrange's Method of Undetermined Multipliers – Differentiation Under the Integral Sign – Liebnitz's Rules.

Ordinary Differential Equations of First Order and First Degree

Formation of ordinary differential equations (ODEs) – Solution of an ordinary differential equation – Equations of the First Order and First Degree – Linear Differential Equation – Bernoulli's Equation – Exact Differential Equations – Equations Reducible to exact equations.

Applications of Differential Equations of First Order

Orthogonal Trajectories – Simple electric (LR & CR) Circuits – Newton's Law of Cooling – Law of Natural growth and decay.

Linear Differential Equations of Higher Order

Solutions of Linear Ordinary Differential Equations With Constant Coefficients – Rules for finding Complimentary Function – Rules for finding the particular integral – Method of variation of parameters – Cauchy's linear equation – Legendre's Linear Equation – Simultaneous linear equations.

Fourier series

Introduction - Euler's Formulae - Conditions for a Fourier Expansion - Functions having points of discontinuity - Change of Interval - Odd and Even Functions - Expansions of Odd or Even Periodic Functions, Half-Range Series - Parseval's Formula.

TEXT BOOK:

1. Scope and Treatment as in "Higher Engineering Mathematics", by Dr. B. S. Grewal, 43rd edition, Khanna Publishers.

REFERENCE BOOKS:

1. Advanced Engineering Mathematics by Erwin Kreyszig, Wiley.
2. A text book of Engineering Mathematics, by N. P. Bali and Dr. Manish Goyal, Lakshmi Publications.
3. Advanced Engineering Mathematics by H. K. Dass, S. Chand Company.
4. Higher Engineering Mathematics by B. V. Ramana, Tata Mc Graw Hill Company
5. Higher Engineering Mathematics by Dr. M. K. Venkataraman, The National Publishing Company.

MATHEMATICS – II
(Common to All Branches)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

COURSE OBJECTIVES:

Students learn

1. The concept of rank, Normal form of a matrix, consistency of a system of linear algebraic equations
2. Eigen values and Eigen vectors of a matrix, the Cayley Hamilton theorem
3. Orthogonal reduction of a quadratic form.
4. Complex matrices and their properties
5. Laplace transform, existence, properties.
6. Inverse Laplace transform, convolution theorem, how to solve Ordinary Differential Equations (ODE) and simultaneous ODEs by the use of Laplace transforms.
7. Difference equations and their methods of solution.
8. Z-transform, its properties, important results and its use to solve difference equations.

COURSE OUTCOMES:

Students will be capable of

1. Utilizing the knowledge of matrices for solving linear simultaneous equations, find Eigen values and Eigen vectors and handle quadratic forms
2. Utilizing the knowledge of Laplace Transforms to find transforms of important functions that arise in applications and also solve ODE
3. Also utilizing the knowledge of Laplace Transforms in courses like Net Works, Signals & Systems and Control Systems
4. Utilizing the knowledge of difference equations and Z-transforms in understanding courses like Discrete Mathematical Structures and also Signals & Systems.

SYLLABUS

Matrices – I

Rank of a matrix - Normal Form – Solutions of Linear System of Equations- Consistency of Linear System of Equations – Rouche’s Theorem(statement) - Direct Methods: Gauss Elimination Method, LU Factorization Method – Eigen Values and Eigen Vectors of a Matrix – Properties - Cayley – Hamilton Theorem – Inverse and Powers of a Matrix using Cayley – Hamilton Theorem.

Matrices – II

Diagonalization of a Matrix – Quadratic Forms – Reduction of Quadratic Form to Canonical Form – Nature of a Quadratic Form – Complex Matrices: Hermitian, Skew-Hermitian and Unitary Matrices and their Properties.

Laplace Transforms-I

Introduction – Existence Conditions – Transforms of Elementary Functions – Properties of Laplace Transforms – Transforms of Derivatives – Transforms of Integrals – Multiplication by ‘t’ – Division by t – Evaluation of Integrals by Laplace Transforms – Laplace Transforms of Unit Step Function, Unit Impulse Function and Periodic Functions.

Laplace Transforms-II

Inverse Laplace Transform – different methods - Convolution Theorem – Applications of Laplace Transforms to Ordinary Differential Equations, Simultaneous Linear Differential Equations with Constant Coefficients.

Difference Equations

Definition - order and solution of a difference equation - Formation of difference equations - Linear difference equations - Rules for finding C.F. - Rules for finding P.I.- Simultaneous difference equations with constant coefficients. Application to deflection of a loaded string.

Z-transforms

Z-transforms – definition - some standard Z-transforms - Linear property, Damping rule - some standard results - shifting rules - initial and final value theorems - convolution theorem - Evaluation of inverse transforms - Applications of Z-transform to difference equation.

TEXT BOOK:

1. Scope and Treatment as in “Higher Engineering Mathematics”, by Dr. B. S. Grewal, 43rd edition, Khanna Publishers.

REFERENCE BOOKS:

1. Advanced Engineering Mathematics by Erwin Kreyszig, Wiley.
2. A text book of Engineering Mathematics, by N. P. Bali and Dr. Manish Goyal, Lakshmi Publications.
3. Advanced Engineering Mathematics by H. K. Dass, S. Chand Company.
4. Higher Engineering Mathematics by B. V. Ramana, Tata Mc Graw Hill Company

CHEMISTRY
(Common to CIVIL, CSE & IT)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

COURSE OBJECTIVES:

1. To develop full pledged laboratories for the analysis of engineering materials.
2. To develop full pledged lab for pollution, corrosion, polymers and electrochemical cells.
3. Extend the infrastructure facilities of the department for the project works of engineering students and also guide the project work under faculty supervision.
4. To develop full pledged lab for analysis of Water, Cement, Hydrogen Peroxide, pollutants and raw materials.

COURSE OUTCOMES:

1. Students learn in-depth about the topics of desalination of sea water, CNG, LPG Biogas, Semiconductors, Liquid crystals, Conducting polymers, fiber reinforced plastics, building materials.
2. Students understand the basic and advanced applied concepts.
3. Students learn to interrelate the theory and with the relevant experiment.
4. Students learn experimental techniques and understand the theory about experiments.

SYLLABUS**Water Chemistry**

Source of water- impurities- Hardness and its determination by EDTA method- Boiler troubles and their removal. Water softening methods- lime soda, zeolite and ion exchange. Municipal water treatment- Break point chlorination. Desalination of sea water – electro dialysis and reverse osmosis methods.

Building Materials

Portland cement: Manufacture-Chemistry involved in setting and hardening of cement – Cement concrete -RCC –Decay of concrete.
Refractories: Classification-Properties and Engineering applications.
Ceramics: Classification-Properties and uses.

Solid State Chemistry

Classification of solids-Types of crystals-properties-Imperfections in crystals. Band theory of solids. Chemistry of semi conductors –Intrinsic, Extrinsic, compound and defect. Organic semi conductors Purification of solids by Zone refining-Single crystal growth – epitaxial growth. Elementary ideas on Liquid crystals.

Corrosion Chemistry

Definition of corrosion- Types of corrosion-chemical & electrochemical corrosion –Pitting, stress corrosion, Galvanic corrosion, Water line corrosion Factors affecting corrosion – Prevention of corrosion- Cathodic protection. Corrosion inhibitors Protective coatings- Metallic coatings, electro plating, electroless plating, chemical conversion coatings- phosphate coatings, chromate coatings, anodizing. Organic coatings-Paints.

Polymers And Plastics

Definition-Types of polymerization – mechanism of free radical polymerization. plastics- Thermosetting and thermoplastic resins cellulose derivatives, vinyl resins, nylon 6,6, bakelite-Compounding of plastics-Fabrication of plastics. Fiber reinforced plastics – conducting polymers -Engineering applications of polymers.

Fuels And Lubricants

Solid fuels: Coal –Analysis of coal – Metallurgical coke- manufacture-Engineering applications.

Petroleum-refining-Knocking and Octane number of gasoline-Cetane number of diesel oil. Synthetic petrol – LPG-CNG–Applications. Rocket fuels – Propellants-classification.

LUBRICANTS: Principles of lubrications, classification of lubricants and properties of lubricants (any five)

TEXT BOOKS:

1. Engineering chemistry by Jain & Jain 15th edition Dhanpatrai publishing company.
2. Engineering chemistry by Dr. K.AnjiReddy and Dr. M.Sita rama reddy, silicon publications, 2015.

REFERENCE BOOKS:

1. A Text Book of Engineering Chemistry by S.S.Dara; S.Chand & Company Ltd.
2. Engineering Chemistry by B.Siva Sankar Mc.Graw Hill Education(India) Pvt.Ltd.
3. A Text Book of Engineering Chemistry by Sashi Chawla, Dhanpatrai & Co.

COMPUTER PROGRAMMING USING C & NUMERICAL METHODS
(Common to CIVIL, CSE & IT)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

COURSE OBJECTIVES:

1. To make the student familiar with programming in C and enable the student to implement the numerical methods described in this course using C as Programming Language.
2. To introduce the programming principles
3. To familiarize the variables construction, declarations, initializations etc.,
4. To introduce the Control statement
5. To introduce the arrays, structures, Unions, Files etc.,
6. To give the overall idea on Pointers
7. To give the idea about dynamic memory allocation
8. The nature of C language is emphasized in the wide variety of examples and applications.
9. To know about some popular programming languages.
10. To solve various numerical Methods
11. Student can write various programs by applying elementary concepts like arrays, structures, Unions, Files etc.,

COURSE OUTCOMES:

1. Student can understand basic terminology used in C programming.
2. Student can write programs by applying elementary algorithms to solve problems in C language.
3. Student can write, compile and debug programs in C language.
4. Student can Write programs to solve numerical methods
5. Student can be familiar with finite precision computation.

SYLLABUS

Introduction to C

Basic structure of C program, Constants, Variables and data types, Operators and Expressions, Arithmetic Precedence and associativity, Type Conversions. Managing Input and Output Operations, Formatted Input, Formatted Output.

Decision Making, Branching, Looping, Arrays & Strings: Decision making with if statement, Simple if statement, The if...else statement, Nesting of if...else statement, the else..if ladder, switch statement, the (?:) operator, the GOTO statement., The while statement, the do statement, The for statement, Jumps in Loops ,One, Two-dimensional Arrays, Character Arrays. Declaration and initialization of Strings, reading and writing of strings, String handling functions, Table of strings.

Functions

Definition of Functions, Return Values and their Types, Function Calls, Function Declaration, Category of Functions: No Arguments and no Return Values, Arguments but no Return Values, Arguments with Return Values, No Argument but Returns a Value, Functions that Return Multiple Values. Nesting of functions, recursion, passing arrays to functions, passing strings to functions, The scope, visibility and lifetime of variables. .

Pointers

Accessing the address of a variable, declaring pointer variables, initializing of pointer variables, accessing variables using pointers, chain of pointers, pointer expressions, pointers and arrays, pointers and character strings, array of pointers, pointers as function arguments, functions returning pointers, pointers to functions, pointers to structures-Program Applications

Structure and Unions

Defining a structure, declaring structure variables, accessing structure members, structure initialization, copying and comparing structure variables, arrays of structures, arrays within structures, structures within structures, structures and functions and unions, size of structures and bit-fields- Program applications.

File handling

Defining and opening a file, closing a file, Input/ Output operations on files, Error handling during I/O operations, random access to files and Command Line Arguments- Program Applications.

Numerical Methods

Solutions of Algebraic and Transcendental Equations: Bisection Method, Newton Raphson Method.

Interpolation - Newton's forward and backward Interpolation, Lagrange's Interpolation in unequal intervals.

Solutions of Linear Equations: Gauss Elimination Method, Jacobi and Gauss Seidel Methods. Numerical Integration: Trapezoidal rule, Simpson's 1/3 rule.

Solutions of Ordinary First Order Differential Equations: Euler's Method, Modified Euler's Method and Runge-Kutta Method.

TEXT BOOKS:

1. Programming in ANSI C, E Balagurusamy, 6th Edition. McGraw Hill Education (India) Private Limited.
2. Introduction to Numerical Methods, SS Sastry, Prentice Hall India Pvt.Ltd.

REFERENCE BOOKS:

1. Let Us C, Yashwant Kanetkar, BPB Publications, 5th Edition.
2. Computer Science, A structured programming approach using C, B.A.Forouzan and R.F.Gilberg, 3rd Edition, Thomson, 2007.
3. The C –Programming Language, B.W. Kernighan, Dennis M. Ritchie, Prentice Hall India Pvt.Ltd
4. Scientific Programming: C-Language, Algorithms and Models in Science, Luciano M. Barone (Author), Enzo Marinari (Author), Giovanni Organtini, World Scientific

HISTORY OF SCIENCE AND TECHNOLOGY
(Common to CIVIL, CSE & IT)

Theory : 2 periods
Exam : 3 Hrs.

Sessionals : 30
Ext. Marks : 70
Credits : 2

COURSE OBJECTIVES:

1. To know the contributions of scientists for the development of society over a period of time.
2. To understand the Science and Technological developments that lead to human welfare.
3. To appreciate the Science and Technological contributions for the development of various sectors of the economy.
4. To identify the technological transfer versus economic progress of the countries.

COURSE OUTCOMES:

1. By the end of this course the students should be able to understand the contribution of Scientific and Technological developments for the benefit of society at large.

SYLLABUS**Historical Perspective of Science and Technology**

Nature and Definitions; Roots of Science – In Ancient Period and Modern Period (During the British Period); Science and Society; Role of Scientist in the Society.

Policies and Plans after Independence

Science and Technology Policy Resolutions; New Technology Fund; Technology Development (TIFAC); Programs aimed at Technological Self Reliance; Activities of Council of Scientific and Industrial Research.

Science and Technological Developments in Critical Areas

Space – The Indian Space Program: India's Geostationary Satellite Services – INSAT System And INSAT Services; **Defense Research and Technology** – Research Coordination, Research efforts and Development of technologies and Spin-off technologies for civilian use; **Nuclear Energy** – Effects of a nuclear explosion and India's safety measures.

Impact of Science and Technology in Major Areas

Ocean Development: Objectives of Ocean Development, Biological and Mineral resources, Marine Research and Capacity Building; **Biotechnology:** Meaning, Biotechnology techniques- Bioreactors, Cell fusion, Cell or Tissue Culture, DNA Fingerprinting, Cloning, Artificial Insemination and Embryo Transfer Technology and Stem Cell Technology; Application of Biotechnology – Medicine, Biocatalysts, Food Biotechnology, Fuel and Fodder and Development of Biosensors.

Technology Transfer and Development

Transfer of Technology – Types, Methods, Mechanisms, Process, Channels and Techniques; **Appropriate Technology** - Criteria and Selection of an Appropriate Technology; Barriers of Technological Change. .

TEXT BOOKS:

1. Kalpana Rajaram, Science and Technology in India, Published and Distributed by Spectrum Books (P) Ltd., New Delhi-58.
2. Srinivasan, M., Management of Science and Technology (Problems & Prospects), East – West Press (P) Ltd., New Delhi.

REFERENCE BOOK:

1. Dr. G.R. Kohli, History of Science and Technology and Environmental Movements in India, Surjeet Publications, New Delhi

CHEMISTRY LAB
(Common to CIVIL, CSE & IT)

Lab : 3 periods
Exam : 3 Hrs.

Sessionals : 50
Ext. Marks : 50
Credits : 2

LIST OF PRACTICAL EXPERIMENTS:

1. Estimation of Sodium hydroxide with HCl(Na_2CO_3 primary standard).
2. Estimation of Iron as Ferrous iron in an Ore Sample.
3. Estimation of Oxalic acid by a redox method
4. Estimation of Calcium in a sample of Portland cement.
5. Estimation of Volume strength of Hydrogen Peroxide.
6. Estimation of Mohr's salt by Potassium dichromate
7. Determination of Hardness of an underground water sample.
8. Estimation of Zinc by EDTA method.
9. Determination of Alkalinity of water sample.

DEMONSTRATION EXPERIMENTS:

10. Determination of Viscosity and Viscosity index of a lubricant.
11. Printed Circuit Board
12. Determination of dissolved oxygen in given water sample.
13. Potentiometric titrations.
14. P^{H} Determination by using P^{H} meter.

REFERENCE BOOKS:

1. Essentials Experimental Engineering Chemistry by Sashi Chawla, Dhanpatrai & Co Pvt. Ltd.
2. Laboratory Manual on Engineering Chemistry by Dr. Sudha Rani, Dhanpatrai Publishing Company.
3. Engineering Chemistry Laboratory Manual by Dr. K.Anji Reddy, Tulip Publications.

COMPUTER PROGRAMMING USING C & NUMERICAL METHODS LAB
(Common to CIVIL, CSE & IT)

Lab	: 3 Periods	Sessionals	: 50
Exam	: 3 Hrs.	Ext. Marks	: 50
		Credits	: 2

LIST OF PROGRAMMES:

1. Write a program to read x, y coordinates of 3 points and then calculate the area of a triangle formed by them and print the coordinates of the three points and the area of the triangle. What will be the output from your program if the three given points are in a straight line.
2. Write a program which generates 100 random numbers in the range of 1 to 100. Store them in an array and then print the array. Write 3 versions of the program using different loop constructs (eg. for, while and do-while).
3. Write a set of string manipulation functions eg. for getting a sub-string from a given position, copying one string to another, reversing a string and adding one string to another.
4. Write a program which determines the largest and the smallest number that can be stored in different data types like short, int, long, float and double. What happens when you add 1 to the largest possible integer number that can be stored?
5. Write a program which generates 100 random real numbers in the range of 10.0 to 20.0 and sort them in descending order.
6. Write a function for transporting a square matrix in place (in place means that you are not allowed to have full temporary matrix).
7. First use an editor to create a file with some integer numbers. Now write a program, which reads these numbers and determines their mean and standard deviation.
8. Implement bisection method to find the square root of a given number to a given accuracy.
9. Implement Newton Raphson Method to determine a root of polynomial equation.
10. Given a table of x and corresponding f(x) values, write a program which will determine f(x) value at an intermediate x value using Lagrange's Interpolation.
11. Write a function which will invert a matrix.
12. Implement Simpson's $1/3^{\text{rd}}$ rule for numerical integration.
13. Implement Trapezoidal rule for numerical integration.
14. Write a program to solve a set of linear algebraic equations.
15. Write a program to solve a differential equation using Runge-Kutta Method.

REFERENCE BOOKS:

1. Programming in ANSI C, E Balagurusamy, 6th Edition. McGraw Hill Education (India) Private Limited.
2. Introduction to Numerical Methods, SS Sastry, Prentice Hall India Pvt.Ltd.

SPORTS
(Common to CIVIL, CSE & IT)

Contact Hrs. 3 Hrs.

This is an Audit Course. It does not carry any Grade or Credit. It will not be included in the Grade Memo / Certificate.

ADD-ON COURSE
TECHNICAL ENGLISH
(Common to All Branches)

Total Hrs: 30

COURSE OBJECTIVES:

1. To improve the language proficiency of the students regarding the technical aspects.
2. To enhance the awareness about the formal speaking and writing skills in different contexts.
3. Motivating the students to improve their personal skills.

COURSE OUTCOMES:

1. Students improve their language skills in formal/ technical contexts.
2. They enhance their understanding of technical terms.
3. They improve their personal skills.

SYLLABUS

1. Defining Technical Writing.
2. Writing Skills
Business letters, E-mails
Report Writing
Writing Proposals & Business Quotations
Brochures, Newsletters
How to write an abstract & research paper
3. Speaking Skills
Technical Descriptions
PPT Presentations
Paper Presentations
4. Vocabulary
Technical Terms
5. Behavioral / Attitudinal Skills

TEXT BOOK:

1. Effective Technical Communication by Ashraf Rizvi: Tata McGraw Hill.

REFERENCE BOOKS:

1. Handbook for Technical Writing by David A. McMurrey and Joanne Buckley: Cengage Learning.
2. Technical Writing: Process and Product by Sharon J. Gerson and Steven M. Gerson: Pearson Publications

COMMON SCHEME OF INSTRUCTION & EXAMINATION
(Regulation R16)

I/IV B.TECH
(With effect from **2016-2017** Admitted Batch onwards)
Under Choice Based Credit System

GROUP – A (CIVIL, CSE & IT) Branches

II-SEMESTER

Code No.	Course	Credits	Lecture Hrs	Tutorial Hrs	Lab Hrs	Total Contact Hrs/Week	Sessional Marks	Exam Marks	Total Marks
B16 ENG 1201	Mathematics-III *	4	3	1	--	4	30	70	100
B16 ENG 1202	Physics	4	3	1	--	4	30	70	100
B16 ENG 1204	Engineering Graphics	4	2	--	3	5	30	70	100
B16 ENG 1206	Professional Ethics & Moral values	2	2	--	--	2	30	70	100
DS	Department Subject #	4	3	1	--	4	30	70	100
B16 ENG 1209	Physics Lab	2	--	--	3	3	50	50	100
B16 ENG 1211	Workshop	2	--	--	3	3	50	50	100
B16 ENG 1213	English Language Lab *	2	--	--	3	3	50	50	100
B16 ENG 1214	NCC/NSS (Audit)	--	--	--	--	3	--	--	--
Total		24	13	3	12	31	300	500	800

Note: Add-On Course: Technology Course-I *

*** Common to both Group-A and Group-B**

Civil: B16CE1208: Building Materials and Building Construction

CSE: B16CS1208: Probability, Statistics & Queueing Theory

IT: B16CS1208: Probability, Statistics & Queueing Theory

Code: B16 ENG 1201

MATHEMATICS – III
(Common to All Branches)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

COURSE OBJECTIVES:

Students learn

1. Equations of a line, shortest distance between skew lines
2. About sphere, cone and cylinder
3. Double integrals, change of order of integration
4. Beta function, Gamma function, error function
5. Applications of double integrals to Areas, Volumes, mass, centre of gravity, Moment of inertia
6. The concept of Fourier Integral, Fourier Transform, properties.
7. Parseval's formulae

COURSE OUTCOMES:

Students will be able to

1. Utilize knowledge of line, sphere etc. in his engineering subjects
2. Utilize the knowledge of Beta and Gamma functions and multiple integrals to evaluate the integrals they come across in their applications
3. Utilize the knowledge of Fourier Transform in courses like Signals and Systems and in the solution of partial differential equations at a later stage

SYLLABUS

Solid Geometry

Equations of a plane, Normal form, Intercept form, Equations of Straight Line – Conditions for a line to lie in a Plane – Coplanar lines – Shortest distance between two skew lines - Intersection of three Planes – Equations of Sphere – Tangent Plane to a Sphere –Cone – Cylinder.

Multiple Integrals-1

Double Integrals - Change of Order of Integration - Double Integrals in Polar Coordinates - Triple Integrals - Change of Variables.

Multiple Integrals-2

Beta Function - Gamma Function - Relation between Beta and Gamma Functions-Error Function - Area enclosed by plane curves - Volumes of solids - Area of a curved surface - Calculation of mass - Center of gravity of a plane lamina- Moment of inertia.

Fourier Transforms

Introduction – definition - Fourier integral - Sine and Cosine integrals - Complex form of Fourier integral - Fourier transform - Fourier Sine and Cosine transforms -Finite Fourier Sine and Cosine transforms - properties of Fourier transforms, Convolution theorem for Fourier transforms - Parseval's identity for Fourier transforms - Fourier transforms of derivatives of a function.

TEXT BOOK:

1. Scope and Treatment as in “Higher Engineering Mathematics”, by Dr. B. S. Grewal, 43rd edition, Khanna Publishers.

REFERENCE BOOKS:

1. Advanced Engineering Mathematics by Erwin Kreyszig, Wiley.
2. A text book of Engineering Mathematics, by N. P. Bali and Dr. Manish Goyal, Lakshmi Publications.
3. Advanced Engineering Mathematics by H. K. Dass, S. Chand Company.
4. Higher Engineering Mathematics by B. V. Ramana, Tata Mc Graw Hill Company
5. Higher Engineering Mathematics by Dr. M. K. Venkataraman, The National Publishing Company.

PHYSICS
(Common to CIVIL, CSE & IT)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

COURSE OBJECTIVES:

1. To understand the basic and advanced concepts of classical and modern physics – Identify, analyze, formulate and solve a wide variety of real world problems.
2. To develop scientific reasoning and problem solving methods.
3. To bring awareness of latest technologies (physics based) – lasers and fiber optics.

COURSE OUTCOMES:

1. Students learn in depth about the topics of Lasers, fiber optics, quantum mechanical Theory and classical theories of thermodynamics and electromagnetism.
2. Students understand the classical and modern concepts.

SYLLABUS**Thermodynamics**

Introduction, Heat and Work, First Law of Thermodynamics and applications, Reversible and Irreversible Process, Carnot Cycle and Efficiency, Second Law of Thermodynamics, Carnot's Theorem, Entropy, Second Law in terms of entropy, Entropy and disorder, Third Law of Thermodynamics (Statement Only).

Electromagnetism

Effect of Magnetic Field on – moving charges, current in long straight wire and rectangular Current Loop, Hall Effect, Biot-Savart's Law, B near a Long Wire, B for a Circular Current Loop, Ampere's Law, B for a Solenoid, Faraday's Law of electromagnetic induction, Lenz's law, Inductance of a solenoid, L-R Circuit, Displacement Current, Maxwell's Equations (integral form) and their significance (without derivation).

Interference

Principle of Super Position – Young's Experiment – Coherence – Inference in thin transparent films, Newton's Rings, Michelson Interferometer and its applications.

Lasers

Introduction, spontaneous and stimulated emissions, requirements of laser device, Ruby Laser, Gas Laser (He-Ne Laser), Semiconductor diode Laser, Characteristics and applications of Lasers.

Optical Fibers

Introduction, Principles of light propagation in optical fiber, Acceptance angle, Numerical aperture, types of fiber, Applications of Optical Fibers, Optical Fiber in Communications, advantages.

Ultrasonics

Definition, Production of Ultrasonics by Magnetostriction and Piezoelectric methods, detection methods, acoustic grating, characteristics and applications of Ultrasonics.

Modern Physics

Introduction, de Broglie concept of matter waves, Properties of matter waves, experimental verification (Davisson-Germer experiment), Heisenberg uncertainty principle, Wave function and its physical significance, Schrodinger time independent wave equation, application to a particle in a box, Band theory of Solids, Kronig - Penney model (qualitative treatment), Origin of energy band formation in solids, Classification of materials into conductors, semi conductors and insulators .

Nanophase Materials

Definition, Synthesis – Synthesis methods, Condensation and Ball milling, Chemical vapour deposition method – sol-gel methods, Characterisation, analysis and applications of nano materials.

TEXT BOOKS:

1. Physics by David Halliday and Robert Resnick – Part I and Part II, Wiley Eastern India (pvt.) Ltd.
2. Engineering Physics by M.N. Avadhanulu & P.G. Kshirasagar; S. Chand & Company Ltd

REFERENCE BOOKS:

1. Modern Engineering Physics by A.S. Vasudeva, S. Chand & Company Ltd
2. University Physics by Young and Freedman, Addison-Wesley
3. Engineering Physics by R.K. Gaur and S.L. Gupta, Dhanpat Rai & CO

ENGINEERING GRAPHICS
(Common to CIVIL, CSE & IT)

Theory	: 2 Periods	Sessionals	: 30
Lab	: 3 Periods	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

COURSE OBJECTIVES:

1. To highlight the significance of universal language of engineers.
2. To impart basic knowledge and skills required to prepare engineering drawings.
3. To impart knowledge and skills required to draw projections of solids in different contexts.
4. To visualize and represent the pictorial views with proper dimensioning and scaling.

COURSE OUTCOMES:

Students will be able to

1. Apply principles of drawing to represent dimensions of an object.
2. Construct polygons and engineering curves.
3. Draw projections of points, lines, planes and solids.
4. Represent sectional views of solids.
5. Develop the surfaces of regular solids.
6. Draw the isometric views of solids and combination of solids.

SYLLABUS

Introduction

Lines, Lettering and Dimensioning. Geometrical Constructions.

Curves

Conic sections: General construction of ellipse, parabola and hyperbola. Construction of involutes. Normal and Tangent.

Projections of Points

Principal or Reference Planes, Projections of a point situated in any one of the four quadrants

Projections of Straight Lines

Projections of straight lines parallel to both reference planes, perpendicular to one reference plane and parallel to other reference plane, inclined to one reference plane and parallel to the other reference plane. Projections of straight line inclined to both the reference planes:

Projections of Planes

Projection of Perpendicular planes: Perpendicular to both reference planes, perpendicular to one reference plane and parallel to other reference plane, perpendicular to one reference plane and inclined to other reference plane. Projection of Oblique planes. Introduction to Auxiliary Planes.

Projections of Solids

Types of solids: Polyhedra and Solids of revolution. Projection of solids in simple positions: Axis perpendicular to horizontal plane, Axis perpendicular to vertical plane and Axis parallel to both the reference planes, Projection of Solids with axis inclined to one reference plane and parallel to other and axes inclined to both the reference planes.

Projections of Section of Solids

Section Planes: Parallel and inclined section planes, Sections and True shape of section, Sections of Solids: Prism, Pyramid, Cylinder and Cone .

Development of Surfaces

Methods of Development: Parallel line development and radial line development. Development of a cube, prism, cylinder, pyramid and cone.

Isometric Views

Introduction to Isometric projection, Isometric scale and Isometric view. Isometric views of simple planes. Isometric view of Prisms, Pyramids, cylinder and cone. Isometric view of a combination of solids.

TEXT BOOK:

1. Elementary Engineering Drawing by N.D.Bhatt, Charotar Publishing House.

REFERENCE BOOK:

1. Engineering Graphics by K.L. Narayana and P. Kannaiah, Tata Mc-Graw Hill.

PROFESSIONAL ETHICS AND MORAL VALUES
(Common to CIVIL, CSE & IT)

Theory : 2 Periods
Exam : 3 Hrs.

Sessionals : 30
Ext. Marks : 70
Credits : 2

COURSE OBJECTIVES:

1. To inculcate Ethics and Human Values into the young minds.
2. To develop moral responsibility and mould them as best professionals.
3. To create ethical vision and achieve harmony in life.

COURSE OUTCOME:

1. By the end of the course student should be able to understand the importance of ethics and values in life and society.

SYLLABUS**Ethics and Human Values**

Ethics and Values, Ethical Vision, Ethical Decisions, **Human Values** – Classification of Values, Universality of Values.

Engineering Ethics

Nature of Engineering Ethics, Profession and Professionalism, Professional Ethics, Code of Ethics, Sample Codes – IEEE, ASCE, ASME and CSI.

Engineering as Social Experimentation

Engineering as social experimentation, Engineering Professionals – life skills, Engineers as Managers, Consultants and Leaders, Role of engineers in promoting ethical climate, balanced outlook on law.

Safety Social Responsibility and Rights

Safety and Risk, moral responsibility of engineers for safety, case studies – Bhopal gas tragedy, Chernobyl disaster, Fukushima Nuclear disaster, Professional rights, Gender discrimination, Sexual harassment at work place.

Global Issues

Globalization and MNCs, Environmental Ethics, Computer Ethics, Cyber Crimes, Ethical living, concept of Harmony in life.

TEXT BOOKS:

1. Govindharajan, M., Natarajan, S. and Senthil Kumar, V.S., Engineering Ethics, Prentice Hall of India, (PHI) Delhi, 2004.
2. Subramainam, R., Professional Ethics, Oxford University Press, New Delhi, 2013.

REFERENCE BOOK:

1. Charles D, Fleddermann, “Engineering Ethics”, Pearson / PHI, New Jersey 2004 (Indian Reprint)

BUILDING MATERIALS AND BUILDING CONSTRUCTION
(For Civil Engineering)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

COURSE OBJECTIVES:

1. To teach students about the Physical and Mechanical properties of construction materials and their respective testing procedure.
2. To teach students about the building materials available in market to be used for many components of building industry.
3. To teach students about the principles and methods to be followed in constructing various components of a building.
4. To teach students about the deterioration and repair of buildings.

COURSE OUTCOMES:

1. Learn and identify the relevant physical and mechanical properties pertaining to the construction industry.
2. Demonstrate the relevant BIS testing procedure to be carried out to ascertain the quality of building materials.
3. Develop ability to choose the modern construction material appropriate to the climate and functional aspects of the buildings.
4. Ability to supervise the construction technique to be followed in brick, stone and hollow block masonry, concreting, flooring, roofing, plastering and painting etc.
5. Learn about the causes of deterioration, crack pattern, and assessment of damages.
6. Learn about the construction techniques in repairing of buildings.

SYLLABUS**Bricks and Clay Products**

Bricks: Classification of Bricks, Manufacture of Bricks, general qualities of Bricks as per IS code, tests for good bricks as per IS code, including field tests, special forms of Bricks and their uses. Clay Products: Various types of tile manufacturing and their uses, Earth-ware, Terra-cotta, stone ware, porcelain, glazing of tiles etc.

Wood, Wood Based Products

Wood: cross section details of trees, their general properties, various types of defects in wood and timber, Methods of seasoning and their importance, various Mechanical Properties of timber, Decay of timber, Wood based Products: Veneers, Plywood and its types, Manufacturing of Plywood, plywood grades as per IS code, Laminated wood, merits of plywood and laminated wood, Lamin Boards, Block Boards, Batten board, Hard board, Particle boards and Composite boards.

Paints, Varnishes

Paints and Varnishes: Constituents and characteristics of paints, types of paint, their uses and preparation on different surfaces, painting defects, causes and remedies. Constituents of varnishes, uses of varnishes, different kinds of varnish, polishes. Painting of interior walls, exterior walls, wooden doors and windows – steel windows – various types of paints (chemistry of paints not included) including distempers; emulsion paints etc., Varnishes wood work finishing types. **Plastic:** definition, classification of plastic and modern

development of plastic, chemistry of plastic, polymerization, manufacturing process and their uses. **Glass:** properties of glass, types of glass and manufacturing. **Asbestos & Asphalt Bitumen & Tar :** Availability and uses of asbestos, properties of asbestos, various types of asbestos, difference between asphalt & bitumen, Types, uses and properties of Asphalt & Bitumen, composition of coal tar, wood tar, mineral tar and Naphtha.

Cements

Natural and artificial cements, types of various artificial cements and their uses. Wet and dry process of manufacturing ordinary Portland cement (OPC), Chemical and Physical analysis of OPC, various field and Laboratory Tests on OPC as per IS code. Storing of cement in the field and godowns

Foundations

Types of Foundations : Strip, Isolated, Strap, Combined Footings, Raft – Mat –Slab and Beam Raft, Box Type Raft, inverted arch foundations, SHELL foundations, Grillage foundations – Minimum depth of Foundation – Bearing capacity of soils **Masonry:** Different types of Stone Masonry – Plan, Elevation, Sections of Stone Masonry Works – Brick Masonry – Different Types of Bonds – Plan, elevation and Section of Brick Bonds upto Two-Brick wall thickness – Partition walls – Different types of Block Masonry – Hollow concrete Blocks – FAL- G Blocks, Hollow Clay Blocks.

Roofing

Mangalore tiled Roof, RCC roof, Madras Terrace, Hollow Tiled Roof, Asbestos Cement, Fibre glass, Aluminium G.I. Sheet roofing's. **Form Work, Scaffolding:** form work, Types of formwork, centering-scaffolding-types of scaffolding. **Trusses:** King Post & Queen Post Trusses – Steel roof Truss for 12m Span with details. **Wooden Doors And Windows:** Parallel – Glazed – Flush shutters, Plywood, Particle Board Shutters – Aluminum, PVC, Steel doors, windows and ventilators, various types of windows, Glazing – different varieties. **Stair Cases:** Stair cases or Stairway design (Architectural design or planning only) various types such as, straight flight – dog legged, quarter landing, open spiral, spiral stairs etc.

Concrete Technology and Mix Design

Polymer Concrete, Types of cement concretes, Ingredients and their characteristics, Cement concrete properties and relevant tests, storage, batching, mixing & transporting, placing & vibrating and curing. Concrete grades & mix designs upto M 20 as per IS code. Introduction to polymer concrete and it's uses.

TEXT BOOKS:

1. "Engineering Materials" [Material Science], by Rangwala 'Charotar Publications'
2. "Building Materials" by S.K. Duggal New Age International Publishers.
3. "Building Construction" by B.C. Punmia. Laxmi Publications.
4. "Concrete Technology Theory & Practice" by M.S.Shetty, 'S.Chand & Company Ltd.'
5. "Engineering Materials", 5th edition, By Surindra Singh,, Konark Publishers Pvt. Ltd., New Delhi.

REFERENCE BOOKS:

1. "Civil Engg. Materials", by Technical Teachers' Training Institute, Chandigarh, Tata-McGraw-Hill Publishing Company Ltd., New Delhi.
2. "Materials of construction", by R.C. Smith, McGraw-Hill Company, New York.
3. "Materials of construction", by D.N. Ghose, Tata-McGraw-Hill.
4. "Engineering Materials", By Sushil Kumar, Metropolitan Book Co., Pvt. Ltd, New Delhi
5. "Building Construction" Vol.II & III By W.B. McKay, E.L.B.S. and Longman, UK.
6. "Construction Technology" by R. Chudly Vols I & II 2nd Edition Longman, UK.

PROBABILITY, STATISTICS AND QUEUING THEORY
(Common to CSE & IT)

Theory	: 3 Periods	Sessionals	: 30
Tutorials	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

COURSE OBJECTIVES:

1. To discuss various definitions of probability and related theorems, understand the concepts of prior and posterior probabilities used in Baye's theorem and their use in decision making.
2. To illustrate the concept of a random variable and their properties
3. To learn the assumptions, p.m.f.'s /p.d.f.'s and analyse various statistical measures of a few discrete/continuous distributions.
4. To understand and compute the correlation coefficient, and estimation techniques from regression lines.
5. To develop a framework for testing of hypothesis in giving inferences about Population Parameter.
6. To study Queuing models and their Characteristics.

COURSE OUTCOMES:

At the end of the course a student able to

1. Handle the situation of uncertainty in decision making in our day-to-day life.
2. Identify the random variable as discrete/continuous and analyse it.
3. Predict the distribution suitable for the given data from its moments.
4. Measure the intensity of association between the variables and to fit a best suitable Curve for the given data.
5. Decide the test applicable for giving inference about Population Parameter based on Sample statistic.
6. Make business decisions about the resources needed to provide a service in day-to-day life applications including telecommunication, traffic engineering, computing and the design of factories, shops, offices and hospitals.

SYLLABUS**Probability:**

Various definitions of probability, Addition theorem, Conditional probability, Multiplication theorem, Baye's theorem of probability and Geometric probability.

Random Variables And Their Properties:

Definition, Distribution function, Properties of Distribution Function, Discrete Random Variable, Probability Mass Function, Discrete Distribution Function, Continuous Random Variable, Probability Density Function, Continuous Distribution Function. Mathematical Expectation of a Random Variable, Expected Value of function of a Random Variable, Addition Theorem and Multiplication Theorem of Expectation (**without proofs**), Statistical Measures like Mean, Variance, Moments and Covariance in terms of Expectations, Moment generating Function, Characteristic Function and Probability generating Function of a Random Variable.

Distributions:

Discrete Distributions:

Binomial distribution and Poisson distribution (Definition, Mean, Variance, m.g.f., Characteristic function, p.g.f., Fitting of distributions)

Continuous Distributions: Uniform distribution, its Mean and Variance. Normal Distribution and its Properties (without proofs), Standard Normal Variate, Importance of Normal distribution. Exponential Distribution, Definition, Mean Variance Memory less property of Exponential distribution

Bivariate Analysis and Curve Fitting:

CORRELATION: Definition, Karl Pearson's Coefficient of Correlation, Limits for correlation coefficient, Rank Correlation, Spearman's formula for rank correlation coefficient.

CURVE FITTING, Method of least Squares, Fitting of a Straight line, Fitting of a Parabola

REGRESSION ANALYSIS: Regression Lines, Regression Coefficients and their properties (without proofs)

Sampling Theory:

Sample, population, statistic, parameter, Sampling distribution, standard error, interval estimation. Testing of Hypothesis: Formulation of Null hypothesis, Alternative hypothesis, Critical region, Critical value, level of significance, Statistical Inference, Type-I-error, Type-II-error, One-tailed/Two-tailed test.

Large Sample Theory

Test of significance of single sample proportion, Test of significance for difference of proportions, Test of significance of single sample mean, Test of significance for difference of means.

Small Sample Theory: Degrees of freedom, Student's-t-distribution: definition, t-test for single mean, t-test for difference of means, Paired t-test for difference of means.

F-distribution: definition, F-test for equality of two population variances. Chi-square distribution definition, Chi-square test for goodness of fit, Chi-square test for single Population Variance.

Queuing Theory: Queue description, Birth and Death Process, Distribution of Inter-arrival times, Distribution of service times, Kendall's representation of a queueing model, Operating characteristics of a queueing model, steady-state solutions of $\{M/M/1: \infty/FCFS\}$ Model and $\{M/M/1; N/FCFS\}$ Model.

TEXT BOOK:

1. Probability, Statistics and Random Processes by T.Veerarajan, Tata Mc Graw Hill Pub.

REFERENCE BOOKS:

1. Probability & Statistics with Reliability, Queueing and Computer Applications by Kishore.S.Trivedi, Prentice Hall of India, 1999.
2. Fundamentals of Mathematical Statistics by S.C.Gupta and V.K.Kapoor, Sultan Chand & Sons Publishers.

PHYSICS LAB
(Common to CIVIL, CSE & IT)

Lab	: 3 Periods	Sessionals	: 50
Exam	: 3 Hrs.	Ext. Marks	: 50
		Credits	: 2

LIST OF EXPERIMENTS

1. Sonometer – verification of laws of transverse vibrations in stretched strings.
2. Melde’s Experiment – Determination of the frequency of an electrically maintained tuning fork.
3. Newton’s Rings – Determination of radius of curvature of a convex lens.
4. Diffraction Grating – Determination of Wavelengths of lines of mercury spectrum using spectrometer by normal incidence method.
5. Determination of Cauchy’s constants of the material of the given prism using Spectrometer and mercury light.
6. Wedge Method – Determination of thickness of a paper by forming parallel interference fringes.
7. Variation of magnetic field along the axis of current carrying circular coil – Stewart and Gee’s apparatus.
8. Carey Foster’s bridge – Verification of laws of resistance.
9. Lee’s Method – Determination of coefficient of thermal conductivity of a bad conductor.
10. Calibration of voltmeter using potentiometer.
11. Calibration of low range Ammeter using potentiometer.
12. Laser – Diffraction.

REFERENCE BOOK:

1. Advanced Practical Physics Vol I & II by SP Singh and MS Chauhan, Pragathi Prakasam Publications

WORKSHOP
(Common to CIVIL, CSE & IT)

Lab	: 3 Periods	Sessionals	: 50
Exam	: 3 Hrs.	Ext. Marks	: 50
		Credits	: 2

COURSE OBJECTIVES:

1. To impart hands-on training on basic engineering trades.

COURSE OUTCOMES:

Student will be able to

1. Use various tools to prepare basic carpentry and fitting joints.
2. Fabricate simple components using tin smithy.

SYLLABUS

Carpentry

Bench Work, tools used in carpentry.

Jobs for Class work – half lap joint, mortise and tenon joint, half – lap dovetail joint, corner dovetail joint, central bridle joint.

Sheet Metal

Tools used in sheet metal work, Laying development of the sheet metal jobs, soldering.

Jobs for class works – Square tray, taper tray (sides), funnel, elbow pipe joint, 60° pipe joint.

Fitting

Tools used in fitting work, Different files, chisels, hammers and bech vice.

Jobs for class work – Square, hexagon, rectangular fit, circular fit and triangular fit.

REFERENCE BOOK:

1. Elements of workshop technology, Vol.1 by S.K. and A.K. Choudary, Media Promoters and Publishers Pvt. Ltd

ENGLISH LANGUAGE LAB
(Common to All Branches)

Lab	: 3 Periods	Sessionals	: 50
Exam	: 3 Hrs.	Ext. Marks	: 50
		Credits	: 2

The Language Lab focuses on the production and practices of sounds of language and familiarizes the students with use of English in everyday situations and contexts.

COURSE OBJECTIVES:

1. To make students recognize the sounds of English through Audio-Visual aids.
2. To help students build their confidence and help overcome their inhibitions and self consciousness while speaking in English. *The focus shall be on fluency.*
3. To familiarize the students with stress and intonation and enable them to speak English effectively.

COURSE OUTCOMES:

1. Students will be sensitized towards recognition of English sound pattern.
2. The fluency in speech will be enhanced.

SYLLABUS

1. English Sound Pattern – Letters
2. Sounds of English
3. Pronunciation
4. Stress and Intonation

Laboratory Practice Sessions:

1. Letters and Sounds, Worksheet-1
2. Interactions-1, Worksheet-2
3. The Sounds of English, Worksheet-3
4. Interactions-2, Worksheet-4
5. Pronouncing Words-Some Important Patterns, Worksheet-5
6. Interactions-3, Worksheet-2
7. Stress and Intonation, Worksheet-2

REFERENCE BOOKS:

1. Speak Well, Board of Editors, Orient Black Swan Publishers, Hyderabad, 2012.
2. Cambridge English Pronouncing Dictionary, Cambridge University Press, India, 2012.
3. A Textbook of English phonetics for Indian students by T. Balasubramanian, Macmillan publisher, 1981.

NCC/NSS
(Common to CIVIL, CSE & IT)

Contact Hrs. 3 Hrs.

This is an Audit Course. Every student should choose either NCC or NSS at the starting of the semester and pursue the same in that semester. It does not carry any Grade or Credit. It will not be included in the Grade Memo / Certificate.

ADD ON COURSE
TECHNOLOGY COURSE – I
(Professional C Programming)
(Common to All Branches)

Total Hrs: 30

COURSE OBJECTIVES:

1. On successful completion of the course, students will be able to reinforce their learning of C, read code, and write good procedural code.

COURSE OUTCOMES:

1. Students will be able to solve a series of graduated problems
2. Students will be able to do projects in 'C' Language.

SYLLABUS

Introduction to Linux

Introduction to C with problem solving

Functions usage and application with solved problems

Introduction to pointers and applications with examples

Structures and Unions and file handling with program applications

Project using C

RESOURCES:

Talent Sprint Problem Bank, Project Euler



SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (AUTONOMOUS)

(Affiliated to Andhra University, Visakhapatnam), (Recognised by AICTE, New Delhi)

Accredited by NAAC with 'A' Grade

Recognised as Scientific and Industrial Research Organisation

CHINNA AMIRAM (P.O):: BHIMAVARAM :: W.G.Dt., A.P., INDIA :: PIN: 534 204

SCHEME OF INSTRUCTION & EXAMINATION

(Regulation R16)

II/IV B.TECH

(With effect from **2016-2017** Admitted Batch onwards)

Under Choice Based Credit System

CIVIL ENGINEERING

I-SEMESTER

Code No.	Course	Credits	Lecture Hrs	Tutorial Hrs	Lab Hrs	Total Contact Hrs/Week	Sessional Marks	Exam Marks	Total Marks
B16 ENG 2101	Mathematics –IV	4	3	1	--	4	30	70	100
B16 CE 2101	Engineering Mechanics	4	3	1	--	4	30	70	100
B16 CE 2102	Mechanics of Solids	4	3	1	--	4	30	70	100
B16 CE 2103	Fluid Mechanics-I	4	3	1	--	4	30	70	100
B16 CE 2104	Surveying	4	3	1	--	4	30	70	100
B16 ENG 2103	Environmental Studies	2	3	1	--	4	30	70	100
B16 CE 2105	Strength of Materials Laboratory	2	--	--	3	3	50	50	100
B16 CE 2106	Surveying Field Work	2	--	--	3	3	50	50	100
B16 ENG 2104	English Proficiency	2	1	1	--	2	50	50	100
B16 CE 2107	Auto CAD	1	--	--	2	2	50	--	50
Total		29	19	7	8	34	380	570	950

Code: B16 ENG 2101

MATHEMATICS – IV
(Common to CIV,ECE,EEE & ME)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

Students learn

1. The concepts of Gradient, Divergence, Curl, Directional derivative, solenoidal and Irrotational fields
2. Green's, Stokes' and Divergence theorems
3. Classification of 2nd order Partial Differential Equations as well as solution of 1-Dimensional Wave equation and 1-Dimensional Heat equation
4. the concept of Analytic function, CR equations
5. Cauchy's Integral Theorem and Integral Formula
6. Taylor and Laurent series, Residues and Residue theorem

Course Outcomes:

Students will be able to

1. Apply the concepts of Gradient, Divergence, Curl, Directional derivative, solenoidal and Irrotational fields
2. Determine scalar potential, circulation and work done
3. Evaluate integrals using Green's, Stokes' and Divergence theorems
4. Obtain the solution of 1-D wave equation and 1-D heat equation
5. Determine the zeroes and poles of functions and residues at poles
6. Evaluate certain real definite integrals that arise in applications by the use of Residue theorem

SYLLABUS

Vector Calculus-1

Definitions of Scalar and Vector point functions, Differentiation of vectors, Vector differential operator del, Del applied to scalar point function – gradient, Del applied to vector point function- divergence and curl, physical interpretation of gradient, divergence and curl(without proof), Del applied twice to a point function, Del applied to product of two functions, Irrotational and Solenoidal Fields, scalar potential

Vector Calculus-2

Integration of vectors, line integral, circulation, work done, surface integral, Flux, Green's, Stokes' and Gauss Divergence Theorems (Without proofs). Introduction to orthogonal curvilinear coordinates, cylindrical polar coordinates and spherical polar coordinates.

Applications Of Partial Differential Equations

Classification of second order partial differential equations, Method of separation of variables, One –dimensional wave equation- vibrations of a stretched string (no derivation)-, one-dimensional heat equation – Heat flow along a long horizontal bar (no derivation) (problems on heat equation involving homogeneous end conditions only), two dimensional Laplace equation in Cartesian coordinates.

Complex Variables-1

Review- Cartesian form and polar form of a complex variable, Real and imaginary parts of z^n , e^z , $\sin z$, $\sinh z$ and $\log z$.

Limit and continuity of a function of the complex variable, derivative, analytic function, properties of Analytic functions, Cauchy- Riemann equations, Harmonic functions and Orthogonal system, application of analytic function to flow problems, geometric representation of $w=f(z)$, conformal mapping – Bilinear transformation only.

Complex Variables-2

Integration of complex functions, Cauchy's theorem, Cauchy's integral formula (statements only) . Taylor and Laurent series expansions of functions (statement of theorems only), zeros and singularities, Residue, calculation of residues, Cauchy's Residue theorem (without proof), Evaluation of real and definite integrals- integration around a unit circle

Text Book:

1. "Higher Engineering Mathematics", by Dr.B.S.Grewal, 43rd Edition, Khanna Publishers.

Reference Books:

1. Advanced Engineering Mathematics, by Erwin Kreyszig, Wiley.
2. A text book of Engineering Mathematics, by N.P.Bali and Dr. Manish Goyal, Lakshmi Publications.
3. Advanced Engineering Mathematics, by H.K.Dass, S.Chand Company.
4. Higher Engineering Mathematics, by B.V.Ramana, Tata Mc Graw Hill Company.
5. Higher Engineering Mathematics, by Dr. M.K.Venkatraman, The National Publishing Company.

ENGINEERING MECHANICS

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives

Students learn

1. The course uses the Laws of Mechanics to predict forces in and motions of machines and structures.
2. The course is the key prerequisite course to sequences of courses dealing with mechanics of machines, stress analysis and design of mechanical systems.
3. Comprehensive, theory based understanding of the underpinning natural and physical sciences and the engineering fundamentals applicable to the engineering discipline.
4. In-depth understanding of specialist bodies of knowledge within the engineering discipline.
5. Application of established engineering methods to complex engineering problem solving.
6. Application of systematic engineering synthesis and design processes.

Course Outcomes

Students will be able to

1. Analyse 2-D and 3-D force systems by scalar and vector approaches .
2. Analyse for forces in different types of Determinate trusses by 'Method of sections' and 'Method of joints'.
3. Apply method of virtual work to statically determinate structures
4. Distinguish between rectilinear, curvilinear motion of particles and plane motion of rigid bodies .
5. Utilise the principles of kinematics and kinetics for physical bodies .
6. Solve engineering problems .

SYLLABUS**Basic Concepts:**

Introduction to Engineering Mechanics – Scalar and Vector quantities – Forces – Characteristics of a force – Definitions and examples of various types of force systems – Definition of resultant – Composition and resolution of forces – Moment of a force – Principles of moments of force – Couples – characteristics of a couple – on Transformations of a couple – Resolution of a force into a force and couple.

Forces in plane only:

Concurrent forces in plane – principles of statics –composition and resolution of forces – equilibrium of concurrent forces in a plane – method of projections – equilibrium of three forces in a plane –method of moments – friction.

Parallel forces in plane :

Two parallel forces – general case of parallel forces in a plane –centre of parallel forces and center of gravity – centroids of composite plane figures and curves – distributed force in a plane.

General case of forces in a plane :

Composition of forces in a plane – equilibrium of force in a plane – Analysis of statically determinate trusses by (a) Method of joints and (b) Method of sections.

Virtual work:

Introduction – concept of virtual work –Equilibrium of ideal system – application of virtual work on beams(simply supported ,cantilever, continuous beams)carrying point load and uniformly distributed loads.

Kinematics of particles:

Introduction of dynamics – rectilinear motion of particles – curvilinear motion of particles. Kinetics of particles – introduction – Newton's second law of motion – linear momentum of a particle – equations of motion – dynamic equilibrium –equations of motion in terms of radial and transverse components
Energy and momentum methods – introduction – kinetic energy of a particle – principle of work and energy – conservation of energy – principle of impulse and momentum – impulsive motion – impact–direct central impact– problems solving energy and momentum.

Kinematics of rigid bodies:

Introduction – translation – rotation about a fixed axis – equation of rotation of fixed body about a fixed axis – general plane motion –plane motion of rigid bodies –equation of motion for a rigid body – D'Alembert's principle of plane motion.
Energy and momentum methods – introduction – principle of work and energy for a rigid body – forces acting on rigid body – kinetic energy of rigid body in plane motion – conservation of energy – principle of impulse and momentum for the plane motion of a rigid body

Text Books:

1. Engineering Mechanics by Timoshenko and D.H. Young., Tata Mcgrawhill
2. Mechanics for Engineers Statics and Dynamics by F.B. Beer and E.R. Johnston , McGraw-Hill Professional

Reference Books:

1. Engineering Mechanics by Singer, BS Publications
2. Engineering mechanics by Bhavikatti, New Age International Publishers
3. Engineering mechanics by J.L Meriam , John Wiley & Sons (Asia) Pvt. Ltd.

MECHANICS OF SOLIDS

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives

Students learn

1. To introduce the concepts of stress, strain and elastic constants and their relations
2. To familiarize with shear force, bending moment and torsion induced and shear stresses and bending stresses developed for different sections of beams and shafts
3. To impart the knowledge on the concept of Principal Stresses and principal strains
4. To Familiarize with stresses and strains induced in columns

Course Outcomes:

Students will be able to

1. Summarise the behaviour of basic materials under the influence of different external loading conditions and support conditions.
2. Determine shear Force and Bending moments in statically determinate Beams and draw the Diagrams.
3. Examine the different methods to find slope and deflection of beams subjected to loads
4. Estimate the principal stresses & strains and torsional stresses in structural members
5. Evaluate the crippling load for columns with different end conditions.

SYLLABUS**Simple stresses & Strains:**

Definitions of stress and strain – types of stresses and strains – Elasticity – Hooke's law – Stress – Strain diagram for Mild steel – working stress- factor of safety- Lateral strain – Poisson's ratio and volumetric strain – Elastic Moduli and the relationship between them – Bars of varying section – composite bars – temperature stresses.

Strain Energy:

Definition – Resilience – SE due to gradually applied; Suddenly applied and impact loads – simple applications.

Shear Force & Bending Moment Diagrams:

Definition of beam – Types of beams – concept of SF and BM – SF & BM diagrams for cantilever, SS and overhanging beams subjected point loads, UDL, Uniformly varying loads and combination of these loads – point of contraflexure – Relationship b/w S.F, BM and rate of loading.

Flexural Stresses:

Theory of simple Bending – Assumptions–Derivation of Bending equation - ($\frac{M}{I} = \frac{F}{Y} = \frac{E}{R}$)

Neutral axis – Determination of bending stresses – section modulus of rectangular, & Circular sections (Solid and Hollow), I,T, channel sections – Design of simple beam sections.

Shear Stresses

Derivation of shear stress formula – shear stress distribution across various beam sections like rectangular, circular, Triangular, I, T, angle sections, built up beams, Definition of shear centre.

Deflections of Beams: (i) Cantilever (ii) Simply supported and (iii) Over hanging beams using (a) Double integration and (b) Macaulay's method.

Principal Stresses and strains:

Introduction-stresses on an inclined section of a bar under axial loading- compound stresses-Normal and tangential stresses on an inclined plane for biaxial stresses-Two perpendicular normal stresses accompanied by a state of simple shear-Mohr's circle of stress-Principal planes and principal stresses- Construction of Mohr's Circle (graphical Method)

Torsion of Circular Shafts:

Theory of pure Torsion – Derivation of Torsion equation $\left(\frac{T}{J} = \frac{\tau_{max}}{R} = \frac{G\theta}{l} \right)$ – Torsional moment of Resistance – polar section Modulus – power transmitted by a shaft – combined bending and torsion.

Springs

Types of springs – springs in series and parallel – close coiled helical springs.

Buckling of columns:

Introduction – short, medium and long columns – axially loaded compression members – crushing load – Buckling load (or) critical load (or) crippling load – Euler's theory for long columns – Assumptions – Derivations of Euler's critical load formula for various end conditions – Effective length of column – slenderness ratio – limitations of Euler's Theory – Rankine formula – for both long and short columns – column subjected to Eccentric loading – Euler's Method and prof. Perry's formula.

Text Books:

1. Strength of materials by Ramamrutham , Dhanpat rai publishing Company
2. Strength of materials by Vazrani and Ratwani , Khanna Publishers

Reference Books:

1. Elements of strength of materials by Timoshenko and Young,,East West press private Ltd
2. Introduction to mechanics of solids by Popov. Englewood cliffs N.J Prentice Hall
3. Strength of materials by Dr R.K.Bansal, Laxmi Publications (p) Ltd
4. Strength of materials by S.S.Bhavikatti, Vikas Publishing house (p) Ltd

FLUID MECHANICS- I

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course objectives

Students learn

1. A foundation in the fundamentals of fluid mechanics
2. Develop an appreciation for the properties of Newtonian fluids
3. Study analytical solutions to variety of simplified problems
4. Understand the dynamics of fluid flows and the governing non-dimensional parameters
5. Apply concepts of mass, momentum and energy conservation to flows.
6. Apply fundamental principles of fluid mechanics for the solution of practical civil engineering problems of water conveyance in pipes, pipe networks

Course Outcomes

Students will be able to

1. Define fundamental concepts of fluid mechanics as applied to civil engineering and environmental problems
2. Discuss and derive the fundamental mathematical equations of fluid mechanics
3. Solve the problems of water conveyance in pipes, orifices, mouthpieces, notches and weirs
4. Apply conservation laws to derive governing equations of fluid flows
5. Compare hydrostatic and hydrodynamic forces
6. Analyze and design simple pipe systems

SYLLABUS**Fluid Properties and Fluid Statics:**

Introduction & Physical Properties of Fluids – Definition of Fluid, Fluid as Continuum; Mass Density, Specific Weight, Specific Gravity, Specific Volume, Bulk Modulus, Compressibility, Vapour Pressure, Cavitation, Viscosity – Newton's Law of Viscosity, Rheological Diagram; Capillarity and Surface Tension.

Fluid Statics, Pressure and its measurement

Forces Acting on a Fluid Element – Pascal's law; Variation of Pressure in Static Fluid; Absolute, Gauge and Total Pressure; Pressure Measurement – Piezometers, Manometers, Micro-manometers, Mechanical Gauges and Pressure Transducers.

Forces on Immersed Bodies in Static Fluids –

Force on a Plane Surface – Centre of Pressure; Pressure Diagram; Forces on Curved Surfaces; Forces on radial Crest Gates and Lock Gates. Buoyancy & Floatation – Archimedes Principle; Stability of Floating Bodies – Centre of Buoyancy, Metacentric Height and its Determination.

Liquids in Relative Motion

Pressure of Liquids in a Container Subjected to Linear Acceleration and Rotation.

Fluid Kinematics and Conservation of Mass:

Types of Fluid Flow & Methods of Fluid Flow Analysis – Methods of Describing Fluid Motion; Types of Flow – Steady & Unsteady Flows, Uniform & Non-uniform Flows, free and forced vortex motions, Laminar & Turbulent Flows; Streamline, Path line, Streak line; Stream Surface – Stream Tube.

Fluid Kinematics

Translation, Deformation and Rotation of a Fluid Element in Motion; Local, Convective and Total Accelerations; One, Two & Three Dimensional Analysis of Flows.

Ideal Fluid Flow – Stream Function, Velocity Potential; Rotational & Irrotational Flows – Vorticity & Circulation; Laplace Equation in terms of Stream Function and Velocity Potential; Flow Nets.

Principle of Conservation of Mass –

Concepts of System and Control Volume; Continuity Equation in three dimensional Cartesian coordinates; Continuity Equation for flow through a Stream tube.

Fluid Dynamics:

Principle of Conservation of Energy – Equation of Motion for Ideal Fluids, Euler's Equation in Streamline Coordinates, Derivation of Energy Equation through integration of Euler's Equation, Bernoulli's Equation, Energy Correction Factor. Flow measuring devices – Flow Measurement in Pipes – Measurement of Static, Stagnation and Dynamic Pressures and Velocity – Pitot Tube, Prandtl Pitot Tube; Measurement of Discharge through a Pipe using Flow Meters – Venturimeter, Flow Nozzle meter and Orifice meter.

Flow through Tanks and Reservoirs

Measurement of Discharge from Tanks and Reservoirs – Steady and Unsteady Flow through Orifices and Mouthpieces – Small & Large Orifices – Different types of Mouthpieces; Discharge from tanks through Drowned Orifices, Time of Emptying Tanks, Discharge from a Tank with Inflow.

Flow Measurement in Channels

Flow Measurement in Open Channels, Flow Past Weirs and Notches, Sharp Crested and Broad Crested Weirs, Weirs with and without end contractions, Ventilation of Weirs, Triangular Notches, Cipolletti Weir.

Principle of Conservation of Momentum

Momentum of Fluids in Motion, Impulse Momentum Equation, Momentum Correction Factor. Application of Momentum Principle – Forces on Pipe Bends and Reducers, Flow through a Nozzle; Angular Momentum of fluid flow – Sprinkler Problems.

Flow through Pipes:

Introduction to Pipe Flow and Laws of Friction – Reynolds Experiment; Steady Turbulent Flow through Pipes; Laws of Friction; Darcy-Weisbach Equation.

Total Energy and Hydraulic Gradient

Energy and Hydraulic Gradient Lines; Minor Losses in Pipes; Pipes in Series and Parallel – Equivalent Length of Pipe.

Flow between Two reservoirs;

Three Reservoir Problems; Distribution Mains; Working Pressures, Design Pressure and Test Procedures; Choice of Pipe Material; Siphon; Pipe Network Analysis by Hardy-Cross Method; Hydraulic Power Transmission through Pipes and Nozzles, Water hammer (only concept).

Text Books:

1. Fluid Mechanics and Hydraulic Machinery, Modi, P.N. and S.M. Seth, Standard Book House.
2. Fluid Mechanics, Jain, A.K., Khanna Publishers.

Reference Books:

1. Engineering Fluid Mechanics Kumar, K.L., S. Chand & Co. Ltd.
2. Engineering Hydraulics, Rouse, H., John Wiley & Sons Inc.
3. Mechanics of Fluids, Shames, I.H., McGraw-Hill Professional

SURVEYING

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

Students learn

Student shall be able to

1. Learn and understand the basic methods of classifications and principles of surveying.
2. Learn how to differentiate the methods of area calculation used in surveying.
3. Differentiate the Angles and Bearings and able to identify direction of a line
4. Describe Plane tabling methods of Surveying
5. Construct the Leveling methods to identify Elevation of the required points
6. Differentiate the usage of Minor Instruments in surveying.

Course Outcomes:

After the completion of the course student should be able to

1. Appreciate the importance of preparation of Map and Plan for required site with suitable scale.
2. Prepare contour Map and Estimate the Quantity of earthwork required for formation level for Road and Railway Alignment.
3. Judge on which type of instrument to be used for carrying out survey for a specific work
4. Describe different modern instruments used in surveying.

SYLLABUS**Introduction:**

Classification and principles of surveying. Triangulation and Trilateration Earth as spheroid, datum, geoid, Azimuth, latitude, longitude, Map projections, scales, plans, & Maps. Chain surveying: Instrumentation for chaining – Errors due to incorrect chain-Chaining on uneven and sloping ground-Errors in chaining-Tape corrections – Problems: Base line measurement-chain Triangulation-Check lines, Tie lines, Offsets. Basic problems in chaining, obstacles in chaining-Problems-Conventional signs

Compass Survey :

(a) Introduction to compass survey Definitions of Bearing. True bearing, True meridian, Magnetic Meridian, Magnetic bearing – Arbitrary Meridian, R.B. & B.B of lines – Designation of bearings – W.C.B. & R.B. – Conversion of bearings from one system to the other Related problems – Calculation of angles for bearings, Calculation of bearing for angles, Related problems – Theory of Magnetic compass (i.e. Prismatic compass) – Magnetic dip-Description of Prismatic compass. Temporary adjustments of compass-Magnetic Declination – Local attraction-Related Problems-Errors in compass survey.

(b) Traverse Surveying : Chain and compass traversing-Free or loose needle method – Fast needle method-Checks in closed and open traverse-Plotting methods of traverse Survey - Closing error-Balancing the traverse-Bowditch's method-Transist method, Gale's Travers table.

Plane table surveying:

Introduction-Advantages, Accessories-Working operations such as fixing the table to tripod, levelling-centering-orientation by back-sighting. Methods of plane tabling-Plane table traversing-Three point problem – Mechanical method – Graphical method – Two point problem-Errors in plane tabling.

Levelling :

Definitions of terms-Methods of levelling-Uses and adjustments of dumpy level-Temporary and permanent adjustments of dumpy level levelling staves-Differential leveling, Profile levelling-Cross sections-Reciprocal levelling. Precise levelling-Definition of BS, IS, FS, HI, TP-Booking and reduction of levels, H.I. methods-Rise and fall method-Checks-Related problems-Curvature and Refraction Related Problems-Correction-Reciprocal levelling-Related problems-L.S & C.S Levelling-Problems in levelling-Errors in levelling.

Minor instruments :

Uses and adjustments of the following minor instruments:

Line Ranger, Optical Square, Abney level, Clinometer, Ceylon Ghattracer, Pantagraph, Sextant and Planimeter.

Contouring:

Definitions-Interval, Characteristics of contours-methods of locating contours. Direct and indirect methods-interpolation of contours-Contour gradient-Uses of contour maps. Contours mapping using computer techniques (surfer, CAD)

Text Books:

1. Surveying By Dr. K.R. Arora, Standard Book House.
2. Surveying Vol. 1,2 and 3 – By Punmia, Standard Book House.

Reference Books:

1. Surveying Vol. 1 and 2 – By S.K. Duggal, Tata Mc.Graw Hill Publishing Co.
2. Surveying: Theory & Practices by James M. Anderson and Edward M. Mikhail

ENVIRONMENTAL STUDIES
(Common to CIV, CSE & IT)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs	Credits	: 2

Course Objectives:

Students learn

1. To develop an awareness and sensitivity to the total environment and its related problems.
2. To participate actively participation in environmental protection and improvement.
3. To develop skills for active identification and development of solutions to environmental problems
4. To evaluate environment programmes in terms of social, economic, ecological and aesthetic factors.
5. To Create a “CONCERN AND RESPECT FOR THE ENVIRONMENT”

Course Outcomes:

Students will be able to

1. Get awareness among the students about the nature and natural ecosystems.
2. Learn sustainable utilization of natural resources like water, land, minerals, air.
3. Learn resource pollution and over exploitation of land, water, air and catastrophic (events) impacts of climate change, global warming, ozone layer depletion, marine, radioactive pollution etc to inculcate the students about environmental awareness and safe transfer of our mother earth and its natural resources to the next generation.
4. Safe guard against industrial accidents particularly nuclear accidents.
5. Learn Constitutional provisions for the protection of natural resources.

SYLLABUS

Global Environmental Crisis:

Environmental Studies - Definition, Scope and importance, Need for public awareness.
Global Environmental Crisis

Ecosystems:

Basic concepts, Forest Ecosystems, Grassland Ecosystems and Desert Ecosystems, Aquatic Ecosystems

Biodiversity:

Introduction to Biodiversity, Value of Bio-diversity, Bio-geographical classification of India, India as a Mega-diversity habitat, Threats to biodiversity, Conservation of Biodiversity: In-situ and Ex-situ conservation of bio-diversity.

Environmental and Natural Resources Management:

Land Resources: Land degradation, soil erosion and desertification, Effects of modern agriculture **Forest Resources:** Use and over exploitation-Mining and Dams-their effects on forest and tribal people, **Water resources:** Use and over utilization of surface and ground water, Floods, droughts, conflict over water, water logging and salinity, dams – benefits and problems

Energy Resources: Renewable and non-renewable energy sources, use of alternate energy sources-impact of energy use on environment.

Environmental Pollution:

Causes, Effects and Control measures of - Air pollution, Water pollution, Soil pollution, Marine Pollution, Thermal pollution, Noise pollution, Nuclear Hazards; Climate change and global warming, acid rain and Ozone layer depletion.

Environmental Problems in India:

Drinking water, Sanitation and Public health, population growth and environment; Water Scarcity and Ground Water Depletion; Rain water harvesting, Cloud seeding and Watershed management.

Text Books:

1. Environmental Studies (From Crisis to Cure) by R. Rajagopalan, Oxford university Press, 2008
2. Environmental Studies by Anubha Kaushik & C.P. Kauskik, New Age International (P) Ltd, New Delhi, 2006

Reference Books:

1. Environmental Sciences by G.Tyler Miller, JR, 10th ed, Thomson publishers, 2004

STRENGTH OF MATERIALS LABORATORY

Lab : 3 Periods
Exam : 3 Hrs

Sessionals : 50
Ext. Marks : 50
Credits : 2

Course Objectives:

Students learn

1. To introduce various strength and strain measuring equipments
2. To determine various physical, mechanical properties and strength of various engineering materials

Course Outcomes:

Students will be able to

1. Illustrate the stress strain relationship for Mild steel/ HYSD bars (K2)
2. Inspect wood samples for compressive strength (K4)
3. Determine modulus of rigidity of spring
4. Measure the hardness of metals by BHN, Rockwell & Vicker's
5. Relate bolts subjected to double shear
6. Estimate the Impact resistance of materials by Charpy & Izod tests
7. Distinguish simply supported beam and cantilever beam and determine the young's modulus of beam material
8. Solve coplanar force system

LIST OF EXPERIMENTS

1. Tension test on Mild/ HYSD bars.
2. Compression test on wood (parallel and perpendicular to grains).
3. Tests on springs for the determination of rigidity of modulus and spring constant.
4. Brinell's and Rockwell hardness tests.
5. Charpy and Izod impact tests.
6. Double shear test on mild steel specimen.
7. Bending test : Load deflection test for the determination of young's modulus on simply supported and cantilever beam for wood and steel.
8. Verification of Maxwell's reciprocal theorem.

Reference Books:

1. P.N. Singer and P.K. Jha , Elementary Mechanics of solids , New Age International Pvt .Ltd

SURVEYING FIELD WORK

Lab : 3 Periods
Exam : 3 Hrs

Sessionals : 50
Ext. Marks : 50
Credits : 2

Course Objectives:

Student shall be able to

1. Understand and apply the basic methods of Chain Surveying.
2. Understand and apply the basic methods of Compass Surveying.
3. Summarize the different methods of Plane Table Surveying
4. Observe and report the different types of Leveling.

Course Outcomes:

After the completion of the course student should be able to

1. Apply the linear measurement in simple Boundary Surveys.
2. Identify direction of any line using compass survey.
3. Judge on which type of instrument to be used for carrying out survey for a specific work
4. Prepare contour maps and estimate the quantities.

LIST OF EXPERIMENTS**1. Chain Surveying**

- a. Introduction of instruments used for chain survey, Folding and unfolding of chain-Line ranging (direct method)-Pacing.
- b. Chain traversing –Preparation of plan of a residential building by making use of chain, ranging rods, by oblique off-set method, introduction of check line.
- c. Preparation of residential building by perpendicular offset, introduction of tie lines.
- d. Finding the distance between inaccessible points by making use of chain, cross staff, tape, ranging rods; Arrows and field problems of obstacles to chaining.

2. Compass Survey.

- a. Introduction to prismatic compass-Temporary adjustments.
- b. Finding the distance between inaccessible points by making use of compass, tape and ranging rods.
- c. Compass traversing-plotting of a residential building.

3. Plane Table Survey.

- a. Introduction to plane table-Use of its accessories: Two & Three Point Problem.
- b. Finding the distance between inaccessible points by making use of plane table, its accessories-Ranging rods and tape.

4. Levelling.

- a. Introduction to dumpy level, levelling staff. Reading of level staff, temporary adjustments of dumpy level.
- b. Introduction to fly levelling-Booking the readings by height of collimation method.
- c. Introduction to fly levelling-Booking the readings by rise and fall method-To find closing error.
- d. Check levelling.- L.S. & C.S. of a road profile.
- e. Preparation of contour plan for an open area by taking level of the site.
1. Field work examination, for sessional marks.

Reference Books:

1. B.C.Punmia , Ashok Kumar jain , Arun kr. Jain , Surveying I & II ,Laxmi publications

ENGLISH PROFICIENCY
(Common to All Branches)

Theory	: 1 Period	Sessionals	: 50
Tutorial	: 1 Period	Ext. Marks	: 50
Exam	: 3 Hrs.	Credits	: 2

AIM:

Enriching the communicative competency of the students by adopting the activity-based as well as the class-oriented instruction with a view to facilitate and enable them to enhance their language proficiency skills.

Course Objectives:

Students be able to

1. Understand the importance of professional communication.
2. Learn language skills and vocabulary in order to improve their language competency.
3. Know and perform well in real life contexts.
4. Identify and examine their self-attributes which require improvement and motivation.
5. Build their confidence and overcome their inhibitions.
6. Improve their strategies in reading skills.

Course Outcomes:

1. Students enhance their vocabulary and use it in the relevant contexts .
2. They improve speaking skills.
3. They learn and practice the skills of composition writing.
4. They enhance their reading and understanding of different texts.
5. They enrich their communication both in formal and informal contexts.
6. They strengthen their confidence in presentation skills.

SYLLABUS

Speaking Skills

PPT

Describing event/place/thing

Picture Description

Extempore

Debate

Telephonic Skills

Analyzing Proverbs

Vocabulary

Affixes

Pairs of Words

Reading Skills

Reading Comprehension

Reading/Summarizing News Paper Article

Writing Skills

Designing Posters

Essay writing

Resume Writing

Reference Books:

1. Interchange (4th edition) Student's books 1&2 by Jack C. Richards, CUP.
2. Fundamentals of Technical Communication by Meenakshiraman, Sangeta Sharma of OUP
3. English and Communication Skills for Students of Science and Engineering, by S.P.
4. Dhanavel, Orient Blackswan Ltd. 2009
5. Enriching Speaking and Writing Skills, Orient Blackswan Publishers
6. The Oxford Guide to Writing and Speaking by John Seely OUP

(***Note: Sessional Marks will be evaluated based on Continuous Comprehensive Evaluation of the students' Performance - 40M, Attendance – 10M and External Marks will be evaluated based on Presentation Skills – 30M, Project 20M)

Auto CAD

Lab	: 2 Periods	Sessionals	:50
Exam	: 3 Hrs.	Credits	: 1

Course Objectives:

1. Increase ability to communicate with the people.
2. Learn to sketch and take field dimensions.
3. Learn basic AutoCAD skills.
4. Learn basic engineering drawing formats.
5. Prepare the student for future Engineering positions.

Course Outcomes:

1. Student's ability to perform basic sketching techniques will improve.
2. Student's ability to use architectural and engineering scales will increase.
3. Student's ability to produce engineered drawings will improve.
4. Student's ability to convert sketches to engineered drawings will increase.
5. Student's will become familiar with office practice and standards.
6. Student's will become familiar with AutoCAD two dimensional drawings.
7. Student's will develop good communication skills and teamwork.

SYLLABUS**Fundamentals of Computers**

- Introduction
- Computer Hardware and Software Concepts
- Introduction of Personal Computer and Operating Systems WINDOWS-XP, Windows-7, File Management

Drawing using AutoCAD

- Starting a New Drawing/Opening an existing drawing
- Drawing Commands
- Hatching Command Text (multi-line & single line) and Formatting Text Styles
- View Commands & Drawing Settings and Aids
- Modify Commands
- Dimension Command Formatting Dimension Style and Multi-leader Style
- Drawing Settings and Aids
- Saving and Plot

Reference Books:

1. Learning Auto CAD 2010 Volume-I , Autodesk.
2. Auto CAD 2013 fundan1entals- Elisen1oss, SDC Publications

(Note: Total Marks will be evaluated based on Continuous Evaluation - 25 Marks, Design Contest -25 Marks)

SCHEME OF INSTRUCTION & EXAMINATION
(Regulation R16)

II/IV B.TECH
(With effect from **2016-2017** Admitted Batch onwards)
Under Choice Based Credit System

CIVIL ENGINEERING

II-SEMESTER

Code No.	Course	Credits	Lecture Hrs	Tutorial Hrs	Lab Hrs	Total Contact Hrs/Week	Sessional Marks	Exam Marks	Total Marks
B16 CE 2201	Analysis of Structures	4	3	1	--	4	30	70	100
B16 CE 2202	Reinforced Concrete Structures	4	3	1	--	4	30	70	100
B16 CE 2203	Fluid Mechanics-II	4	3	1	--	4	30	70	100
B16 CE 2204	Building Planning & Design	4	2	--	3	5	30	70	100
B16 CE 2205	Advanced Surveying Methodologies	4	3	1	--	4	30	70	100
B16 CE 2206	Engineering Geology	4	3	1	--	4	30	70	100
B16 CE 2207	Total station and Geomatics lab	2	--	--	3	3	50	50	100
B16 CE 2208	Engineering Geology Lab	2	--	--	3	3	50	50	100
B16 CE 2209	Fluid Mechanics Lab-I	2	--	--	3	3	50	50	100
B16 CE 2210	Industry Oriented Technology Lab	1	--	--	2	2	50	--	50
Total		31	17	5	14	36	380	570	950

Code: B16 CE 2201

ANALYSIS OF STRUCTURES

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives

Students learn

1. To familiarize with the deflection of simple determinate beams
2. To Familiarize with stresses and strains induced in Thin& Thick cylinders
3. To impart knowledge on various energy methods and theories of failure
4. To analyze the propped cantilever and fixed beams
5. To determine the deflections of determinate beams using energy methods
6. To draw the influence lines and moving loads for statically determinate beams
7. To analyze the statically indeterminate structures using energy methods

Course Outcomes:

Student will be able to :

1. Determine deflections in determinate beams by different methods(K5)
2. Evaluate the strain energy for structural members subjected to different loads(K5)
3. Analyse different indeterminate beams for BM and SF by different methods of analysis(K4)
4. Determine reactions, BM & SF in beams subjected to moving loads. (K5)
5. Distinguish between thin and thick cylinders and understand different failure theories.(K4)

SYLLABUS

Deflections of Beams using (i) Moment area method, (ii) Conjugate beam method, (iii) Unit load method, (iv) Castigliano's theorem-1.

Strain- energy due to (i) Axial load (ii) Bending Moment (iii) Shear force and (iv) Torque

Deflections of Statically Determinate Structures: (a) Single storey, single bay rectangular portal frames using (i) Unit load method, (ii) Castigliano's theorem-1. (b) Trusses (having 9 members or less) using (i) Unit load method, (ii) Castigliano's theorem-1.

Propped Cantilevers:

Analysis of propped cantilever by method of consistent deformation.

Fixed Beams:

Fixed end moments for beams of uniform section for different types of loading; Effect of sinking of support; effect of Rotation of a support; BMD for fixed beam.

Analysis of continuous beams by

1. Theorem of three moments
2. Slope deflection method
3. Moment distribution method
4. Kani's method.

Influence Lines:

Definition – Influence line for Reaction, SF and BM-Load position for Max SF at a section – Load position for max BM at a section- Single point load, U.D.L longer than the span, U.D.L. shorter than the span- Focal length.

Moving Loads:

Introduction – Max SF and BM at a given section and absolute Max SF and BM due to single concentrated load, U.D.L. longer than the span, U.D.L. shorter than the span, two point loads with fixed distance between them and several point loads, Maximum Bending moment at a section under a wheel load and absolute maximum Bending moment in the case of several wheel loads- Equivalent uniformly Distributed load.

Thin cylinders :

Calculation of longitudinal and hoop stresses in thin cylinders subjected to internal pressure, Wire wound thin cylinders.

Thick cylinders- Lamé's theory, Compound tubes.

Theories of failures (i) Maximum Principal stress theory, (ii) Maximum Principal strain theory, (iii) Maximum shear theory (iv) Maximum strain energy theory and (v) Maximum distortion theory.

Text Books:

1. Structural Analysis Volume _II By Vazrani and Ratwani , Khanna Publishers
2. Strength of Materials – Ramamrutham, Dhanpat Rai Publishing Company

Reference Books:

1. Elementary strength of materials – Timoshenko and Young , East west press Pvt Ltd
2. Strength of materials by Dr. R.K. Bansal ,Laxmi Publications (p) Ltd
3. Strength of materials – Volume – I by S.S.Bhavikatti , Vikas Publishing house (p) Ltd

REINFORCED CONCRETE STRUCTURES

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

Students learn to

1. Explain concept of limit state of collapse in flexure, shear, torsion & compression. (K2)
2. Estimate ultimate moment of resistance of singly, doubly reinforced rectangular beams and flanged beams by using Limit state Method. (K3)
3. Design flexural members such as singly, doubly, flanged, reinforced rectangular beams, one way and two way rectangular slabs and staircase by using Limit state Method. (K4)
4. Design and detailing of members subjected to flexure, bond, shear and torsion by using Limit state Method. (K4)
5. Design axially loaded compression members and combined action of axial load with uniaxial bending and bi-axial moments isolated square and rectangular footing by using Limit state Method. (K4)

Course Outcomes:

1. Student should be able to understand and Design the Super structure and sub structure elements.

General : Loading standards as per IS 875, Grades of steel and cement, Stress- Strain characteristics of concrete and steel, Introduction to working stress method and Limit State Method (L.S.D.) of design.

Limit State of Collapse of in Flexure: Introduction and Principles of L.S.D., Characteristic load and strengths, Design values, Partial safety factors, Factored loads.

Limit State of Collapse: Under reinforced, Balanced and over reinforced sections. Compression stress block, Analysis of singly reinforced rectangular section, Doubly reinforced rectangular section and singly reinforced flanged sections. Guide lines for choosing width, depth and percentage of reinforcements in beams. Design of singly reinforced rectangular section, Doubly reinforced rectangular section and singly reinforced flanged sections. Curtailment of flexural tension reinforcement.

Shear, Torsion and Bond: Limit state of collapse in shear, Modes of cracking, shear transfer mechanisms, shear span - depth ratio, shear failure modes. Nominal shear stress, critical sections for shear design, types of shear reinforcement. Truss analogy. General procedure for design of rectangular beams for shear. Limit state of collapse in torsion, Torsional shear stress in rectangular and flanged sections. Reinforcement in member subjected to torsion in RC beams. Design of RC beams subjected to combined effect of bending, shear and torsion. Concept of bond, development length, anchorage, bond, flexural bond.

Design of slabs: One way and Two-way action of slabs, Choosing slab thickness. Design of one way slab. Design of restrained and unrestrained Two way slabs as per I.S. code provision. Shear forces in uniformly loaded Two-way slabs.

Columns: Define short and long columns, estimation of effective length of a column. Code requirements on slenderness limits, minimum eccentricity and reinforcement. Design of short column under axial compression with lateral ties and helical reinforcement. Design of short columns subjected to combined axial load and uniaxial moment. Design of short columns subjected to combined axial load and biaxial moment. Design of isolated square and rectangular footing.

Text Books :

1. Limit State of Design of Reinforced Concrete – P. C. Vergheese ,PHI Learning (P) Ltd
2. Reinforced Concrete Limit state Design – A.K. Jain , Nemchand & bros ,Roorke
3. R.C.C Design – Unnikrishna Pillai and Vasudeva Menon , McGrawhill professional

Reference Books:

1. Reinforced Concrete Limit state Design, P. Dayaratnam , oxford & IBH publish & co Pvt Ltd
2. Reinforced Concrete Structures by R Park and Paulay ,John Wiley & Sons (p) Ltd.

FLUID MECHANICS-II

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives

Students learn

1. To understand Laminar ,turbulent flows and concepts of boundary layer theory
2. To know the concepts of the discharge through canals and closed conduits
3. To understand the principles of open channel flows.

Course Outcomes:

Students will be able to

1. Solve Turbulent Flow problems.
2. Explain development of boundary layer in external and internal flows
3. Identify variation of lift and drag coefficients with variation in flow
4. Develop an expression for the discharge of uniform flow in open channels
5. Interpret the specific energy diagram for a free surface flow
6. Analyse practical problems in varied flow

SYLLABUS**Viscous Effects on Fluid Motion:**

Equation of Motion for Real Fluids – Modifications in Equation of Motion, Stress Strain Relationships, Tangential Stress Terms.

Navier-Stokes Equations (No Derivation) – N.S. equations for standard cases of Plane two Dimensional and Axisymmetric Flows.

Plane Two- dimensional Flows – Steady Flow between Parallel Plates, Couette and Poiseuille Flows; Axisymmetric Flows – Flow through a Circular Annulus, Flow without and with Pressure Gradient – Hagen-Poiseuille Equation; Relationship between Friction factor and Reynolds Number for Laminar Flow through Pipes; Stokes' law.

Boundary Layer Theory: Theory of Boundary Layer – Characteristics of Laminar Boundary Layer – Boundary Layer growth over a Flat Plate (without pressure gradient) – Boundary Layer Thickness and its Characteristics – Displacement, Momentum and Energy Thicknesses; Stability Parameter; Laminar and Turbulent boundary layers.

Boundary Layer Separation – Mechanism of Separation, Control of B.L. Separation; Boundary Layer on rough surfaces - Laminar Sublayer, Shear friction velocity; Friction Drag.

Turbulent Flow: Critical Reynolds Number – Characteristics of Turbulent Flow – Mean and Fluctuating Components of Velocity, Quantitative Description of Turbulence, Statistical Nature of Turbulent Flow, Isotropic and Homogeneous Turbulence.

Analysis of Turbulent Flows – Shear Stress due to turbulence – Semi-empirical Theories, Boussinesq Eddy Viscosity Model, Prandtl Mixing Length Concept; Velocity distribution for hydrodynamically smooth and rough pipes; Variation of Friction Factor in turbulent flow;

Friction Factor for commercial pipes – Moody diagram.

Drag, Lift & Propulsion:

Concepts of Drag and Pressure Distribution over Immersed Bodies – Drag and Lift – Deformation Drag, Friction Drag, Form Drag – Drag coefficient.

Distribution of Fluid Pressure on immersed bodies – Pressure Distribution for flow past a circular disk, sphere; Effects of eddy pattern in two dimensional flow – Distribution of pressure for two dimensional flow past a cylinder – von Kármán vortex trail, Eddy shedding; Drag of immersed bodies – Variation of Drag Coefficient with Reynolds Number – Drag on Cylinder – Resistance diagram for bodies of revolution; Drag Coefficient of Practical Bodies.

Lift & Propulsion

Effect of Circulation in Irrotational Flow, Generation of Lift around a Cylinder, Magnus Effect, Computation of Lift Force; Lift on Airfoil – Lift Coefficient and its Variation with Angle of Attack, Joukowski Profile, Polar Diagram, Stall; Induced Drag.

Open Channel Flows:

Basic Concepts – Introduction, Classification of Open Channels – Classification of Flow; Channel Geometry – Geometric Elements of a Channel Section; Velocity Distribution in a Channel Section; Wide Open Channel; Measurement of Velocity; Velocity Distribution Coefficients; Pressure Distribution in a Channel Section – Effect of Slope on Pressure Distribution; Basic Equations – Chezy's Equation, Manning's Equation.

Uniform Flow Computation;

Conveyance of a Channel Section – Section Factor and Hydraulic Exponent. Flow Characteristics in a Closed Conduit with Open Channel Flow; Determination of Normal Depth and Velocity; Design of Channels for Uniform Flow; Design of Non-erodible Channels; Best Hydraulic Section; Determination of Section Dimensions for Uniform Flow; Most Economical Channel Sections – Rectangular, Trapezoidal, Circular and Triangular Channel Sections; Critical Flow – Computation of Critical Flow, Section Factor for Critical Flow.

Application of Energy Principle in Open channels –

Definition of Specific Energy, Specific Energy Diagram, Critical depth, Critical Velocity, Conjugate or Alternate Depths, Sub-critical, Critical and Super-critical Flows, Froude Number, Relationship between Critical depth and Specific Energy for Rectangular, Trapezoidal Sections; Application of Momentum Principle in Open channels – Specific Force; Canal Transitions – Change of Depth in Channels with Change in Cross-section and Hump in the Bed; Control Sections; Venturi Flume and Parshall Flume.

Varied Flow in Open Channels:

Analysis & computation of G.V.F: Definition of G.V.F. and Derivation of Governing Equation – Mild, Steep, Critical, Horizontal and Adverse Slopes – Backwater and Drawdown Curves – G.V.F. Profiles for Channels with Changing Slopes; Computation of G.V.F. Profiles – Method of Direct Integration (Procedures Only), Direct Step Method – Computation of G.V.F. Profiles in rectangular channels using Direct and Single Step methods (Simple Slope cases only).

Rapidly Varied Flow – Hydraulic jump, Types of jump, Hydraulic jump in horizontal rectangular Channels; Surges.

Text Books:

1. Fluid Mechanics and Hydraulic Machinery, Modi, P.N. and S.M. Seth, Standard Book House.
2. Fluid Mechanics, Jain, A.K., Khanna Publishers.

Reference Books:

1. Engineering Fluid Mechanics, Kumar, K.L., S. Chand & Co. Ltd.
2. Flow in Open Channels, Subramanya, K., Tata McGraw-Hill Publishing Co. Ltd.
3. Flow through Open Channels, Ranga Raju, K.G., Tata McGraw-Hill Publishing Co. Ltd.
4. Open Channel Hydraulics, Chow, V.T., McGraw-Hill Ltd.

BUILDING PLANNING & DESIGN

Theory	: 2 Periods	Sessionals	: 30
Lab	: 3 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

Student shall be able to

1. Learn and understand the Conventional Signs.
2. Differentiate the Types of Residential Buildings.
3. Learn the different influencing parameters and factors in the design of Residential Building.
4. Design and Draw the Site Plan, Dimensional Plan, Front Elevation and Cross Section Elevations.
5. Develop Residential Plan using Auto Cad.

Course Outcomes:

After the completion of the course student should be able to

1. Use the Conventional Signs in Design
2. Design Different Types of Residential Buildings
3. Appreciate influencing parameters in the design of Residential Building
4. Develop Site Plan, Dimensional Plan, Front Elevation and Cross Section Elevations.
5. Use the Auto Cad in the Design of Residential Buildings

SYLLABUS

Residential Buildings : Different types of Residential Buildings Selection of Site for Residential Building. Brief Information of Housing Colonies for Different Income Groups in India-Sizes of Plots - Public Spaces, Evolutionary Housing Concept.

Climatology: Elements of Climate : Sun, Wind, Relative Humidity, Temperature effects, Comfort Conditions for House, various types of Macro Climatic Zones. Design of Houses and Layouts with Reference to Climatic Conditions. Orientation of Buildings. Solar Charts, Ventilation. Principles of Planning Anthropometric Data

Preliminary Drawings : (a) Conventional signs of materials various equipment used in a Residential Building (copying exercise) (b) Plan section and Elevation of a small House (one room and varandah) (copying exercise) (c) Plan section and Elevation of Two Bed Room House (copying exercise) (d) (e) (f) Plan section and Elevation of three bed room house in Hot and Humid zone, Hot and Arid zone, cold zone (copying exercises)
(a) Design of Individual rooms with particular attention to functional and furniture requirements. Building regulations and Byelaws of Residential Buildings;
(b) Auto Cad drawing of residential building (only for internal assessment)

Drawing the Plan Section and Elevation of Houses with given Functional requirements and

climatic data. (Emphasis may be given to Hot and Humid zones.)

Text Books:

1. Building Planning and Drawing by Dr.N. Kumara Swamy and A.Kameswara Rao, Charotar Publishing House.
2. Building Planning Drawing and Scheduling by Gurucharansingh and Jagadish Singh, Standard Publishers Distributors.

Reference Books:

1. Civil Engineering Drawing Series 'B' by R.Trimurty, M/S Premier Publishing House.
2. Building Drawing with an integrated approach to Built environment by M.G.Shah, C.M.Kale and S.Y.Patki, McGraw-Hill Publishing Company Limited, New Delhi.

ADVANCED SURVEYING METHODOLOGIES

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objective:

Student shall be able to

1. Learn and understand the different methods of Theodolite surveying.
2. Differentiate the Concepts of Tachometry Surveying.
3. Learn the different methods of Curve Setting.
4. Explore the methods of RS and GIS
5. Describe the procedures available in Total Station Instrument for surveying.

Course Outcome:

After the completion of the course student should be able to

1. Appreciate the importance of Theodolite in Surveying
2. Apply Concepts of Tachometry in Surveying.
3. Construct the Curves in Highways, road construction and canal works.
4. Use the RS and GIS in designing
5. Use the Total Station in Surveying.

SYLLABUS

Theodolite- Types of Theodolite – Temporary Adjustments, Measurement of horizontal angle – Method of repetition, Method of reiteration – Uses of theodolites – Errors in theodolite or Permanent adjustments of a theodolite – Identification – Rectifying the errors.

Theodolite traversing – Open and closed traverse – Closing errors, Balancing the error – Bowditch method – Transit method, Omitted measurements – Gales traverse table or Trigonometric levelling – Elevation of top of the tower - same plane - Different planes – Axis signal correction.

Tacheometry – Principle of tacheometry – Stadia methods – Fixed hair method – Movable hair method – Tangential method – Subtense bar – Beaman's stadia, Arc – Reduction diagrams or Triangulation – Classification-intervisibility of station – Signals and towers-base line measurements – Corrections – Satellite station and Reduction to centre – Basenet.

Curves – Simple curves – Elements of simple curves – Methods of setting simple curves – Rankine's method – Two theodolite method – Obstacles in curve setting – Compound curves – Elements of compound curves or Reverse curves – Elements of reverse curve – Determination of various elements – Transition curves – Ideal shape – Spiral transition curves - length of transition curve - Setting out methods.

Total Station Surveying: Electronic Theodolite, Electronic Distance Measurements, Total Station, Errors in measurements, Advantages, Disadvantages, Applications; Contour

mapping, determination of height of remote point, position of hidden point, free station, Area measurement, volume measurement.

Modern surveying and mapping: GPS survey's – Introduction, Errors in GPS, Positioning methods, classification of GPS surveying, applications, advantages and disadvantages, photogrammetric surveying; sensors & platforms, aerial photogrammetry, Satellite images resolution, concept of stereo models, photogrammetric products, rectified images, orthophotography, topographic map, digital maps, DEM, GIS, Advantages & Disadvantages of photogrammetric surveying.

Text Books:

1. Surveying By Dr. K.R. Arora, Standard Book House.
2. Surveying Vol. 1,2 and 3 – By Punmia, Standard Book House.

Reference Books :

1. Surveying Vol. 1 and 2 – By S.K. Duggal, Tata Mc.Graw Hill Publishing Co.
2. Principles of GIS for land resource assessment by P.A. Burrough – Clarendon Press, Oxford.

ENGINEERING GEOLOGY

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

Students learn

1. To identify the formation of minerals
2. To understand the mega-scopic identification of rocks and minerals
3. To understand the importance of geophysical methodologies
4. To understand the geological maps

Course outcomes:

After completion of the lab, Students will be able to

1. Elucidate the mega-scopic identification of rocks
2. Categorize the rocks according to mega-scopic description
3. Interpret geological maps
4. Estimate the types of subsurface formation by using geophysical methods

SYLLABUS

Introduction to General Geology:

Importance of geology from civil engineering point of view. Branches of Geology. Weathering – types and its engineering importance; Erosion, Soils: Soil profile, soil formation, types of Indian soils. Land forms produced by, running water and glaciers. Land forms produced by Wind, Sea Waves and Currents. Ground Water: origin, groundwater table, porosity and permeability. Aquifers, Groundwater Moment and Water, Bearing Properties of Rocks.

Mineralogy & Petrology:

Mineralogy: Mineral definition, physical properties of minerals. Study of important rock forming minerals: Silicate structures, Quartz, Feldspars, Pyroxenes, Amphiboles, Micas and Clays.

Petrology: Definition of rock. Types of rocks - Ingenious rocks: Granite, Synite, Dolerite, Gabro, Diorite, Basalt. Sedimentary rocks: Breccia, Conglomerate, Sandstone, Shale, Limestone. Metamorphic rocks: Gneiss, Khondalite, Schist, Slate, Marble, Quartzite, Charnokite. Engineering properties of rocks.

Statigraphy & Structural Geology:

Stratigraphy: Geological Time scale, Major geological formations of India and their geological importance - Achaeans, Cuddapahs, Vindhyans, Gondwanas and Deccan Traps. Mineral resources of Andhra Pradesh.

Structural Geology: Elements of Structural Geology- Strike, Dip, Plunge. Working principles of Clinometer compass and Brunton Compass and their use in Civil Engineering. Study of Geological Structures - Folds, Faults and Joints.

Remote Sensing and Geophysical Methods:

Remote Sensing: Introduction, Electromagnetic Spectrum, Aerial Photographs: types of aerial photos and flight planning. Aerial mosaics. Elements of photo interpretation. Satellite Remote Sensing: Satellites, Sensors and Data Products. Principles of Geographical Information Systems. RS and GIS applications to Civil Engineering.

Geophysical Methods: Principles of Geophysical Methods, Electrical, Seismic, Gravity and Magnetic. Principle of Resistivity method and configurations. Applications of Resistivity Method: Soil Profile, Hard rock and Ground Water Table. Principles of Seismic refraction and reflections methods and their applications to Civil Engineering problems.

Geological Investigations:

Role of Engineering Geologist in planning, design and construction and post construction stages in Civil Engineering works. Geological investigations for Dams and Reservoirs and Tunnels, Case Studies – Nagarjuna Sagar, Bhakra Nangal, Jawahar Tunnel. Geological investigations for bridges and multistoried structures. Geological investigations for highways, air fields and railway lines. Geological investigations for Coastal structures and Environmental Geology.

Text Books:

1. Engineering and General Geology by Parbin Singh – Katson Publishing House
2. Engineering Geology by N.Chennakesavulu, Mc-Millan, India Ltd. 2005
3. Principles of Engineering Geology by KVGK Gokhale. B.S Publications-2005
4. Principles and Applications of Photo Geology by Pandey, Willey Eastern Limited
5. Engineering Geology by K.M.Bangar

Reference Books:

1. Engineering Geology and Geo-techniques by F.G. Bell
2. Fundamentals of Remote Sensing by George Joseph. University Press (India) Private Limited.
3. Fundamentals of Engineering Geology by F.G. Bell, Button Wortus Landon

Code: B16 CE 2207

TOTAL STATION AND GEOMATICS LAB

Lab : 3 Periods
Exam : 3 Hrs.

Sessionals : 50
Ext. Marks : 50
Credits : 2

Course Objectives:

Student shall be able to

1. Study and identify the different methods of Theodolite surveying.
2. Distinguish the Concepts of Tachometry Surveying.
3. Learn the different methods of Curve Setting.
4. Explore the methods of RS and GIS
5. Describe the procedures available in Total Station Instrument for surveying.

Course Outcomes:

After the completion of the course student should be able to

1. Relate the importance of Theodolite in Surveying
2. Apply Concepts of Tachometry in Surveying.
3. Construct the Curves in Highways, road construction and canal works.
4. Use the RS and GIS in designing
5. Use the Total Station in Surveying.

LIST OF EXPERIMENTS

1. Measurement of Horizontal Angles by Repetition & Reiteration, Measurement of Vertical Angles, Heights & Distances
2. Distance between two in-accessible points by theodolite
3. Tachometry
4. Setting out curve by deflection angle method by two theodolites
5. Point positioning using GPS
6. Contour mapping using total station
7. Height of remote point using total station
8. Position of hidden point using total station
9. Area & volume measurement using total station
10. GIS related surveying applications

Reference Books:

1. B.C.Punmia , Ashok Kumar jain , Arun kr. Jain , Surveying I & II ,Laxmi publications

ENGINEERING GEOLOGY LAB

Lab : 3 Periods
Exam : 3 Hrs.

Sessionals : 50
Ext. Marks : 50
Credits : 2

Course Objectives: Students learn

1. To identify the formation of minerals
2. To understand the mega-scopic identification of rocks and minerals
3. To understand the importance of geophysical methodologies
4. To understand the geological maps

Course Outcomes:

After completion of the lab, Students will be able to

1. Elucidate the mega-scopic identification of rocks
2. Categorize the rocks according to mega-scopic description
3. Interpret geological maps
4. Estimate the types of subsurface formation by using geophysical methods

LIST OF EXPERIMENTS

1. Study of physical properties and identification of minerals
2. Identification of rocks and their Engineering properties
3. Description and Identification of Geomorphological models
4. Description and Identification of Structural models
5. Geophysical methods – Electrical Resistivity & Seismic Methods
6. Simple Structural Geology problems

Lab Examination Pattern:

1. Description and identification of SIX minerals
2. Description and identification of SIX rocks (Igneous, Sedimentary and Metamorphic rocks)
3. Problem on geophysical method
4. Problem on Strike and Dip

Reference Books:

1. Engineering and General Geology by Parbin Singh – Katson Publishing House

FLUID MECHANICS LAB-I

Lab : 3 Periods
Exam : 3 Hrs.

Sessionals : 50
Ext. Marks : 50
Credits : 2

Course objectives

Students learn

1. To verify the principles of channel flow in laboratory by conducting experiments

Course Outcomes:

Students will be able to

1. Define and Measure Fluid Properties.
2. Illustrate Flow Measuring Devices used in pipes, channels and Tanks
3. Analyze characteristics of broad crested weir.
4. Illustrate the characteristics of surface profiles in free and forced vibrations.
5. Compare sharp crested full width and contracted weirs

LIST OF EXPERIMENTS

1. Study of Small orifice, by constant head method and Time of emptying a tank through a small orifice.
2. Study of Cylindrical mouthpiece by constant head method and Time of emptying a tank through a cylindrical mouthpiece.
3. Study of floating body and determination of Meta-centric Height.
4. Study of surface profiles in Free and Forced Vortex motions.
5. Study of Venturimeter.
6. Study of Orifice meter.
7. Study of Flow nozzle meter.
8. Study of Sharp – crested full width and contracted weirs.
9. Study of V-notch and Trapezoidal notch.
10. Study of Broad-crested weir.
11. Study of Frictional Resistance in pipes.
12. Study of types of flow in pipes

Reference Books:

1. Fluid Mechanics and Hydraulic Machinery, Modi, P.N. and S.M. Seth, Standard Book House.

INDUSTRY ORIENTED TECHNOLOGY LAB

Lab	: 2 Periods	Sessionals	: 50
Exam	: 3 Hrs.	Credits	: 1

**ADVANCED COMPUTATION SURVEYING / GEOINFORMATICS / GEOMATICS
ENGINEERING****Course Objectives:**

This course is design to fulfill the following objectives

1. To prepare the student to plan and conduct field work and application of scientific methodology in handling field samples.
2. To interpret the False Color Composite Images of Satellite data.
3. To Create Fundamental Knowledge on GIS software.
4. To learn the identification of Geographic Coordinates by GPS.
5. To Study the land use/ Land cover dynamics of certain region.

Course Outcomes:

Students who successfully complete this course will be able to:

1. Fully equipped with various surveying concepts and methods using advanced ground survey equipment's.
2. Carry out profiling and grid leveling, for generation of profiles, contour maps, and earth works computations.
3. Handle the Satellite images and interpret the satellite data.
4. The interpret data can be used to prepare plan for urban development/town planning.
5. Prepare the candidates with National Global employability.

LIST OF EXPERIMENTS

1. Transferring and Drafting the collected raw data from total station survey using AutoCAD
2. Computation on drafted data using AutoCAD.
3. Developing Contour using raw data from total station using surfer software.
4. Visual Interpretation of standard FCC (False color composite).
5. Digitization of physical features on a map / image using GIS software.
6. Coordinates measurement using GPS.
7. Field data collection under national land use / land cover mapping on 1:120000 scale using temporal AWIFS data.
8. Asset Mapping of village using Bhuvans Panchayat Moblie App.

Reference Books:

1. A M Chandra : Higher surveying
2. T M Lillesand et al: Remote sensing & Image Interpretation
3. B.Bhatta : Remote sensing&GIS
4. M.Anjireddy: Remote sensing & GIS,BS Publications

5. N K Agarwal : essentials of GPS,Spatial Networks, Hyderabad.

(Note: Total Marks will be evaluated based on Continuous Evaluation - 25 Marks, Record/Report-10 Marks, Exam-10 Marks and Attedndance-5 Marks)



SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (AUTONOMOUS)

(Affiliated to Andhra University, Visakhapatnam), (Recognised by AICTE, New Delhi)

Accredited by NAAC with 'A' Grade

Recognised as Scientific and Industrial Research Organisation

CHINNA AMIRAM (P.O):: BHIMAVARAM :: W.G.Dt., A.P., INDIA :: PIN: 534 204

SCHEME OF INSTRUCTION & EXAMINATION

(Regulation R16)

III/IV B.TECH

(With effect from 2016-2017 Admitted Batch onwards)

Under Choice Based Credit System

CIVIL ENGINEERING

I-SEMESTER

Code No.	Course	Credits	Lecture Hrs	Tutorial Hrs	Lab Hrs	Total Contact Hrs/Week	Sessional Marks	Exam Marks	Total Marks
B16 CE 3101	Structural Analysis	4	3	1	--	4	30	70	100
B16 CE 3102	Reinforced Concrete Structures - II	4	3	1	--	4	30	70	100
B16 CE 3103	Steel Structures	4	3	1	--	4	30	70	100
B16 CE 3104	Geotechnical Engineering - I	4	3	1	--	4	30	70	100
B16 CE 3105	Environmental Engineering - I	4	3	1	--	4	30	70	100
#ELE-I	Elective-I	4	3	1	--	4	30	70	100
B16 CE 3111	Environmental Engineering Lab	2	--	--	3	3	50	50	100
B16 CE 3112	Geotechnical Engineering Lab I	2	--	--	3	3	50	50	100
B16 ENG 3102	Verbal & Quantative Aptitude-I	2	5	--	--	5	100	--	100
#M-I	MOOCS -I	2	4	--	--	4	100	--	100
#M-II	MOOCS -II	2	4	--	--	4	100	--	100
Total		34	31	6	6	43	580	520	1100

#ELE-I	B16 CE 3106	Concrete Technology
	B16 CE 3107	GIS & Remote Sensing
	B16 CE 3108	Pre Stressed Concrete
	B16 CE 3109	Building Services and Maintenance
	B16 CE 3110	Watershed Management
#M-I	B16 CE 3113A	Geo Environmental Engineering
	B16 CE 3113B	Higher Surveying
	B16 CE 3113C	Integrated Waste Management for a Smart City
	B16 CE 3113D	Introduction to Accounting and Finance for Civil Engineers
#M-II	B16 CE 3114A	Fire protection, Services & Maintenance Management for Building
	B16 CE 3114B	Modern Construction Materials
	B16 CE 3114C	Principles of Construction Management
	B16 CE 3114D	Reinforced Concrete Road Bridges

STRUCTURAL ANALYSIS

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

Students learn

1. Understand the basic configuration and classification of structures.
2. To analyze indeterminate structures.

Course Outcomes:

Students will be able to

1. Analyze indeterminate trusses and frames
2. Evaluate indeterminate trusses and its application in the field.
3. Analyze two and three hinged arches and its application.

SYLLABUS

Analysis of statically indeterminate trusses (having not more than 7 members and 3 supports) containing (a) external redundant supports (b) internal redundant members using (i) method of consistent deformation of unit load method (ii) Castigliano's theorem – II.

Analysis of statically indeterminate frames (single storey, single bay portal frames only) using (i) slope-deflection method (ii) moment distribution method (iii) Kani's method

Arches: Introduction, Geometrical properties, Arch action, Normal thrust, radial shear and bending moment in three hinged and two hinged parabolic and segmental arches. Effects of yielding of supports, rib-shortening and temperature changes.

Cables and Suspension Bridges: Stresses in loaded cables with supports at the same and different levels. Length of cable; Three hinged stiffening girders.

Introduction to matrix methods of structural analysis (very elementary treatment only) static indeterminacy, kinematic indeterminacy, Stiffness and flexibility method for two span continuous beams only.

Text Books:

1. Statically indeterminate structures – C.K. Wang, Mc Graw Hill Education PVT.LTD
2. Mechanics of structures Vol. II- S.B.Junnarkar and Dr.H.J.shah, Charotar Publishing House.
3. Structural analysis – Devdas Menon, Narosa Publishing House PVT.LTD

Reference Books:

1. Basic structural Analysis- C S Reddy McGraw Hill Education PVT.LTD
2. Structural analysis – A matrix approach – G.S. Pandit and S.P. Gupta.
3. Indeterminate Structures by R.I. Jindal

REINFORCED CONCRETE STRUCTURES -II

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

Students learn

1. To learn design of retaining walls.
2. To learn design of piles and pile caps
3. To learn basic concept of bridges and design

Course Outcomes:

Students will be able to

1. Distinguish between behaviour of cantilever and counter fort retaining walls and analyse for maximum bending moment and shear force.
2. Determine the stresses in various parts of underground and over head water tanks including design.
3. Analyze and design solid slab and T-beam bridges
4. Importance of pile foundations and their design
5. Importance of Bunker& Silos and their design.
6. Apply IS 456 and bridge codal provisions to RC structures.

SYLLABUS

Retaining Walls: Types of retaining walls, forces on retaining walls, Rankine and Coulomb earth pressure theories (c and ϕ soils). Passive earth pressure, Drainage of retaining walls, Stability requirements, Preliminary proportioning of cantilever retaining walls, Design of cantilever and counterfort retaining walls.

Water Tanks: Stress in concrete and steel in water tanks, Modular ratio, Impermeability requirements, Under ground rectangular tanks, Elevated circular tanks, Design of these tanks for strength and cracking, Design of staging of circular tanks.

Bridges: Components of a bridge in sub structure and super structure. Classification of bridges. Highway loading standards, kerbs, footpaths, railings, parapet loadings, Impact, wind, longitudinal forces. Design of solid slabs (casual reference to MOST drawings)

Piles and Pile caps: Design of bored cast in situ piles (bearing and friction types), under reamed piles. Pile Caps design by bending action and truss action methods.

Bunkers and silos (sessionalsonly): Difference between bunker and silos, Design of square or rectangular bunkers, Design of a battery of bunkers; Design of silos (circular).

Text Books:

1. Design of reinforced concrete structures by N. Subramanian, published by OUP India
2. Design of reinforced concrete structures by N. Krishnam Raju, CBS Publishers
3. Reinforced concrete structures by Robert Park, Willey India

Reference Books:

1. Limit State of Design of Reinforced Concrete - P.C. Varghese
2. Reinforced Concrete Limit State Design - A.K. Jain.
3. Design of reinforced Concrete Structures - P. Dayaratnam
4. Advanced Reinforced Concrete - Krishnam Raju
5. Reinforced Concrete Structures - R. Park T. Paulay

STEEL STRUCTURES

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

Students learn

1. To learn the properties of steel sections and design basics and code provisions Design of connections
2. To design steel members subjected to tension and compression members
3. Design steps involved in beams, built up beams and connections in beam column etc.
4. To design plate girders, gantry girders and light gauge section

Course Outcomes:

Students will be able to

1. Know the properties and fundamentals of steel sections.
2. Know the connections in steel structures
3. Identify the possible failure modes in compression members
4. Importance of roof trusses and their design

SYLLABUS

Fundamental Concepts of limit state design of structures, Different types of rolled steel sections available to be used in steel structures. Stress – Strain relationship for mild steel.

Bolted connections: Behaviour of bolted joints, Design strength of ordinary black bolts, high strength friction grip bolts, Simple connections.

Welded Connections: Advantages of welding, Types and properties of welds, Types of joints, weld specifications, Design of welded joints subjected to axial load.

Tension members: Types of tension members, slenderness ratio, displacement of tension members, behaviour of tension members, modes of failure, factors affecting strength of tension members, design of tension members, Lug angles.

Compression members: Possible failure modes, behaviour of compression members, Effective length, radius of gyration and slenderness of compression members, Allowable stresses in compression, Design of axially loaded compression members, built up compression members with Laced and Battened columns.

Column bases: Allowable stress in bearing, Design of Slab base, Design of Gusset base.

Text Books:

1. Design of Steel structures by N. Subramanian, Oxford University Press.
2. Limit State Design of steel structures – Ramchandra and Virendra Gehlot, Scientific Publishers (India)

3. Limit State Design of steel structures by S.K.Duggal, McGraw Hill Education Private Ltd.

Reference Books:

1. Design of steel structures by K.S.Sai Ram, Pearson Education India.
2. Design of steel structures by Limit State Method as per IS: 800-2007 – S.S. Bhavikatti, IK International Publishing House, Bangalore – 560 001.

GEOTECHNICAL ENGINEERING - I

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

The objectives of this course are to make the student aware of

1. Fundamental relationships between different parameters of a soil mass.
2. Different types of soils and identify their properties.
3. Processes of compaction and consolidation and apply them to field problems.
4. Stress distribution, settlement and strength of different soils in different conditions.
5. Importance of Soil Mechanics in solving the engineering problems.

Course Outcomes:

Students will be able to

1. Know the fundamental relationships between different parameters of a soil mass.
2. Classify different types of soils and identify their properties.
3. Appreciate the processes of compaction and consolidation and apply them to field problems.
4. Estimate stress distribution and settlement of different soils in different conditions.
5. Identify shear strength parameters for field conditions.
6. Apply the knowledge of Soil Mechanics in solving the engineering problems

SYLLABUS

Introduction: Historical development, Soil Formation, Minerals in clays and sand, Soil Structure, Physical properties of Soil: Void ratio, Porosity, Degree of Saturation, Water content, Unit Weights, Specific Gravity, weight –volume Relationships, Relative density, Consistency limits: Determination and consistency indices, Activity.

Mechanical analysis and Soil Classification: Sieve analysis, stoke's law, hydrometer and Pipette Analysis. Unified soil classification and Indian Standard Soil Classification Systems, Field Identification of Soils.

Soil Hydraulics: Types of soil water, capillary rise and surface tension, Darcy's law and its limitations, constant head and variable head permeability tests, pumping tests, Factors effecting Coefficient of permeability, permeability of stratified soils. Total, neutral and effective stresses, Effective Stress Principle, Upward flow conditions, quick sand conditions, critical hydraulic gradient.

Stress Distribution in Soils: Bousinesq's theory for determination of vertical stress, assumptions and validity, rectangular and circular loaded areas, Pressure Bulb and Influence diagrams, westergaard's theory, Newmarks influence chart - construction and use, 2:1 approximate method, contact pressure distribution beneath footings.

Compaction: Mechanism of compaction, Factors effecting compaction: water content, compactive effort, Type of soil. IS Light and IS Heavy compaction tests, Effect of compaction on soil Properties, Field compaction: compaction Equipment and Evaluation of field Compaction.

Consolidation: Basic Definitions: compression index, coefficient of compressibility and coefficient of volume decrease. Terzaghi's one dimensional consolidation theory - assumption, Derivation of differential equation and Solution, Oedometer Test, Determination of coefficient of consolidation by time fitting methods, initial compression, primary compression and secondary compression, determination of preconsolidation pressure. Normally consolidated, over consolidated and under consolidated clays.

Shear Strength of Soils: Stress at a point, Mohr circle of stress, Mohr coulomb failure theory, shear parameters, laboratory shear tests – shear box, triaxial and unconfined compression tests, laboratory and field vane shear tests, Sensitivity of clays, Types of shear tests on drainage conditions, shear strength of sands, critical void ratio and dilatancy, Factors affecting shear strength of clays and sands, Total stress analysis and Effective stress analysis.

Text Books :

1. Soil Mechanics and Foundation Engineering by K.R. Arora.
2. Basic and Applied Soil Mechanics by Gopal Rajan and A.S.R. Rao.
3. Geotechnical Engineering by P. Purushothama Raj

Reference Books :

1. Geotechnical Engineering by C. Venkataramiah, New age International Pvt. Ltd. (2002).
2. Geotechnical Engineering Handbook By Das - J Ross Publishing.
3. Geotechnical Engineering Principles and Practices by Cuduto, PHI International.
4. Geotechnical Engineering by Manoj Dutta & Gulati S. K - Tata Mc.Graw Hill Publishers New Delhi.
5. Soil Mechanics and Foundation Engg. By K. R. Arora, Standard Publishers and Distributors, Delhi.

ENVIRONMENTAL ENGINEERING - I

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

Student shall be able to:

1. Explain the fundamentals of chemistry involved in water
2. Understand the analysis of water quality parameters
3. Understand the working principles of various unit operations of water treatment plant
4. Understand the importance of modern membrane techniques employed for water treatment
5. Design the water treatment unit operations

Course Outcomes:

By the end of the course, the student should be able to:

1. Explain the quality of water.
2. Analyze the water quality parameters and compare with the permissible limits.
3. Summarize the working principles of conventional unit operations of a water treatment plant.
4. Determine the sizes of different unit operations in a water treatment plant.
5. Assess the suitability of conventional methods and latest membrane processes for different water bodies.
6. Design a conventional water treatment plant with given specifications for given capacity.

SYLLABUS

Necessity and objectives of protected water supply system - Flow chart of public water supply system – Role of environmental agencies

Water Demand and Quantity studies: Estimation of water demand for a town or city, Types of water demands, Per capita Demand, Factors affecting the Per Capita Demand, Variations in the Demand, Design Period, Factors affecting the Design period, Population Studies, Population Forecasting Studies.

Hydrological Concepts: Hydrological Cycle - Types of Precipitation - Measurement of Rainfall. Surface sources of water: Lakes, Rivers, Impounding Reservoirs - Capacity of storage reservoirs - Mass curve analysis. Groundwater sources of water: Types of water bearing formations – Springs - Wells and Infiltration galleries, Yields from wells and infiltration galleries.

Collection of Water: Factors governing the selection of the intake structure - Types of Intakes - Conveyance of Water: Gravity and Pressure conduits - Types of Pipes - Pipe Materials, Pipe joints - Design aspects of pipe lines - Laying of pipe lines.

Quality and Analysis of Water: Characteristics of water – Physical, Chemical and Biological. Analysis of Water – Physical, Chemical and Biological - Impurities in water - Water borne diseases - Drinking water quality standards.

Treatment of Water: Flowchart of water treatment plant, Treatment methods (Theory and Design) - Sedimentation, Coagulation, Sedimentation with Coagulation, Filtration, Chlorination and other Disinfection methods

Softening of Water, Defluoridation and Removal of odours.

Distribution of Water: Methods of Distribution system - Components of Distribution system - Layouts of Distribution networks - Pressures in the distribution layouts - Analysis of Distribution networks -Water connection to the houses.

Text Books:

1. “ Water Supply engineering” by S.K.Garg, 17th Edition, Khanna Publishers.

Reference Books:

1. Environmental Engineering by Peavy, Rowe & Tchobanoglous, McGraw Hill Publishers
2. Elements of Environmental Engineering by K.N. Duggal, McGraw Hill Publishers

CONCRETE TECHNOLOGY (Elective-I)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

Students learn

1. To learn the concepts of Concrete production and its behaviour in various environments.
2. To learn the test procedures for the determination of properties of concrete.
3. To understand durability properties of concrete in various environments.

Course Outcomes:

Students will be able to

1. Understand the basic concepts of concrete.
2. Realise the importance of quality of concrete.
3. Familiarise the basic ingredients of concrete and their role in the production of concrete and its behaviour in the field.
4. Test the fresh concrete properties and the hardened concrete properties.
5. Evaluate the ingredients of concrete through lab test results.
6. Design the concrete mix by BIS method.
7. Familiarise the basic concepts of special concrete and their production and applications.
8. Understand the behaviour of concrete in various environments.

SYLLABUS

Cements &Admixtures: Portland cement – Chemical composition –Hydration, Setting of cement, Fineness of cement, Structure of hydrate cement – Test for physical properties – Different grades of cements –Admixtures – Mineral and chemical admixtures – accelerators, retarders, airentrainers, plasticizers, super plasticizers, fly ash and silica fume.

Aggregates: Classification of aggregate – Particle shape & texture – Bond, strength & other mechanical properties of aggregates – Specific gravity, Bulk density, porosity, adsorption & moisture content of aggregate – Bulking of sand –Deleterious substance in aggregate – Soundness of aggregate – Alkali aggregate reaction – Thermal properties – Sieve analysis – Fineness modulus – Gap graded and well graded aggregate as per relevant IS code – Maximum aggregate size.

Fresh Concrete: Steps in Manufacture of Concrete–proportion, mixing, placing, compaction, finishing, curing – including various types ineach stage. Properties of fresh concrete-Workability – Factors affecting workability – Measurement of workability by different tests, Setting times ofconcrete, Effect of time and temperature on workability – Segregation &bleeding – Mixing and vibration of concrete, Ready mixed concrete,Shotcrete.

Hardened Concrete: Water / Cement ratio – Abram's Law – Gelspaoe ratio – Nature of strength of concrete – Maturity concept – Strength in tension & compression – Factors affecting strength – Relation between compression & tensile strength – Curing, Testing of Hardened Concrete: Compression tests – Tension tests – Factors affecting strength – Flexure tests – Splitting tests – Non-destructive testing methods – code provisions for NDT.

Elasticity, Creep & Shrinkage– Modulus of elasticity – Dynamic modulus of elasticity – Poisson's ratio – Creep of concrete – Factors influencing creep – Relation between creep & time – Nature of creep– Effects of creep – Shrinkage – types of shrinkage.

Mix Design: Factors in the choice of mix proportions – Durability of concrete – Quality Control of concrete – Statistical methods – Acceptance criteria – Concepts Proportioning of concrete mixes by various methods – BIS method of mix design.

Special Concretes: Ready mixed concrete, Shotcrete -Light weight aggregate concrete – Cellular concrete – No-fines concrete, High density concrete, Fibre reinforced concrete – Different types of fibres – Factors affecting properties of F.R.C, Polymer concrete – Types of Polymer concrete– Properties of polymer concrete, High performance concrete – Self consolidating concrete, SIFCON, self healing concrete.

Text Books:

1. Concrete Technology by M.S.Shetty. – S.Chand & Co.; 2004.
2. Concrete Technology by M.L. Gambhir. – Tata Mc. GrawHill Publishers, New Delhi.

Reference Books:

1. Properties of Concrete by A.M.Neville – PEARSON – 4th edition.
2. Concrete Technology by A.R. Santha Kumar, Oxford University Press, New Delhi.

GIS & REMOTE SENSING (Elective-I)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

Students learn

1. To learn the principles and concepts of GIS and remote sensing
2. To learn the importance and classification of digital elevation models
3. To understand GIS analysis applications and their implementations.

Course Outcomes:

Students will be able to

1. Understand the basic concepts of GIS.
2. Explain the sensor systems used in remote sensing
3. Realize the importance of remote sensing in civil engineering.
4. Classify the GIS analysis functions.
5. Apply the knowledge in solving engineering issues by using GIS

SYLLABUS

Introduction to remote sensing : Introduction, A brief history of RS, Energy sources and radiation principles, sensor systems used in RS, RS satellites, land sat, spot, IRS etc., RS data products, RS analysis examples – measurement analysis – classification. RS in civil engineering projects : Topographic mapping : Geometric characteristics, digital elementary model, Cartographic requirements of satellite data, Mapping using SLAR.

Resource Mapping: Geometric and hydrographic features. Soil mapping and characteristics. Application in water resource engineering. Environmental pollution monitoring. Regional and urban mapping, planning systems and waste disposal sites.

Introduction to GIS: Introduction, GIS overview, Engineering of GIS applications, GIS components. Data Structures in Thematic maps: Data structures for GIS, Data base structures, Data models, H,N,R query languages for data models. The nature of geographic data, spatial data models, Raster data models, Vector data models, Data base management for GIS, Data structures for Thematic maps. The choice between Raster and vector.

Digital Elevation Models : Importance of DEM, Methods of DEM, Image methods, Data sources and sampling methods for DEM. DATA INPUT, VERIFICATION, STORAGE AND OUTPUT : Data input, Data verification, Classification, and storage data output.

Data Quality, Errors and Natural Variation: Components of data quality, sources of errors, nature of boundaries, static nature of boundaries, combining attributes from overlaid maps.

GIS Analysis Functions: Introduction, Organization of data analysis, Classification of functions, maintenance and analysis of spatial data, Maintenance & analysis of nonspatial attribute data, integrated analysis of spatial & nonspatial data, output formatting, cartographic modeling.

Choosing and Implementing a GIS - Awareness, need for GIS, Developing system requirements, evaluation of alternative systems, system justification and development of an implementation plan, operational system.

Text Books:

1. Principles of Geographical information systems for land resource assessment – P. A. Burrough (Clarendon Press Oxford).
2. Geographic Information systems a management perspective Stan Aronoff (WDL Publications, Ottawa, Canada).

Reference Books:

1. Remote sensing in civil engineering – Kennie, J.J.M., Matthews, M.C.
2. Remote sensing principles and interpretation – Floyd F. Sabins, Jr. W.H. Freeman & Co.

PRESTRESSED CONCRETE
(Elective-I)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

Students learn

1. Analysis and design of prestressed concrete members and connections.

Course Outcomes:

Students will be able to

1. Understand the general mechanical behavior of prestressed concrete.
2. Analyze and design prestressed concrete flexural members.
3. Analyze and design for vertical and horizontal shear in prestressed concrete.
4. Analyze transfer and development length as well as prestress losses.
5. Analyze and design for deflection and crack control of prestressed concrete members.
6. Analyze and design simple connections of prestressed concrete members.

SYLLABUS

Introduction, Basic concepts of prestressing, need for high strength steel and concrete, advantages of prestressed concrete. Materials for prestressed concrete, high strength concrete and high strength steel. Prestressing systems (1) Fressinet System (2) Gifford Udall (3) Magnel Blatan System, Tensioning devices, anchoring devices. (d) Pre tensioning and Post tensioning.

Prestressing losses, Elastic shortening, loss due to shrinkage, loss due to creep, loss due to friction, loss due to curvature etc.I.S. code provisions.

Analysis of pre stress members, assumptions, pressure, or thrust line concept of load balancing, cable profile, kern distance, stress in tendons as per IS 1343, cracking moment.

Limit state design of flexural members, stress, I.S. code provisions, design of symmetrical beams, design of pre stressed concrete poles, design for shear, I.S. code provisions.

Transfer of prestress (Pretensioned members), Transmission length, bond stress, Transversetensile stress, End Zone reinforcement, flexural bond stress, I.S. Code Provisions.

Anchorage zone in post tensioned members, stress distribution in end block, Guyon's method of approach of analysis of end block (Not more than 2 cables).

Text Book :

1. Prestressed Concrete by N.Krishna Raju.

Reference Books :

1. Prestressed Concrete by N.Rajagopalan.
2. Prestressed Concrete by P. Dayaratnam.
3. Design of Prestressed Concrete Structures by T.Y. Lin and Ned. H. Burns.

BUILDING SERVICES AND MAINTENANCE (Elective-I)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

Students learn

1. To know the problems posses in the buildings.
2. To learn the maintenance activities to be done in buildings.
3. To understand special repairs to counteract the problems posses in the buildings.

Course Outcomes:

Students will be able to

1. Understand the basic issues occurring in the buildings.
2. Realise the importance of maintenance of buildings.
3. Apply the knowledge of repair techniques for rehabilitation of structures.

SYLLABUS

Maintenance of Building: Introduction, various problems occurs in buildings, White washing, colour washing and distempering, painting, replacement of glass panels, re-polishing of terrazo and mosaic, replacement of decayed timber, easing of doors and windows, repairs to damaged part of the flooring, cleaning of fire chimneys and gutters, removal of stains from concrete and terrazzo floor, anti termite treatment (in building, foundations, floors and wood work), repairing of plumbing, drain and sanitary works. Repair of water storage sumps and tanks, repair of any joints i.e. wall-beam joint leak, beam-column, slab-beam, etc.

Special Repairs: Strengthening of foundation and foundation soils, rectification of leaking roof and concrete cover spalled roof, repairs to crack in masonry wall, repairs to leakage at window sill, special repairs to joinery work at roof level, providing D.P.C. to the exciting buildings, repairs to expansion and contraction joints, repairs to ramped floors. Repair of electrical installation system, repair of fire services system, repair of gas supply system. Repair of broad cast reception installation system, repair of security system, etc.

Repairs of Multistory Structures: Cracks in concrete, possible damages to the structural element-beams,slab, Column, Footings, etc., Repairing techniques like Jacketing, Grouting, External prestressing

Text Book:

1. Building Construction by V.N. Vazirani and S.P. Chandola, by Khanna Publishers.

Reference Books:

1. General Specification for Building Maintenance Works in Residential Buildings, prepared by Building Surveying Division, HKIS
2. Building Maintenance Management, Second Edition by Barrie Chanter and Peter Swallow, Balckwell publishing

WATERSHED MANAGEMENT (Elective-I)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

Students learn

1. Introduce the concept of watershed management
2. Understand the watershed characteristics
3. Learn the principles of soil erosion and measures to control erosion
4. Appreciate various water harvesting techniques.
5. Learn land management practices for various land use/land cover.
6. Introduce concepts of watershed modelling.

Course Outcomes:

Students will be able to

1. Calculate watershed parameters and analyse watershed characteristics to take appropriate management action.
2. Quantify soil erosion and design control measures.
3. Apply land grading techniques for proper land management .
4. Suggest suitable harvesting techniques for better watershed management.
5. Apply appropriate models for watershed management.

SYLLABUS

Principles of Watershed Management: Basics concepts, Hydrology and water availability, Surface water, Groundwater, Conjunctive use, Human influences in the water resources system, Water demand, Integrated water resources system

River basins Watershed Management Practices in Arid and Semi-arid Regions, Watershed management through wells, Management of water supply - Case studies, short term and long term strategic planning

Conservation of Water: Perspective on recycle and reuse, Waste water reclamation

Social Aspects of Watershed Management: Community participation, Private sector participation, Institutional issues, Socio-economy, Integrated development, Water legislation and implementations, Case studies

Sustainable Watershed Approach: Sustainable integrated watershed management, natural resources management, agricultural practices, integrated farming, Soil erosion and conservation

Water Harvesting: Rainwater management - conservation, storage and effective utilisation of rainwater, Structures for rainwater harvesting, roof catchment system, check dams, aquifer storage

Applications of Geographical Information System and Remote Sensing in Watershed Management, Role of Decision Support System in Watershed Management.

Text Book:

1. Murthy, J.V.S., Watershed Management in India, Wiley Eastern, New Delhi, 1994.

Reference Books:

1. Murty, J.V.S., Watershed Management, New Age Intl., New Delhi 1998.
2. Allam, G.I.Y., Decision Support System for Integrated Watershed Management, Colorado State University, 1994.
3. Vir Singh, R., Watershed Planning and Management, Yash Publishing House, Bikaner, 2000.
4. American Society of Civil Engineers, Watershed Management, American Soc. of Civil Engineers, New York, 1975.

ENVIRONMENTAL ENGINEERING LAB

Lab : 3 Periods
Exam : 3 Hrs.

Sessionals : 50
Ext. Marks : 50
Credits : 2

Course Objectives:

The objectives of this course are to make the student aware of

1. Tests to identify the physical properties of water.
2. Tests to identify the p^H and conductivity of water.
3. Tests to identify concentration of acidity and alkalinity of water.
4. Tests to identify COD and BOD of water.

Course Outcomes:

Students will be able to

1. Determine physical properties of water.
2. Determine the turbidity and hardness of water.
3. Determine COD and BOD of water.
4. Estimate concentration of acidity and alkalinity.
5. Estimate chloride content of water.

SYLLABUS

1. (a) p^H .
(b) Conductivity.
2. (a) Turbidity.
(b) Jar Test .
3. Hardness.
4. Acidity estimation.
5. Alkalinity estimation.
6. Available Chlorine & Residual Chlorine.
7. Fluorides.
8. Iron Estimation.
9. Estimation of Total Solids : Settleable Solids : Suspended solids, dissolved solids.
10. D.O.
11. B. O. D.
12. C. O. D.
13. Chlorides.

Reference Books:

1. Environmental engineering lab manual, Department of civil engineering, S. R. K. R Engg College.
2. Environmental Engineering by Peavy, Rowe & Tchobanoglous, McGraw Hill Publishers

GEOTECHNICAL ENGINEERING LAB - I

Lab : 3 Periods
Exam : 3 Hrs.

Sessionals : 50
Ext. Marks : 50
Credits : 2

Course Objectives:

The objectives of this course are to make the student aware of

1. Tests to identify the physical properties of soil.
2. Tests to identify the types of soil.
3. Tests to identify compaction characteristics and permeability of soil.
4. Tests to identify in-situ characteristics of soil.

Course Outcomes:

Students will be able to

1. Determine physical properties of soil.
2. Classify various types of soil.
3. Determine the permeability of soil.
4. Determine compaction characteristics of soils.
5. Estimate in-situ density of soil.

SYLLABUS

1. Atterberg limits
2. Field density by Core Cutter and Sand replacement method.
3. Grain size analysis
4. Hydrometer/pipette analysis.
5. Specific gravity by pycnometer/density bottle method.
6. Permeability of soil – Constant and variable head tests.
7. IS light compaction.

DEMONSTRATION EXPERIMENTS:

1. Consolidation test.
2. Quick sand model and others if any.

Reference Books:

1. Geotechnical engineering lab manual, Department of civil engineering, S. R. K. R Engg College.
2. Soil Mechanics and Foundation Engg. By K. R. Arora, Standard Publishers and Distributors, Delhi.

VERBAL & QUANTITATIVE APTITUDE – I
(Common to All Branches)

Theory	: 5 Periods(VA-3+QA-2)	Sessionals	: 100
Exam	: 3 Hrs.	Credits	: 2

Part-A: Verbal and Soft Skills-I

Course objectives:

1. To introduce concepts required in framing grammatically correct sentences and identifying errors while using Standard English.
2. To familiarize the learner with high frequency words as they would be used in their professional career.
3. To inculcate logical thinking in order to frame and use data as per the requirement.
4. To acquaint the learner of making a coherent and cohesive sentences and paragraphs for composing a written discourse.
5. To familiarize students with soft skills and how it influences their professional growth.

Course Outcomes:

The student will be able to

1. Detect grammatical errors in the text/sentences and rectify them while answering their competitive/ company specific tests and frame grammatically correct sentences while writing.
2. Answer questions on synonyms, antonyms and other vocabulary based exercises while attempting CAT, GRE, GATE and other related tests.
3. Use their logical thinking ability and solve questions related to analogy, syllogisms and other reasoning based exercises.
4. Choose the appropriate word/s/phrases suitable to the given context in order to make the sentence/paragraph coherent.
5. Apply soft skills in the work place and build better personal and professional relationships making informed decisions.

SYLLABUS

Grammar: (VA)

Parts of speech(with emphasis on appropriate prepositions, co-relative conjunctions, pronouns-number and person, relative pronouns), articles(nuances while using definite and indefinite articles), tenses(with emphasis on appropriate usage according to the situation), subject – verb agreement (to differentiate between number and person) , clauses(use of the appropriate clause , conditional and relative clauses), phrases(use of the phrases, phrasal verbs) to-infinitives, gerunds, question tags, voice, direct & indirect speech, degrees of comparison, modifiers, determiners, identifying errors in a given sentence, correcting errors in sentences.

Vocabulary: (VA)

Synonyms and synonym variants(with emphasis on high frequency words), antonyms and antonym variants(with emphasis on high frequency words), contextual meanings with regard to inflections of a word, frequently confused words, words often mis-used, multiple meanings of the same word (differentiating between meanings with the help of the given context), foreign phrases, homonyms, idioms, pictorial representation of words, word roots, collocations.

Reasoning: (VA)

Critical reasoning (understanding the terminology used in CR- premise, assumption, inference, conclusion), Analogies (building relationships between a pair of words and then identifying similar relationships), Sequencing of sentences (to form a coherent paragraph, to construct a meaningful and grammatically correct sentence using the jumbled text), odd man (to use logical reasoning and eliminate the unrelated word from a group), YES-NO statements (sticking to a particular line of reasoning Syllogisms).

Usage: (VA)

Sentence completion (with emphasis on signpost words and structure of a sentence), supplying a suitable beginning/ending/middle sentence to make the paragraph coherent, idiomatic language (with emphasis on business communication), punctuation depending on the meaning of the sentence.

Soft Skills:

Introduction to Soft Skills – Significance of Inter & Intra-Personal Communication – SWOT Analysis –Creativity & Problem Solving – Leadership & Team Work - Presentation Skills Attitude – Significance – Building a positive attitude – Goal Setting – Guidelines for Goal Setting – Social Consciousness and Social Entrepreneurship – Emotional Intelligence - Stress Management, CV Making and CV Review.

Text Books:

1. Oxford Learners's Grammar – Finder by John Eastwood, Oxford Publication.
2. R S Agarwal's books on objective English and verbal reasoning
3. English Vocabulary in Use- Advanced , Cambridge University Press.
4. Collocations In Use, Cmbridge University Press.
5. Soft Skills & Employability Skills by Samina Pillai and Agna Fernandez, Cambridge University Press India Pvt. Ltd.
6. Soft Skills, by Dr. K. Alex, S. Chand & Company Ltd., New Delhi

Reference Books:

1. English Grammar in Use by Raymond Murphy, CUP
2. Websites: Indiabix, 800score, official CAT, GRE and GMAT sites
3. Material from 'IMS, Career Launcher and Time' institutes for competitive exams.
4. The Art of Public Speaking by Dale Carnegie
5. The Leader in You by Dale Carnegie
6. Emotional Intelligence by Daniel Golman
7. Stay Hungry Stay Foolish by Rashmi Bansal
8. I have a Dream by Rashmi Bansal

Part-B: Quantitative Aptitude -I

Course objectives:

The objective of introducing quantitative aptitude-1 is:

1. To familiarize students with basic problems on numbers and ratio's problems.
2. To enrich the skills of solving problems on time, work, speed, distance and also measurement of units.
3. To enable the students to work efficiently on percentage values related to shares, profit and loss problems.
4. To inculcate logical thinking by exposing the students to reasoning related questions.
5. To expose them to the practice of syllogisms and help them make right conclusions.

Course Outcomes:

1. The students will be able to perform well in calculating on number problems and various units of ratio concepts.
2. Accurate solving problems on time and distance and units related solutions.
3. The students will become adept in solving problems related to profit and loss, in specific, quantitative ability.
4. The students will present themselves well in the recruitment process using analytical and logical skills which he or she developed during the course as they are very important for any person to be placed in the industry.
5. The students will learn to apply Logical thinking to the problems of syllogisms and be able to effectively attempt competitive examinations like CAT, GRE, GATE for further studies.

SYLLABUS

Numbers, LCM and HCF, Chain Rule, Ratio and Proportion

Importance of different types of numbers and uses of them: Divisibility tests, Finding remainders in various cases, Problems related to numbers, Methods to find LCM, Methods to find HCF, applications of LCM, HCF. Importance of chain rule, Problems on chain rule, Introducing the concept of ratio in three different methods, Problems related to Ratio and Proportion.

Time and work, Time and Distance

Problems on man power and time related to work, Problems on alternate days, Problems on hours of working related to clock, Problems on pipes and cistern, Problems on combination of the some or all the above, Introduction of time and distance, Problems on average speed, Problems on Relative speed, Problems on trains, Problems on boats and streams, Problems on circular tracks, Problems on polygonal tracks, Problems on races.

Percentages, Profit Loss and Discount, Simple interest, Compound Interest, Partnerships, shares and dividends

Problems on percentages-Understanding of cost price, selling price, marked price, discount, percentage of profit, percentage of loss, percentage of discount, Problems on cost price, selling price, marked price, discount. Introduction of simple interest, Introduction of compound interest, Relation between simple interest and compound interest, Introduction of partnership, Sleeping partner concept and problems, Problems on shares and dividends, and stocks.

Introduction, number series, number analogy, classification, Letter series, ranking, directions

Problems of how to find the next number in the series, Finding the missing number and related sums, Analogy, Sums related to number analogy, Ranking of alphabet, Sums related to Classification, Sums related to letter series, Relation between number series and letter series, Usage of directions north, south, east, west, Problems related to directions north, south, east, west.

Data sufficiency, Syllogisms

Easy sums to understand data sufficiency, Frequent mistakes while doing data sufficiency, Syllogisms Problems.

Text Books:

1. Quantitative aptitude by RS Agarwal
2. Verbal and non verbal reasoning by RS Agarwal.
3. Puzzles to puzzle you by shakunataladevi

References:

1. Barron's by Sharon Welner Green and Ira K Wolf (Galgotia Publications pvt. Ltd.)
2. Websites: m4maths, Indiabix, 800score, official CAT, GRE and GMAT sites
3. Material from 'IMS, Career Launcher and Time' institutes for competitive exams.
4. Books for cat by arunsharma
5. Elementary and Higher algebra by HS Hall and SR knight.

Websites:

1. www.m4maths.com
2. www.Indiabix.com
3. www.800score.com
4. Official GRE site
5. Official GMAT site

**GEO ENVIRONMENTAL ENGINEERING
(MOOCS-I)**

Theory	: 4 Periods	Sessionals	: 100
Exam	: 3 Hrs.	Credits	: 2

Course Objectives:

Students learn

1. Concepts and principles of Geo environmental Engineering.
2. Geotechnical aspects of planning and design of MSW and Hazardous waste Landfills
3. Geotechnical aspects of planning and design of slurry ponds - ash ponds and tailing ponds.
4. Geotechnical aspects of detection & monitoring of subsurface contamination and control & remediation of contaminated sites.
5. Rehabilitation of waste dumps and geotechnical re-use of waste.

Course Outcomes:

At the end of the course, Students will be able to:

1. Understand the concepts of Integrated SWM & Geo environmental Engineering
2. Learn the Principles and Planning of Landfills
3. Explain Geotechnical Properties of Slurry Deposited Waste
4. Learn Environmental Control Measures at Slurry Ponds, Geotechnical Reuse of Waste

SYLLABUS

- 1 Introduction, Sources & Impact of Contamination and Soil-Waste Interaction
- 2 Concepts of Integrated SWM & Geo environmental Engineering
- 3 Principles and Planning of Landfills
- 4 Liners for Landfills
- 5 Landfill Covers, Generation and Control of Leachate and Gas from Landfills
- 6 Stability of Slopes and Settlement of Landfills
- 7 Solved examples, Monitoring and Detection of Subsurface Contamination
- 8 Costs, Construction Aspects and Site Selection of Landfills
- 9 Control, Rehabilitation of Old Dumps and Contaminated Sites
- 10 Slurry Deposited Waste and their Geotechnical Properties
- 11 Planning & Design, Incremental Raisings and Failures of Slurry Ponds
- 12 Environmental Control Measures at Slurry Ponds, Geotechnical Reuse of Waste.

WEB LINK:

https://onlinecourses.nptel.ac.in/noc18_ce33/preview

**HIGHER SURVEYING
(MOOCS-I)**

Theory	: 4 Periods	Sessionals	: 100
Exam	: 3 Hrs.	Credits	: 2

Course Objectives:

Students learn

1. The modern techniques of advanced surveying, their fundamental concepts, data acquisition, data processing, and applications.
2. Advanced mapping techniques

Course Outcomes: At the end of the course, Students will be able to:

1. Understand Need and pre-requisites for Higher Surveying
2. Be aware of Coordinate and datum transformations for 3D coordinates on Earth surface
3. Recognize Geolocation process
4. Learn Mapping with RADAR technique
5. Know about the Applications of Higher Surveying techniques for Archeological Surveys, forestry, etc.

SYLLABUS

1. What is Higher Surveying? Need and pre-requisites for Higher Surveying, Connection of Basic Surveying with Higher Surveying, detailed course content, books, notes, and other sources. Fundamental requirements of higher surveying: platform requirements, type of surveys, type of sensors, scanning mechanisms, scale and resolution of survey, planning of survey, data acquisition and data management Coordinate systems in mapping for Earth surface.
2. Coordinate and datum transformations for 3D coordinates on Earth surface; Introduction to Adjustments Computations.
3. Methods of Adjustments Computations Accuracy assessment of parameters using Adjustments Computations.
4. Positioning and direction survey for navigation. Introduction to Photogrammetric survey.
5. Physical models of photogrammetry Collinearity and coplanarity equations. Space resection, space intersection, and aerial triangulation for photogrammetry . Bundle adjustment, camera calibration, relative and absolute orientation, direct georeferencing

6. Mathematical models of photogrammetry Affine, Conformal, Rational functional model(RFM), and direct linear transformations. Close range and terrestrial photogrammetry for mapping.
7. Introduction to LiDAR survey and fundamental concepts Flight planning for airborne LiDAR data acquisition.
8. Geolocation process Error propagation and accuracy assessment for LiDAR 3. LiDAR data processing for DTM, DSM, BEM, and DEM generation.
9. LiDAR data processing for DTM,DSM, BEM, and DEM generation Introduction to Sounding and Bathymetry survey and fundamental concepts.
10. Data acquisition planning and execution for Sounding surveys Introduction to Radar interferometry survey and fundamental concepts Mapping with RADAR technique.
11. Mapping with RADAR technique Compatibility of various Higher Surveying techniques Applications of Higher Surveying techniques for Archeological Surveys.
12. Applications of Higher Surveying techniques for Cadastral survey and building detection and extraction Applications of Higher Surveying techniques for forestry Applications of Higher Surveying techniques for measurements of surface deformation and plate tectonic movement Applications of Higher Surveying techniques for 3D mapping and virtual model development. Applications of Higher Surveying techniques for Geomorphological features mapping

WEB LINK:

https://onlinecourses.nptel.ac.in/noc18_ce37/preview

**INTEGRATED WASTE MANAGEMENT FOR A SMART CITY
(MOOCS-I)**

Theory	: 4 Periods	Sessionals	: 100
Exam	: 3 Hrs.	Credits	: 2

Course Objectives: Students learn

1. The issues of Municipal Solid Waste (MSW) management, Construction and Demolition (C&D) Waste and Electronic Waste Management
2. generation rates and waste composition; Integrated waste management issues, collection, recovery, reuse, recycling, energy-from-waste, and landfilling
3. Biological treatment of the organic waste fraction - direct land application, composting, and anaerobic digestion

Course Outcomes: At the end of the course, Students will be able to:

1. Understand The environmental impact of waste management and its relationship on the sustainable development and smart city development
2. Familiarize the role of MSW management within the various initiatives of the Govt. of India
3. Recognize the challenges of waste management for smart cities
4. Make acquainted with the Construction and Demolition (C&D) Waste and Electronic Waste (E-Waste) management issues in India in general and for the smart cities in particular.

SYLLABUS

- 1: Introduction to Solid Waste Management
- 2: Municipal Solid Waste Characteristics and Quantities
- 3: MSW Rules 2016, Swachh Bharat Mission and Smart Cities Program
- 4: Municipal Solid Waste Collection, Transportation, Segregation and Processing
- 5: Disposal of Municipal Solid Waste: Landfill
- 6: Biochemical Processes and Composting
- 7: Energy Recovery from Municipal Solid Waste
- 8: Current Issues in Solid Waste Management and Review of MSW Management Status in First List of 20 Smart Cities in the Country
- 9: Construction and Demolition (C&D) Waste Management - Overview
- 10: C&D Waste – Regulation, Beneficial Reuse of C&D Waste Materials
- 11: Electronic Waste (E-Waste) Management – Issues and Status in India and Globally
- 12: E-Waste Management Rules 2016 and Management Challenges

WEB LINK:

https://onlinecourses.nptel.ac.in/noc18_ce25/preview

**INTRODUCTION TO ACCOUNTING AND FINANCE FOR CIVIL ENGINEERS
(MOOCS-I)**

Theory	: 4 Periods	Sessionals	: 100
Exam	: 3 Hrs.	Credits	: 2

Course Objectives: Students learn

1. Basics of accounting and terminology related to financing
2. Concepts of finance and general Principles pertaining to it
3. Concepts of Balancing sheet, ledgers, cash flow statements
4. Analysis of financial statements-leverage, ratios and applications

Course Outcomes: At the end of the course, Students will be able to:

1. Learn the Basics of accounting and terminology related to financing
2. Understand the Concepts of finance and general Principles of accounting
3. Know the Concepts of Balancing sheet, ledgers, journal
4. Analyze the financial statements-financial ratios

SYLLABUS

1. Basic Accounting and concepts in finance Book keeping: definitions, objectives, elements, journal and ledger
2. Accounting & Concepts in Finance I: definitions, objectives, characteristics, limitations, basic terms, GAAP(Generally Accepted Accounting Principles)
3. Accounting & Concepts in Finance II: Systems of accounting, cash book, bankbook, depreciation, provisions, reserves, accounting equation, journal & ledger entries, trial balance, profit & loss account, balance sheet, cash flow statement)
4. Analysis of financial statements I: Financial leverage, financial ratios
5. Analysis of financial statements II: Significance and applications
6. Financial planning including capital budgeting I: Definition, financial planning options and objectives, time value of money
7. Financial planning including capital budgeting II: simple and compound interest, rule of 72, methods of capital budgeting - payback period
8. Financial planning including capital budgeting III: Accounting rate of return(ARR), net present value (NPV), internal rate of return (IRR)

WEB LINK:

https://onlinecourses.nptel.ac.in/noc18_ce39/preview

**FIRE PROTECTION, SERVICES AND MAINTENANCE MANAGEMENT FOR BUILDING
(MOOCS-II)**

Theory	: 4 Periods	Sessionals	: 100
Exam	: 3 Hrs.	Credits	: 2

Course Objectives: Students learn

1. The concepts functional design of building for thermal aspects and energy efficiency
2. performing fenestration design for natural ventilation and day lighting & design of space for external and internal noise control

Course Outcomes: At the end of the course, Students will be able to:

1. Understand the Concepts of fire resistant and severity, Effect of fire on materials
2. Be aware of Building Services as a system and HVAC System
3. Recognize role of building maintenance in construction process Maintenance generators
4. Know the effect of design on maintenance, Diagnosis, appraisal, structural defects& various methods of repair.

SYLLABUS

1. Fire Protection: Process of combustion in fire, Effect of fire load & ventilation condition on enclosure fire, growth and decay of fire in enclosure
2. Concepts of fire resistant and severity, Effect of fire on materials. Simple Design of elements for given fire resistance.
3. Planning, Fire detection & suppression systems, Smoke venting
4. Lifts & Vertical Transportation: arrangement of lifts and Design for optimum service condition.
5. Building Services as a system, Capacity of storage and sizing, control systematic. & intelligent building.
6. HVAC System: Design Consideration. Basic psychometrics, Air conditioning process & system. Methods of Air Conditioning.
7. Water Supply, Hydraulic design, Storage Distribution, Component of cold & hot water supply system.
8. Waste water & Drainage systems: Fixture units & Design of system and elements of electrical services.
9. Definition, Role of building maintenance in construction process Maintenance generators, Expression of Standards, selection of level of maintenance and fixing standards.
10. Planned maintenance: Planning vis-a-vis adhoc maintenance, schedule& contingency maintenance, levels of planning, planned inspection, etc
11. Maintenance cycle, maintenance profile, repair & replacement models, statistical methods, decision models, optimal renewal cycle, budgeting etc

12. Effect of design on maintenance, Diagnosis, appraisal, structural defects& various methods of repair.

WEB LINK:

https://onlinecourses.nptel.ac.in/noc18_ce30/preview

**MODERN CONSTRUCTION MATERIALS
(MOOCS-II)**

Theory	: 4 Periods	Sessionals	: 100
Exam	: 3 Hrs.	Credits	: 2

Course Objectives: Students learn

1. The scientific basis for the understanding and development of construction materials
2. The Science, Engineering and Technology of Materials

Course Outcomes: At the end of the course, Students will be able to:

1. Learn about atomic bonding and structure of solids
2. Understand about the Movement of Atoms, Development of Microstructure
3. Study the Failure Theories, Fracture Mechanics
4. Familiarize with wood products, metals, bituminous materials, concrete and polymers.

SYLLABUS

Prologue – Intro. to the course, Science, Engineering and Technology of Materials- 1&2, Atomic Bonding-1, Atomic Bonding-2, Structure of Solids-1, Structure of Solids-2&3, Movement of Atoms, Development of Microstructure-1, Development of Microstructure-2, Surface Properties, Response to Stress-1, Response to Stress-2&3, Failure Theories, Fracture Mechanics-1, Fracture Mechanics-2, Rheology & Thermal properties, Review of Const. Materials & Criteria for Selection, Wood and Wood Products-1, Wood and Wood Products-2, Wood and Wood Products-3, Polymers, Fibre Reinforced Polymers-1&2, Metals-1, Metals-2, Metals-3, Bituminous Materials-1, Bituminous Materials-2, Concrete-1&2, Concrete-3, Concrete-4, Concrete-5, Glass, Social Perception of Const. Materials, Waterproofing Materials, Polymer Floor Finishes, Anchors.

WEB LINK:

https://onlinecourses.nptel.ac.in/noc18_ce22/preview

**PRINCIPLES OF CONSTRUCTION MANAGEMENT
(MOOCS-II)**

Theory	: 4 Periods	Sessionals	: 100
Exam	: 3 Hrs.	Credits	: 2

Course Objectives: Students learn

1. The Estimation of project cost and construction economics
2. Planning scheduling and Execution phases of a project
3. Safety-Quality management and legal aspects of a construction project

Course Outcomes: At the end of the course, Students will be able to:

1. Learn about the Estimation of project cost and construction economics.
2. Understand about the Planning scheduling and Execution phases of a project.
3. Study the Safety management and Quality management aspects.
4. Learn various legal aspects of a construction project.

Though the course primarily targets students of civil engineering in colleges, other engineering students may also find it interesting. The course seeks to present a rounded view of the divers issues involved in the management of construction projects, and includes aspects like construction economics, quality and safety management, and contract management, apart from time management and scheduling, estimation. It is hoped that engineers working in contracting, consulting and other organizations related to construction projects will also find the course useful.

SYLLABUS

- 1: General overview and project organization
- 2: Estimation of project cost
- 3: Construction Economics
- 4: Planning and scheduling: part-1
- 5: Planning and scheduling: part-2
- 6: Quality management
- 7: Safety Management
- 8: Legal aspects of a construction project

WEB LINK:

https://onlinecourses.nptel.ac.in/noc18_ce15/preview

**REINFORCED CONCRETE ROAD BRIDGES
(MOOCS-II)**

Theory	: 4 Periods	Sessionals	: 100
Exam	: 3 Hrs.	Credits	: 2

Course Objectives: Students learn

1. General design considerations, design limit states
2. Different loading conditions, introduction to IRC codes
3. Behavior of concrete
4. Design of slab bridges and RC T beam bridges

Course Outcomes: At the end of the course, Students will be able to:

1. Familiarize with design considerations, loads and IRC codes
2. Design Solid slab bridge
3. Design T-beam bridge

SYLLABUS

- 1: Introduction, design considerations, loads and IRC codes
- 2: Flexural and shear strength of reinforced concrete members
- 3: Solid slab bridge design
- 4: T-beam bridge design

WEB LINK:

https://onlinecourses.nptel.ac.in/noc18_ce23/preview

SCHEME OF INSTRUCTION & EXAMINATION
(Regulation R16)

III/IV B.TECH
(With effect from **2016-2017** Admitted Batch onwards)
Under Choice Based Credit System

CIVIL ENGINEERING
II-SEMESTER

Code No.	Course	Credits	Lecture Hrs	Tutorial Hrs	Lab Hrs	Total Contact Hrs/Week	Sessional Marks	Exam Marks	Total Marks
B16 CE 3201	Advanced Steel Structures	4	3	1	--	4	30	70	100
B16 CE 3202	Geotechnical Engineering- II	4	3	1	--	4	30	70	100
B16 CE 3203	Fluid Mechanics - III	4	3	1	--	4	30	70	100
B16 CE 3204	Estimation and Quantity Surveying	4	3	1	--	4	30	70	100
B16 CE 3205	Environmental Engineering - II	4	3	1	--	4	30	70	100
#ELE-II	Elective-II	4	3	1	--	4	30	70	100
B16 CE 3211	Geotechnical Engineering Lab - II	2	--	--	3	3	50	50	100
B16 CE 3212	Concrete Lab	2	--	--	3	3	50	50	100
B16 CE 3213	Fluid Mechanics Lab - II	2	--	--	3	3	50	50	100
B16 ENG 3202	Verbal & Quantative Aptitude-II	2	5	--	--	5	100	--	100
B16 CE 3214	Mini Project	2	--	--	3	3	50	--	50
#M-III	MOOCS-III	2	4	--	--	4	100	--	100
Total		36	27	6	12	45	580	570	1150

#ELE-II	B16 CE 3206	Ground Improvement Techniques
	B16 CE 3207	Environmental Impact Assessment
	B16 CE 3208	Marine Structures
	B16 CE 3209	Urban Hydrology
	B16 CE 3210	Finite Elements Methods of Analysis
# M-III	B16 ENG 3203	Basic Coding
	B16 CE 3215A	Energy Efficiency, Acoustics And Day lighting In Building
	B16 CE 3215B	Electronic Waste Management - Issues And Challenges
	B16 CE 3215C	Principles And Applications Of Building Science

ADVANCED STEEL STRUCTURES

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

Students learn

1. Design the laterally supported and unsupported beams as per I.S code.
2. Understand various components of a plate girder, design components of a plate girder by using I.S code, use of I.S code provisions for designing the vertical stiffeners, bearing stiffeners etc.,
3. Understand design of beam-column connections subjected to eccentric shear connections.
4. Understand the designing of elevated circular steel tanks in working stress method.

Course Outcomes:

Student should be able to

1. Understand the behavior of steel structures, in particular the various forms of failure for members and connections under tension, compression, bending and combined actions.
2. Apply the principles, procedures and current code requirements to the analysis and design of steel tension members, beams, plate girders and water tanks
3. Understand eccentric shear connections.

SYLLABUS

Beams: Beam types, section classifications, lateral stability of beams, Allowable stress in bending, Shear and Bearing stresses, Effective length of compression flange, Laterally supported and unsupported beams.

Plate Girders (Bolted and Welded): Components of a plate girder, Economical depth, proportioning of web and flanges, shear buckling resistance of web by simple post critical and tension field methods, connection of flange angles to web and flange angles to flange plates.

Web stiffeners: Design of bearing stiffeners. End panel design, design of intermediate stiffeners, connections.

Water tanks: Introduction, permissible stresses in water tanks, Design of circular steel water tank: forces acting over the tanks, stresses in elevated circular tanks, stresses in segmental and spherical bottoms, stresses in conical bottom, Design of circular girder (ring beam), design for circular steel tanks, stresses in columns, wind bracings.

Eccentric shear connections: Introduction, beam-columns connections, connections subjected to eccentric shear: welded seat connections: unstiffened seat angle connection and stiffened seat angle connections.

Text Books:

1. Design of Steel structures by N. Subramanian, Oxford University Press.
2. Limit State Design of steel structures – Ramchandra and VirendraGehlot, Scientific Publishers (India)
3. Limit State Design of steel structures by S.K.Duggal, McGraw Hill Education Private Ltd.

Reference Books:

1. Design of steel structures by K.S.Sai Ram, Pearson Education India.
2. Design of steel structures by Limit State Method as per IS: 800-2007 – S.S. Bhavikatti, IK International Publishing House, Bangalore – 560 001.

GEOTECHNICAL ENGINEERING- II

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

The objectives of this course are to make the student aware of

1. Different techniques of soil exploration.
2. Various methods for estimating bearing capacity of different types of foundations.
3. Estimating Load capacity of single piles and groups of piles.
4. Settlement analysis of foundations.
5. Theoretical aspects of well foundations.
6. Concepts of stability of finite and infinite slopes.
7. Earth pressures on retaining walls following Rankine's and Coulomb's theories.

Course Outcomes:

Students will be able to

1. Plan a detailed soil exploration programme.
2. Apply various methods for estimating bearing capacity of different types of foundations.
3. Perform the settlement analysis of footings.
4. Estimate load capacity of single piles and groups of piles.
5. Know the theory aspects of well foundations.
6. Analyze stability of finite and infinite slopes.
7. Calculate earth pressures on retaining walls following Rankine's and Coulomb's theories

SYLLABUS

Subsoil Exploration: Methods of subsoil exploration Direct, semi direct and indirect methods, Soundings by Standard, Dynamic cone and static cone penetration tests, Types of Boring, Types of samples, Criteria for undisturbed samples, Transport and preservation of samples, Bore logs, planning of exploration programmes, report writing.

Bearing Capacity: Safe bearing capacity and allowable bearing pressure, General and local shear failures, Terzaghi's bearing capacity equations its modifications for square, rectangular and circular foundations, Factors affecting bearing capacity of Soil, Effect of water table on bearing capacity, IS Code method for Bearing capacity of footings, Allowable bearing pressure based on N-values. Bearing capacity from plate load tests.

Shallow Foundations: Factors effecting locations of foundation and design considerations of shallow foundations, choice of type of foundations.

Settlement Analysis: Causes of settlement, Computation of immediate and consolidation settlement, allowable settlement. Measures to reduce settlement.

Pile Foundations : Types, Construction, load carrying capacity of single piles in sands and clays (α -method) Dynamic Formula, Static formula, Pile load tests, Load carrying capacity of pile groups.

Caissons: Types of caissons, pneumatic caissons, Different shapes of well foundations, Relative advantages and disadvantages, Different Components of well and their function, Grip length, problems in well sinking and remedial measures.

Stability Analysis of Slopes : Infinite and Finite Slopes, Stability Analysis of Infinite Slopes, different factors of safety, Types of Slope Failures – Toe slope and Base failure, , Stability Analysis of Finite slopes – Swedish Circle method, Friction Circle method, Fellenius method for location of Critical Slip Circle, Taylor's stability number.

Earth Pressure: Types of Earth pressure, Rankine's Active and passive earth pressure, Smooth Vertical wall with horizontal and inclined backfills. Coulomb's wedge theory, Culmann's and Rebhann's graphical method for active earth pressure, Retaining walls: Types and Stability Analysis.

Text Books:

1. Basic and Applied Soil Mechanics by Gopal Ranjan and A.S.R. Rao
2. Foundation Analysis and Design – J. E. Bowles.
3. Soil Mechanics and Foundation Engineering – By K.R. Arora.

Reference Books:

1. Analysis and Design of Substructures — Swami Saran, Oxford and IBH Publishing company Pvt Ltd 1998.
2. Geotechnical Engineering by S. K. Gulhati & Manoj Datta — Tata Mc.Graw Hill Publishing company New Delhi. 2005.
3. Teng, W.O — Foundation Design , Prentice Hall, New Jersey.
4. Bowles, J.E., (1988) Foundation Analysis and Design — 4th Edition, McGraw-Hill Publishing company, New York.

FLUID MECHANICS - III

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

Students learn

1. The concepts of Gradient, Divergence, Curl, Directional derivative, solenoidal and Irrotational fields
2. Green's, Stokes' and Divergence theorems
3. Classification of 2nd order Partial Differential Equations as well as solution of 1-Dimensional Wave equation and 1-Dimensional Heat equation
4. the concept of Analytic function, CR equations
5. Cauchy's Integral Theorem and Integral Formula
6. Taylor and Laurent series, Residues and Residue theorem

Course Outcomes:

Students will be able to

1. Apply the concepts of Gradient, Divergence, Curl, Directional derivative, solenoidal and Irrotational fields
2. Determine scalar potential, circulation and work done
3. Evaluate integrals using Green's, Stokes' and Divergence theorems
4. Obtain the solution of 1-D wave equation and 1-D heat equation
5. Determine the zeroes and poles of functions and residues at poles
6. Evaluate certain real definite integrals that arise in applications by the use of Residue theorem

SYLLABUS

Dimensional Analysis and Similitude: Fundamental Concepts of Dimensional Analysis – Importance of Dimensional Analysis & Model Study; Units and Dimensional Formulae for Various Engineering Quantities; Fourier Concept of Dimensional Homogeneity.

Methods of Arriving at Dimensionless Groups – Non-dimensional Parameters; Rayleigh's Method; Buckingham π method – Buckingham modified method; Omitted and Superfluous variables.

Examples in Dimensional Analysis – Capillary Rise, Drag on Cylinder, Resistance of a Ship, Discharge over a Sharp Crested Weir, Fall Velocity of a Sphere, Head Characteristics of a Pump, Thrust on a Propeller,

Similarity and Similarity Laws – Concepts of Similarity – Geometric, Kinematic and Dynamic Similarities; Modeling Criteria; Similarity Laws – Important Dimensionless Numbers – Reynolds Number, Froude Number, Mach Number, Euler Number, Weber Number.

Application of Similarity Laws to Practical Problems – Bodies Completely Submerged in Fluids, Bodies subjected to Gravity and Viscous Forces, River Models – Manning's Law; Distorted Models – Depth distortion and slope distortion; Problems related to Modeling of Spillways, Ships and Pumps & Turbines.

Impact of jets: Force exerted by fluid jet on stationary and moving flat and curved vanes, Torque and Work done by series of Moving Vanes.

Hydraulic Machines– Turbines: Introduction and Classification of Turbines – Function of Prime movers and Pumps, Hydraulic Turbines, Classification Based on Head, Discharge, Hydraulic Action – Impulse and Reaction Turbines, Differences between Impulse and Reaction Turbines; Choice of Type of Turbine – Specific Speed.

Working of Impulse Turbines & Design Principles – Components & Working Principles of a Pelton Turbine – Work Done; Hydraulic and Overall Efficiencies; Design of Pelton Turbine – Working Proportions; Governing Mechanism for a Pelton Turbine.

Working of Reaction Turbines & Design Principles – Components & Working Principles of a Francis Turbine – Work Done; Hydraulic and Overall Efficiencies; Design of Francis Turbine– Working Proportions; Governing Mechanism for a Francis Turbine. Draft Tube Theory – Functions and Types of Draft Tubes in Reaction Turbines, Efficiency of Draft Tube; Kaplan turbine and working proportions of Kaplan turbine.

Performance & characteristics of Turbines: Unit Quantities, Specific Speed and its importance; Model Relationships; Operating Characteristic Curves; Cavitation problem in Turbines – Thoma's Cavitation Factor.

Hydraulic Machines – Centrifugal Pumps Functions of a Pump – Types of Pumps – Selection Criterion – Rotodynamic and Positive Displacement Pumps – Comparison between Centrifugal & Reciprocating Pumps.

Centrifugal Pumps – Components & Working principles of Centrifugal Pumps; Classification of Centrifugal Pumps – Impellers based on Shape and Type of Casing, Pump with Volute Casing, Pump with Vortex Chamber & Pump with Guide vanes, Closed, Semi-closed & Open Impellers, Axial, Radial & Mixed Flow Impellers; Working Head and Number of Stages, Single & Double Suction. Work done by Centrifugal Pumps – Pressure Change in a Pump, Manometric and Static Head – Velocity triangles – Effect of Vane Shape; Pump Losses and Efficiency – Pressure Rise in the Impeller – Minimum Starting Speed of pump – Multistage Pumps; Pumps in Parallel and Series.

Performance Characteristics of Pumps – Similarity Relations and Specific speed of Pumps – Dimensionless characteristics – Constant efficiency curves of Centrifugal Pumps

Hydraulic Machines – Reciprocating Pump & Hydraulic Ram: Reciprocating Pumps – Fundamental concepts, Component Parts and Working principle of Single Acting and Double Acting Reciprocating Pumps – Discharge Coefficient, Volumetric Efficiency and Slip; Work done by Reciprocating pumps – Work Done and Power Input – Indicator Diagram – Effect of Acceleration and Friction on Indicator Diagram – Maximum Speed of Rotation of Crank; Air

Vessels and their principles – Modified Indicator Diagram in the presence of Air Vessels, Work Saved due to Presence of Air Vessel, Flow into and from Air Vessel. Hydraulic Ram – Working Principle of Hydraulic Ram.

Text Books:

1. Fluid Mechanics and Hydraulic Machinery, Modi, P.N. and S.M. Seth, Standard Book House.
2. Fluid Mechanics, Jain, A.K., Khanna Publishers.

Reference Books:

1. Fluid Mechanics and Hydraulic Machinery, Modi, P.N. and S.M. Seth, Standard Book House.
2. Fluid Mechanics, Jain, A.K., Khanna Publishers.
3. Engineering Fluid Mechanics Kumar, K.L., S. Chand & Co. Ltd.
4. Hydraulic Machines, Jagadish Lal, Metropolitan Book Company.

ESTIMATION AND QUANTITY SURVEYING

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objective:

To provide the student with the ability to estimate the quantities of item of works involved in buildings, water supply and sanitary works, road works and irrigation works, and also to equip the student with the ability to do rate analysis, valuation of properties and preparation of reports for estimation of various items

Course Outcomes:

Students will be able to

1. List out the various components and units of measurements of different works.
2. Explain various types of estimates and general, detailed specifications of various items of work
3. Apply the method of building estimate to find out the quantities of various items of work
4. Determine the rate per unit of various items of work
5. Select various methods to find out the valuation of a property

SYLLABUS

Introduction: Standard units, Units of measurement of different items of work. Meaning of estimating. Errors in estimation, Different types of estimates. Contingencies and related terms in the estimate, different types of approvals. Plinth area and related terms used in the estimation of various structures, rules and methods of measurements of different works.

Specifications : Meaning, purpose, types of specifications, general specification, detailed specifications of different items of buildings and other structures – Rate analysis –Data sheet for materials and various items of work in buildings and other structures, schedule of rates, abstract estimate of buildings.

Detailed estimate of buildings: Different items of work in building; Principles of taking out quantities, detailed measurement form; long wall and short wall method of building estimate, Centre line method of building estimate. Estimate of RCC building, slope roof buildings; G.I. and A.C. Sheet, Detailed estimate of different types of doors and windows, electricity and water supply. Sanitation works etc.

Estimate of earth work; different formulae for calculations, estimate of metalled road, Tar road, concrete road, Railway track, Estimate of culverts and bridges etc. Valuation of buildings; purpose, different method of building valuation; different terms used in valuation and their meaning.

Text Books:

1. Estimating and Costing by B.N.Dutta, UBBS Publishers, 2000.
2. Estimating and Costing by G.S.Birdie.

Reference Books :

1. Estimation, Costing, Specifications and Valuation in civil Engineering by M.Chakraborti.
2. Estimating and Costing in Civil Engineering by B.N. dutta.
3. Textbook of estimating and costing by G.S. Birdie.
4. Textbook on Estimating, Costing and Accounts by D.D. Kohli and R.C. Kohli.

ENVIRONMENTAL ENGINEERING - II

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

COURSE OBJECTIVES:

Student shall be able to:

1. Explain the fundamentals of chemistry and biology involved in sewage treatment
2. Understand the principles and working of each process that involve physical and biological reactions
3. Design the treatment unit operations

COURSE OUTCOMES:

By the end of the course, the student should be able to:

1. Compare water and waste water [K2]
2. Explain principles of conventional treatment process and miscellaneous treatment techniques [K2]
3. Examine the operational differences of each unit process [K4]
4. Interpret the feasible technique required for particular waste water [K5]
5. Determine the size of unit operations using working principles of each [K5]
6. Design a sewage treatment plant assuming whatever data is required [K6].

SYLLABUS

Sanitation – systems of sanitation– sewerage systems – Estimation of sewage flow and storm water – design of sewers –sewer types – Layout of sewer network – materials for sewers – cleaning and ventilation methods – sewer appurtenances

Pumping – necessity– pumping station – its location – functions – types of pumps- sewage and water pumping requirements- House drainage systems – traps – classification – drain pipes – plumbing systems – Layout of building drainage – Storm water drainage

Quality of sewage –Sampling and analysis - decomposition- cycles of decomposition – BOD – COD – Layout of sewage treatment plant – F/M importance – preliminary treatment – screens – grit chamber – skimming tanks – Primary treatment – Sedimentation tanks

Secondary treatment – Attached growth process – contact beds – intermittent sand filters – trickling filters – Suspended growth process – activated sludge process- Miscellaneous methods for sewage treatment – oxidation ponds - oxidation ditches - RBC's - sewage lagoons- extended aeration process – septic tanks – imhoff tanks

Sewage disposal methods – natural and artificial methods –Sludge generation- characteristics - treatment- sludge disposal methods.

Text Books:

1. “Sewage disposal and Air pollution Engineering”, by S.K. Garg, Khanna Publishers

Reference Books

1. “Environmental Engineering” by Peavy, H.S. Rowe and Tchobanoglous, Mc. Graw Hill publishers
2. “Water Supply and Sanitary Engineering” by S.C.Rangwala, Charotar Publishers

GROUND IMPROVEMENT TECHNIQUES (Elective-II)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

Students learn

1. Various techniques of insitu ground modification.
2. To make the student learn the concepts, purpose and effects of grouting.
3. Knowledge of reinforcement to soils in the form of geo textiles and other synthetic materials.
4. Basic knowledge on various ground improvement techniques and their suitability for various types of soil conditions.

Course Outcomes:

Students will be able to

1. Apply in-situ densification methods for improving cohesive and cohesion less soil deposits. Design stone column for improving soft clays.
2. Apply grouting technique for improving soils.
3. Understand the concepts of reinforced earth.
4. Understand various soil stabilization techniques.

SYLLABUS

In-situ densification Methods: granular soils – Introduction of Vibration at the ground surface, Impact at the Ground surface, Vibration at depth, Impact at depth, compaction control; cohesive soils - introduction, preloading or dewatering, drain walls, sand drains, sand wicks, geodrains/band drains, stone and lime columns, forced vacuum pre consolidation, thermal methods.

Stone columns, introduction, construction practice, design principles, design of stone columns, vibrofloatation techniques and other techniques like dynamic replacement etc.

Grout injections, suspension and solution grouts, grouting equipment and methods, Applications.

Reinforced Earth: Principles, components of reinforced earth, functions, determination of angle of interfacial friction, factors effecting angle of interfacial friction.

Stabilization: Mechanical stabilization, Soil aggregate mixture, properties and proportioning techniques, soft aggregate stabilization, compaction, field compaction control. Cement stabilization, Mechanism, factors affecting and properties, use of additives, design of soil cement mixtures, construction techniques. Lime and Bituminous Stabilization: Types of admixtures, mechanism, factors affecting, construction methods.

Text Books:

1. Construction and Geotechnical Methods in Foundation Engineering, Robert M. Koerner : McGraw Hill.
2. Purushothama Raj. P, “Ground Improvement Techniques”, 2nd ed., Laxmi Publications (p) Ltd., New Delhi, 1998.

Reference Books:

1. Ground improvement techniques by Purushotham Raj, Laxmi Publications
2. Principles and practices of ground improvement by Jie Han, Wiley

ENVIRONMENTAL IMPACT ASSESMENT (Elective-II)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

Student shall be able to:

1. Learn the guidelines to prepare EIA and EIS report
2. Acquire knowledge in the methodologies and various attributes
3. Analyze the impact of urbanization and industrialization on the environment.

Course Outcomes:

By the end of the course, the student should be able to:

1. Define the terms related to EIA [K1]
2. List out the elements of EIA and guidelines to prepare EIS [K1,K4]
3. Identify the environmental attributes to be considered for the study [K3]
4. Explain the methodologies for EIA and review the relief and rehabilitation works [K5]
5. Discuss the case studies of EIA [K6]
6. Adapt the suitable measures to control the environmental impact [K6]

SYLLABUS

EIA and EIS concepts – Elements of EIA – Guidelines in USA, preparation of EIS, Elements of EIA

Agency Activities, Environmental Setting, Environmental attributes, air, water, soil, ecology, noise, Socioeconomic aspects, Culture and Human aspects (Human settlements – rehabilitations)

Environmental impacts, Identification measurement, Aggregation, Secondary and Cumulative Impacts

Criteria for selection of methodology, impact assessment methodologies, procedure for reviewing environment impact statement.

Case studies, Economic Impact analysis energy production impact analysis, cost benefit analysis, Environmental impact mitigation and control measures.

Text Book:

1. “Environmental Impact Analysis” by L.Canter, Mc Graw Hill Publishers

Reference Books:

1. Environmental Impact Assessment Methodologies by Y. Anjaneyulu and ValliManikkam. BSP Books PVT Ltd.
2. Environmental impact assessment by Anji Reddy. BSP Books PVT Ltd

MARINE STRUCTURES (Elective-II)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

Students learn

1. Introduce waves, tides, tsunamis, storm surge and currents.
2. Classification of various offshore structures.
3. Calculate the loading on offshore structures.

Course Outcomes:

Students will be able to

1. Design of break waters and jetties.
2. Understand the principles of rubble mound and composite breakwaters.
3. Evaluate various loads on offshore structures.

SYLLABUS

Introduction: Waves, tides, tsunamis, storm surge and currents. Wave forces on small and large cylinders, Sea walls, Design of break waters and jetties.

Break waters – Types – Selection of site and type – effects on the beach – Design principles of Rubble mound and composite Breakwaters – Stability of Rubble Structures, Wharves and Jetties.

Introduction: Offshore definition, Purpose of Offshore Structures, Classification and Examples, Various types of Offshore Structures – Jacket Platforms, Semi submersibles, Tension Leg Platforms, Gravity Platforms Guyed Towers, Articulated Towers.

Load Calculations:

I. Environmental loads on off shore structures due to

a) Wind b) Wave c) Current d) Ice e) Earth quake

II. Functional loads;

III. Buoyant Forces;

IV. Installation forces, Soil structure interaction.

Preliminary design aspects of offshore structures. Construction, Towing and installation procedure of Jacket platforms and Gravity platforms.

Text Books:

1. J.S. Mani, “Coastal Hydrodynamics” PHI Learning
2. Hydrodynamics of Offshore Structures by S.K. Chakrabarti, SpringerVerlag
3. Handbook of Offshore Engineering by S.K. Chakrabarti, Elseviers, 2005.
4. Offshore Structural Engineering by: Thomas H Dawson.
5. Mechanics of Wave Forces on Offshore structures by: TurgutSarpkaya& M. Issacson.

Text Books:

1. Chaudhary, G.K and Dover, W.D. 1985. Fatigue analysis of offshore platforms subjected to sea wave loading, Int. J. Fatigue.
2. Clauss, G. T. et al. 1992. Offshore Structures, Vol1 - Conceptual Design and Hydro mechanics: Springer, London.

Reference Books:

1. Dawson, T. H., 1983. Offshore Structural Engineering: Prentice-Hall Inc.
2. Gerwick, B.C.Jr. 1986. Construction of Offshore Structures: John Wiley, New York.
3. Graff, W.J. 1981. Introduction to offshore structures: Design, fabrication and installation: Gulf Publishing Co, Tokyo.

URBAN HYDROLOGY (Elective -II)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

Students learn

1. Appreciate the impact of urbanization on catchment hydrology.
2. Understand the importance of short duration rainfall runoff data for urban hydrology studies.
3. Learn the techniques for peak flow estimation for storm water drainage system design.
4. Understand the concepts in design of various components of urban drainage systems.
5. Learn some of the best management practices in urban drainage.
6. Understand the concepts of preparation master urban drainage system.

Course Outcomes:

Students will be able to

1. Develop intensity duration frequency curves for urban drainage systems.
2. Develop design storms to size the various components of drainage systems.
3. Apply best management practices to manage urban flooding.
4. Prepare master drainage plan for an urbanized area.

SYLLABUS

Introduction: Urbanisation and its effect on water cycle – urban hydrologic cycle – trends in urbanisation – Effect of urbanisation on hydrology.

Precipitation Analysis: Importance of short duration of rainfall and runoff data, methods of estimation of time of concentration for design of urban drainage systems, Intensity-Duration - Frequency (IDF) curves, design storms for urban drainage systems.

Approaches to urban drainage: Time of concentration, peak flow estimation approaches , rational method, NRCS curve number approach, runoff quantity and quality, wastewater and stormwater reuse , major and minor systems.

Elements of drainage systems: Open channel, underground drains, appurtenances, pumping, source control.

Analysis and Management: Stormwater drainage structures, design of stormwater network-Best Management Practices–detention and retention facilities, swales, constructed wetlands, models available for stormwater management.

Master drainage plans: Issues to be concentrated upon – typical urban drainage master plan, interrelation between water resources investigation and urban planning processes , planning objectives, comprehensive planning , use of models in planning.

Text Books:

1. 'Manual on Drainage in Urbanised area' by Geiger W. F., J Marsalek, W. J. Rawls and F. C. Zuidema, (1987 – 2 volumes), UNESCO.
2. 'Urban Hydrology' by Hall M J (1984), Elsevier Applied Science Publisher.

Reference Books:

1. 'Hydrology – Quantity and Quality Analysis' by Wanielista M P and Eaglin (1997), Wiley and Sons.
2. 'Urban Hydrology, Hydraulics and Stormwater Quality: Engineering Applications and Computer Modelling' by Akan A.O and R.L. Houghtalen (2006), Wiley International.

FINITE ELEMENTS METHODS OF ANALYSIS
(Elective-II)

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

Students learn

1. To learn basic principles of finite element analysis procedure.
2. To learn the theory and characteristics of finite elements that represent engineering structures.
3. To learn and apply finite element solutions to structural problem to develop the knowledge and skills needed to effectively evaluate finite element analyses performed by others.
4. Learn to model complex geometry problems and solution techniques.

Course Outcomes:

Students will be able to

1. Understand the concepts behind variational methods and weighted residual methods in FEM.
2. Identify the application and characteristics of FEA elements such as bars, beams, 2-D element and axisymmetric element.
3. Develop element characteristic equation procedure and generation of global stiffness equation will be applied.
4. Able to apply Suitable boundary conditions to a global structural equation, and reduce it to a solvable form

SYLLABUS

Introduction to finite element method, Equilibrium equations, strain – displacement relations, stress – strain relations, Compatibility equations, Variational and weighted residual methods, concept of potential energy, one dimensional problems.

Discretization of domain, element shapes, discretization procedures, node numbering, mesh generation, interpolation functions, local and global coordinates, treatment of boundary conditions, assembly of stiffness matrix.

Basic component: One dimensional FEM single bar element, Beam element: Derivation of stiffness matrix, Assembly of stiffness, Matrix boundary conditions, shape functions for 1 D elements, Initial strain and temperature effects, and trusses under axial forces.

Two dimensional FEM: Different types of elements for plane stress and plane strain analysis – Comparison of CST and LST elements, Derivation of CST stiffness matrix and equations, Initial strain and temperature effects, Problems on plane stress and plane strains in CST elements.

Axisymmetric: Axisymmetric formulation, shape functions, strain-Displacement matrix, stress-strain relationship matrix, Element stiffness matrix, Problems on Axisymmetric.

Text Books:

1. Finite Elements Analysis by Dr. S. Senthil, R. Panneerdhass- Lakshmi Publications Chennai.
2. Finite Elements Methods in Engineering by Tirupati.R.Chandrupatla and AshokD.Belegundu - Pearson Education Publications.
3. A first course in the Finite element method by Daryl L. Logan, Cengage learning India.

Reference Books:

1. Concepts and Applications of Finite Element Analysis by Robert D.Cook, David S. Malkus and Michael E. Plesha. John Wiley & Sons.
2. Finite Element analysis – Theory & Programming by C. S. Krishna Murthy- Tata Mc.Graw Hill Publishers.
3. Text book of Finite Element analysis by P.Seshu – Prentice Hall of India.

GEOTECHNICAL ENGINEERING LAB - II

Lab	: 3 Periods	Sessionals	: 50
Exam	: 3 Hrs.	Ext. Marks	: 50
		Credits	: 2

Course Objectives:

The objectives of this course are to make the student aware of

1. Tests for field identification of soils.
2. Tests to determine shear strength of soil.
3. Tests to determine relative density of soil.
4. Tests to determine of CBR of soil.
5. Tests to determine of swell characteristics of soil.

Course Outcomes:

Students will be able to

1. Classify the types of soil deposits.
2. Determine the shear strength parameters of soils by various methods.
3. Estimate the California Bearing Ratio (CBR) of a soil.
4. Determine the relative density of a coarse-grained soil.
5. Determine the swelling characteristics of expansive soils.

SYLLABUS

1. Field identification & classification of soils
2. Unconfined compression test
3. CBR test/plate bearing test
4. Triaxial compression test (u-u test)
5. Direct shear test
6. Vane shear test
7. Relative density.
8. Differential free swell and swell pressure test.

Reference Books:

1. Geotechnical engineering lab manual, Department of civil engineering, S. R. K. R Engg College.
2. Soil Mechanics and Foundation Engg. By K. R. Arora, Standard Publishers and Distributors, Delhi.

CONCRETE LAB

Lab : 3 Periods
Exam : 3 Hrs.

Sessionals : 50
Ext. Marks : 50
Credits : 2

Course Objectives:

The objectives of this course are to make the student aware of

1. Tests to identify the physical properties of cement, sand and aggregate.
2. Tests to identify the initial and final setting of cement.
3. Tests to identify workability of concrete.
4. Tests to identify mechanical characteristics of concrete.

Course Outcomes:

Students will be able to

1. Determine physical properties of cement, sand and aggregate.
2. Classify fine aggregate and coarse aggregate as per IS 383
3. Determine workability of concrete.
4. Determine mechanical properties of concrete.

SYLLABUS

1. Fineness of cement
2. Specific gravity of cement
3. Normal consistency of cement
4. Initial and final setting time
5. Compressive strength of cement for different grades of cement
6. Specific gravity and unit weight of coarse and fine aggregates
7. Sieve analysis of coarse and fine aggregates and classification as per IS 383.
8. Bulking characteristics of sand
9. Workability tests on fresh concrete by using:
 - (a) Slump cone
 - (b) Compaction factor apparatus
 - (c) Flow table
 - (d) Vee-Bee consistometer
10. Tests on hardened concrete
 - (a) Compressive strength
 - (b) Split tensile strength
 - (c) Flexural strength

Reference Books:

1. Concrete lab manual, Department of civil engineering, S. R. K. R Engg College.
2. Concrete Technology by M.S.Shetty. – S.Chand& Co.; 2004.
3. Properties of Concrete by A.M.Neville – PEARSON – 4th edition.
4. Concrete Technology by M.L. Gambhir. – Tata Mc. Graw Hill Publishers, New Delhi.

FLUID MECHANICS LAB - II

Lab : 3 Periods
Exam : 3 Hrs.

Sessionals : 50
Ext. Marks : 50
Credits : 2

Course Objectives:

The objectives of this course are to make the student aware of

1. Tests to identify the different types of flow measuring devices to measure flow in tanks, pipes and open channels.

Course Outcomes:

Students will be able to

2. Analyze the flow through a pipe with friction and determine the friction factor in Darcy-Weisbach equation.
3. Determine the coefficient of impact on a flat plate and curved vane by comparing the theoretical and actual forces by impact.
4. Analyze the working of the centrifugal pump and develop the characteristics of power input, head and efficiency under various discharges and plot the characteristic curves.
5. Analyze the working of the reciprocating pump and develop the characteristics of power input and discharge and efficiency under various heads and plot the characteristic curves.
6. Determine the performance characteristics of pelton wheel turbine and develop the characteristic curves of unit discharge, unit power and unit head under varying unit speed.
7. Determine the performance characteristics of Francis turbine and develop the characteristic curves of unit discharge, unit power and unit head under varying unit speed.

SYLLABUS

1. Study of Characteristics of a hydraulic jump – To measure and draw $(E_1 - E_2)/E_1$ vs F_1 and L_j / y_2 vs F_1 , and compare with theoretical results wherever possible.
2. Study of Rugosity coefficients in an open channel flow.
3. Study of major losses in pipes – Pipe friction – To compute Darcy-Weisbach friction factor.
4. Study of Drag characteristics of a circular cylinder with its axis normal to the direction of flow.
 - (a) To measure the pressure distribution on the surface of a cylinder and plot the dimensionless pressure variation around the cylinder and compute the pressure drag.
 - (b) To measure the velocity variation in the wake of the cylinder, velocity of approach, and compute the total drag by momentum principle.
5. Study of performance characteristics of a centrifugal pump – To measure the discharge, head developed, and power input at various discharges for centrifugal pump and draw the performance characteristics.
6. Study of performance characteristics of a reciprocating pump – To measure the discharge, head developed, and power input at various discharges for reciprocating pump and calculate percentage slip and efficiency.
7. Study of performance characteristics of a Pelton turbine – To measure the discharge, head difference across the turbine, the brake load, speed of turbine for various discharges and draw the performance characteristics.

8. Study of performance characteristics of a Francis turbine – To measure the discharge, head difference across the turbine, the brake load, speed of turbine for various discharges and draw the performance characteristics.
9. Study of impact of a jet on flat and curved vanes.

Reference Books:

1. Fluid mechanics lab manual, Department of civil engineering, S. R. K. R Engg College
2. Fluid Mechanics and Hydraulic Machinery, Modi, P.N. and S.M. Seth, Standard Book House.
3. Fluid Mechanics, Jain, A.K., Khanna Publishers.

VERBAL & QUANTITATIVE APTITUDE – II
(Common to All Branches)

Theory	: 5 Periods(VA-3+QA-2)	Sessionals	: 100
Exam	: 3 Hrs.	Credits	: 2

Part-A: Verbal Aptitude and Soft Skills-II

Course objectives:

1. To expose the students to bettering sentence expressions and also forming equivalents.
2. To instill reading and analyzing techniques for better comprehension of written discourses.
3. To create awareness among the students on the various aspects of writing, organizing data, preparing reports, and applying their writing skills in their professional career.
4. To inculcate conversational skills, nuances required when interacting in different situations.
5. To build/refine the professional qualities/skills necessary for a productive career and to instill confidence through attitude building.

Course Outcomes:

The students will be able to

1. Construct coherent, cohesive and unambiguous verbal expressions in both oral and written discourses.
2. Analyze the given data/text and find out the correct responses to the questions asked based on the reading exercises; identify relationships or patterns within groups of words or sentences
3. Write paragraphs on a particular topic, essays (issues and arguments), e mails, summaries of group discussions, reports, make notes, statement of purpose(for admission into foreign universities), letters of recommendation(for professional and educational purposes).
4. Converse with ease during interactive sessions/seminars in their classrooms, compete in literary activities like elocution, debates etc., raise doubts in class, participate in JAM sessions/versant tests with confidence and convey oral information in a professional manner.
5. Participate in group discussions/group activities, exhibit team spirit, use language effectively according to the situation, respond to their interviewer/employer with a positive mind, tailor make answers to the questions asked during their technical/personal interviews, exhibit skills required for the different kinds of interviews (stress, technical, HR) that they would face during the course of their recruitment process.

SYLLABUS

UNIT -I (VA)

Sentence Improvement (finding a substitute given under the sentence as alternatives), Sentence equivalence (completing a sentence by choosing two words either of which will fit in the blank), cloze test (reading the written discourse carefully and choosing the correct options from the alternatives and filling in the blanks), summarizing and paraphrasing .

UNIT- II (VA)

Types of passages (to understand the nature of the passage), types of questions (with emphasis on inferential and analytical questions), style and tone (to comprehend the author's intention of writing a passage), strategies for quick reading(importance given to skimming, scanning), summarizing ,reading between the lines, reading beyond the lines, techniques for answering questions related to vocabulary (with emphasis on the context), supplying suitable titles to the passage, identifying the theme and central idea of the given passages.

UNIT- III (VA)

Punctuation, discourse markers, general Essay writing, writing Issues and Arguments(with emphasis on creativity and analysis of a topic), paragraph writing, preparing reports, framing a 'Statement of purpose', 'Letters of Recommendation', business letter writing, email writing, writing letters of complaints/responses. picture perception and description, book review.

UNIT-IV (VA)

Just a minute sessions, reading news clippings in the class, extempore speech, telephone etiquette, making requests/suggestions/complaints, elocutions, debates, describing incidents and developing positive non verbal communication, story narration, product description.

UNIT-V (SS)

Employability Skills – Significance — Transition from education to workplace - Preparing a road map for employment – Getting ready for the selection process, Awareness about Industry / Companies – Importance of researching your prospective workplace - Knowing about Selection process - Resume Preparation: Common resume blunders – tips, Resume Review, Group Discussion: Essential guidelines – Personal Interview: Reasons for Rejection and Selection.

Reading/ Listening material:

1. Guide to IELTS, Cambridge University Press
2. Barron's GRE guide.
3. Newspapers like 'The Hindu', 'Times of India', 'Economic Times'.
4. Magazines like Frontline, Outlook and Business India.
5. News channels NDTV, National News, CNN

Text Books:

1. Objective English and Verbal Reasoning by R S Agarwal.
2. Communication Skills by Sanjay Kumar and PushpaLatha, Second Edition, OUP.
3. Business Correspondence and Report Writing – A Practical Approach to Business and Technical Communication by R C Sharma and Krishna Mohan.
4. Soft Skills & Employability Skills by Samina Pillai and Agna Fernandez, Cambridge University Press India Pvt. Ltd.
5. Soft Skills, by Dr. K. Alex, S. Chand & Company Ltd., New Delhi

Reference Books:

1. Oxford Guide to Effective Writing and Speaking by John Seely.
2. Collins Cobuild English Grammar by Collins
3. The Art of Public Speaking by Dale Carnegie
4. The Leader in You by Dale Carnegie
5. Emotional Intelligence by Daniel Golman
6. Stay Hungry Stay Foolish by Rashmi Bansal

Part-B: Quantitative Aptitude-II

Course objectives:

The objective of introducing quantitative aptitude-II is:

1. To refine concepts related to quantitative aptitude. – SOLVING PROBLEMS OF DI and accurate values using averages, percentages.
2. To inculcate logical thinking by exposing the students to puzzles and reasoning related questions.
3. To familiarize the students with finding out accurate date and time related problems.
4. To enable the students solve the puzzles using logical thinking.
5. To expose the students to various problems based on geometry and menstruation.

Course Outcomes:

1. The students will be able to perform well in calculating different types of data interpretation problems.
2. The students will perform efficaciously on analytical and logical problems using various methods.
3. Students will find the angle measurements of clock problems with the knowledge of calendars and clock.
4. The students will skillfully solve the puzzle problems like arrangement of different positions.
5. The students will become good at solving the problems of lines, triangulars, volume of cone, cylinder and so on.

SYLLABUS

UNIT I: Averages, mixtures and allegations, Data interpretation

Understanding of AM,GM,HM-Problems on averages, Problems on mixtures standard method. Importance of data interpretation: Problems of data interpretation using line graphs, Problems of data interpretation using bar graphs, Problems of data interpretation using pie charts, Problems of data interpretation using others.

UNIT II :Puzzle test, blood Relations, permutations, Combinations and probability

Importance of puzzle test, Various Blood relations-Notation to relations and sex making of family Tree diagram, Problems related to blood relations, Concept of permutation and combination, Problems on permutation, Problems on combinations, Problems involving both permutations and combinations, Concept of probability-Problems on coins, Problems on dice, Problems on cards, Problems on years.

UNIT III :Periods, Clocks, Calendars, Cubes and cuboids

Deriving the formula to find the angle between hands for the given time, finding the time if the angle is known, Faulty clocks, History of calendar-Define year, leap year, Finding the day for the given date, Formula and method to find the day for the given date in easy way, Cuts to cubes, Colors to cubes, Cuts to cuboids, Colors to cuboids.

UNIT IV: Puzzles

Selective puzzles from previous year placement papers, sitting arrangement, problems-circular arrangement, linear arrangement, different puzzles.

UNIT V:Geometry and Menstruation

Introduction and use of geometry-Lines, Line segments, Types of angles, Intersecting lines, Parallel lines, Complementary angles, supplementary angles, Types of triangles-Problems on triangles, Types of quadrilaterals-Problems on quadrilaterals, Congruent triangles and properties, Similar triangles and its applications, Understanding about circles-Theorems on circles, Problems on circles, Tangents and circles, Importance of menstruation-Introduction of cylinder, cone, sphere, hemi sphere.

Text Books:

1. Quantitative aptitude by RS Agarwal
2. Verbal and non verbal reasoning by RS Agarwal.
3. Puzzles to puzzle you by shakunataladevi
4. More puzzles by shakunataladevi
5. Puzzles by George summers.

Reference Books:

1. Barron's by Sharon Welner Green and Ira K Wolf (Galgotia Publications pvt. Ltd.)
2. Websites: m4maths, Indiabix, 800score, official CAT, GRE and GMAT sites
3. Material from 'IMS, Career Launcher and Time' institutes for competitive exams.
4. Books for cat by arunsharma
5. Elementary and Higher algebra by HS Hall and SR knight.

MINI PROJECT

Lab : 3 Periods
Sessionals : 50

Credits : 2

Course Objectives:

The objectives of this course are to make the student

1. Know the concrete material properties and their behavior.
2. Know the various problems related to environmental, societal and economical.
3. Learn the performance and durability aspects of construction materials.

Course Outcomes: Students are able to

1. Examine classical theories of civil engineering and their practical applications.
2. Interpret their knowledge on the latest developments in civil engineering.
3. Develop concepts related to environmental, societal, economical and sustainable aspects of any civil engineering structure.
4. Develop teamwork and lifelong learning skills

BASIC CODING
(Common to CE & ME)
(MOOCS-III)

Theory : 4 Periods
Exam : 3 Hrs.

Sessionals : 100
Credits : 2

Course Objectives:

1. To develop programming skills among the students.
2. To familiarize the student with Control Structures, Loop Structures.
3. To familiarize the student with Basic searching and sorting Methods.
4. To familiarize the student with Functions, Recursions and Storage Classes.
5. To familiarize the student with Structures and Unions.
6. To familiarize the student with Operating System concepts.
7. To familiarize the student with Networking concepts.

Course Outcomes:

At the end of the course students will be able to

1. Know about Control Structures, Loop Structures and branching in programming.
2. Know about various searching and sorting methods.
3. Know about Functions, Recursions and Storage Classes.
4. Know about Structures and Unions.
5. Know different Operating System concepts.
6. Differentiate OSI Model Vs. TCP/IP suite.

SYLLABUS

UNIT I Review of Programming constructs

Programming Environment, Expressions formation, Expression evaluation, Input and Output patterns, Control Structures, Sequential branching, Unconditional branching, Loop Structures, Coding for Pattern Display.

UNIT II Introduction to Linear Data, strings and pointers

Structure of linear data, Operation logics, Matrix forms and representations, Pattern coding, Working on character data, Compiler defined methods, Substitution coding for defined methods, Row Major representation, Column Major representation, Basic searching and sorting Methods.

UNIT III Functions, Recursions and Storage Classes

Functions – Introduction to modular programming – Function Communication - Pass by value, Pass by reference – Function pointers – Recursions – Type casting – Storage classes

Practice: programs on passing an array and catching by a pointer, function returning data, comparison between recursive and Iterative solutions.

Data referencing mechanisms: Pointing to diff. data types, Referencing to Linear data, Runtime-memory allocation, Named locations vs pointed locations, Referencing a 2D-Matrix

UNIT IV User-defined datatypes, Pre-processor Directives and standard storage

Need for user-defined data type – structure definition – Structure declaration – Array within a Structure – Array of Structures – Nested Structures - Unions – Declaration of Union data type, StructVs Union - Enum – Pre-processor directives , Standard storage methods, Operations on file, File handling methods, Orientation to Object oriented programming

Practice: Structure padding, user-defined data storage and retrieval programs

UNIT V Operating system principles and Database concepts

Introduction to Operating system principles, Process scheduling algorithms, Deadlock detection and avoidance, Memory management, Networking: Introduction to Networking, OSI Model Vs. TCP/IP suite, Datalink layer, Internet layer, DVR Vs. LSR, Transport Layer, Application Layer

References:

1. Computer Science, A structured programming approach using C, B.A.Forouzan and R.F.Gilberg, 3rd Edition, Thomson, 2007.
2. The C –Programming Language, B.W. Kernighan, Dennis M. Ritchie, Prentice Hall India Pvt.Ltd
3. Scientific Programming: C-Language, Algorithms and Models in Science, Luciano M. Barone (Author), Enzo Marinari (Author), Giovanni Organtini, World Scientific .
4. Object Oriented Programming in C++: N. Barkakati, PHI.
5. Object Oriented Programming through C++ by Robat Laphore.
6. <https://www.geeksforgeeks.org/>.
7. <https://www.tutorialspoint.com/>

**ENERGY EFFICIENCY, ACOUSTICS AND DAYLIGHTING IN BUILDING
(MOOCS-III)**

Theory	: 4 Periods	Sessionals	: 100
Exam	: 3 Hrs.	Credits	: 2

Course Objectives: Students Learn

1. Various Environmental factors and human response to the Environment
2. The Thermal comforts of a building along with heat flow patterns.
3. Day lightening, Noise and building acoustics planning
4. The concepts of functional design of building for thermal aspects and energy efficiency

Course Outcomes: At the end of the course, Students will be able to:

1. Understand Various Environmental factors and human response to the Environment
2. Know the thermal comforts and strategies pertaining to it.
3. Design the lightening, noise, acoustics and various aspects of a building

SYLLABUS

1. Environmental Factors: Factors and their representation, tropical environments and site environments, etc.
2. Human response to environment: Factors affecting human comfort, Human response to thermal environment, noise, visual environment etc.; Comfort indices
3. Response of building to thermal environment: Processes of heat exchange of building with environment; Effect of solar radiation; Thermal properties of material and sections and their influence
4. Steady and periodic heat transfer in buildings
5. Heat flow computations: Transmission matrix, Admittance method, etc.-1
6. Heat flow computations: Transmission matrix, Admittance method, etc.-2
7. Structural control and design for energy efficiency: Selection of envelope elements, Orientations, shape, Glasses and shading devices
8. Natural ventilation: Purpose of ventilation, Mechanisms, Fenestration Design for natural ventilation
9. Noise and Building: Basic acoustics and noise, Planning, Sound in free field, protection against external noise
10. Internal noise sources and protection against air borne & structure borne noise.
11. Day lighting: Lighting principles and fundamentals
12. Sky, Indian sky, daylight prediction and design of fenestration.

WEB LINK:

https://onlinecourses.nptel.ac.in/noc18_ce06/preview

**ELECTRONIC WASTE MANAGEMENT-ISSUES AND CHALLENGES
(MOOCS-III)**

Theory	: 4 Periods	Sessionals	: 100
Exam	: 3 Hrs.	Credits	: 2

Course Objectives: Students learn

1. Various sources of E-waste and their impact on environment
2. The E-waste management rules of India.
3. Case-studies around the world

Course Outcomes: At the end of the course, Students will be able to:

1. Know the sources of E-waste and their impact on environment
2. Know the Rules pertaining to E-waste management
3. Understand Various Environmental hazards of E-waste

SYLLABUS

1. Overview of the course
2. Exposure pathway of pollutants emitted from Recycling of E-Waste
3. E-Waste Management Rules of India (2011 and 2016 Rules)
4. E-waste Management: Case Studies and Unique Initiatives from around the World

WEB LINK:

https://onlinecourses.nptel.ac.in/noc18_ce07/preview

Code: B16 CE 3215C

PRINCIPLES AND APPLICATIONS OF BUILDING SCIENCE
(MOOCS-III)

Theory	: 4 Periods	Sessionals	: 100
Exam	: 3 Hrs.	Credits	: 2

Course Objectives: Students learn

1. The principles of building geometry and climatic responses in building design.
2. Energy efficiency, glazing systems and bio climatic design .
3. Fundamentals of building acoustics, visual quality and effective day lighting

Course Outcomes: At the end of the course the student will be able to:

1. Apply the concept of climatology for housing layouts and principles of planning
2. Know the acoustics, acoustic materials and quality indicators
3. Design of day-lighting in buildings and integrated design approach

SYLLABUS

1. Solar geometry, climate responsive building design, thermal comfort
2. Bio climatic design, building envelop, glazing systems, energy efficiency
3. Fundamentals of building acoustics, Quality indicators, Acoustic materials, Noise control
4. Visual quality in built environment, Effective day lighting design, Integrated design approach

WEB LINK:

https://onlinecourses.nptel.ac.in/noc16_ce11/preview



SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (AUTONOMOUS)

(Affiliated to Andhra University, Visakhapatnam), (Recognised by AICTE, New Delhi)

Accredited by NAAC with 'A' Grade

Recognised as Scientific and Industrial Research Organisation

CHINNA AMIRAM (P.O):: BHIMAVARAM :: W.G.Dt., A.P., INDIA :: PIN: 534 204

SCHEME OF INSTRUCTION & EXAMINATION

(Regulation R16)

IV/IV B.TECH

(With effect from **2016-2017** Admitted Batch onwards)

CIVIL ENGINEERING

I-SEMESTER

Code No.	Name of the Subject	Credits	Lecture Hrs	Tutorial Hrs	Lab Hrs	Contact Hrs/ Week	Internal Marks	External Marks	Total Marks
B16 CE 4101	Water Resource Engineering-I	4	3	1	--	4	30	70	100
B16 CE 4102	Transportation Engineering-I	4	3	1	--	4	30	70	100
B16 CE 4103	Project planning and Management	4	3	1	--	4	30	70	100
B16 CE 4104	Computer Applications in Civil Engineering lab	2	--	--	3	3	50	50	100
B16 CE 4105	Transportation Engineering Lab	2	--	--	3	3	50	50	100
B16 CE 4106	Project Phase-I	2	--	--	3	3	50	--	50
Total		18	9	3	9	21	240	310	550

WATER RESOURCES ENGINEERING-I

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

Student may know the Major hydrologic components & apply key concepts to several practical areas of engineering hydrology & related design aspects, aquifer parameters & yield of wells, surface & subsurface investigation to locate ground water, storage capacity & life of reservoirs, the irrigation needs of crops and unlined & lined irrigation canals.

Course Outcomes: Student will be able to

1. Choose major hydrologic components & apply key concepts to several practical areas of engineering hydrology & related design aspects (K3)
2. Determine aquifer parameters & yield of wells (K3)
3. Carry out surface & subsurface investigation to locate ground water (K2)
4. Determine storage capacity & life of reservoirs (K3)
5. Assess the irrigation needs of crops (K5)
6. Design of unlined & lined irrigation canals (K6)

SYLLABUS

Introduction And Hydrological Aspects: Water Resources in India, Hydrology in water Resources Planning – Hydrologic Planning –Water budget equation; Precipitation – Types, Measurement of rainfall; Average depth of rainfall over an area, Mean annual rainfall, Analysis of Rainfall Data – Consistency of rainfall record, Double mass curve, Infiltration – Factors affecting and its determination, Infiltrometers; Evaporation and Evapotranspiration – Pan Evaporation; Runoff –Factors affecting Runoff, Methods of determination of Runoff, Hydrograph Analysis, Base flow separation, Unit Hydrographs, Hydrograph of different durations, Applications of Unit Hydrograph; S-hydrograph, Synthetic Unit Hydrograph; Stream flow measurement – Gauge discharge curves.

Ground Water Flow: Mechanics of interstitial flow, definitions, subsurface distribution of water, ground water movement; Darcy's law; Permeability – Intrinsic permeability; Well hydraulics – Steady flow in different types of aquifers and wells; Determination of hydraulic properties of aquifer; Well losses; Specific capacity of well; Well efficiency – Pumping tests – Recuperation test method for determination of well yield.

Methods of construction of open well-yield of an open well – Methods of construction of Tube Wells, Well shrouding and Well development, Spacing of tube wells, Design of tube well; Pumping requirements, Centrifugal and bore hole type pumps; Collector wells.

Reservoir Planning: Types of reservoir – Investigations for reservoir planning, Selection of site for a reservoir, Zones of storage in a reservoir; Purpose of reservoir, Design studies, Reservoir regulation, Reservoir yield, Mass curve and Demand curve, Determination of reservoir capacity, Yield from a reservoir of given capacity; Operating schedules – Rule Curve for reservoir operation; Economics of Water resources Projects –Apportionment of total cost of a Multi-Purpose project, Reservoir Losses –Measures to reduce evaporation loss in reservoirs sedimentation, Control of reservoir sedimentation.

Irrigation: Definition of irrigation, Types of irrigation systems – Direct and Indirect, Lift and Inundation irrigation Systems, Methods of irrigation – Surface and Sprinkler methods, Trickle or Drip Irrigation, Soil moisture Constants, Depth of water held by soil in different zones, Water extraction – Quality of irrigation water, Irrigation efficiencies – Soil moisture –Irrigation relationship – Water requirements of crops, Duty, Delta and Base period – Their relationship, Crops – Seasons, Factors affecting duty and methods of improving duty, Consumptive use of water –Determination of evapotranspiration – Blaney-Criddle and Penman of reservoir, Assessment of irrigation water charges.

Canal Systems: Classification of irrigation canals – Canal alignment, Design of unlined canals, Regime theories – Kennedy's and Lacey's theories, Critical tractive force method, Design problems – Balancing depth – L.S. of a channel – Design according to I.S : 7112, 1975; Schedule of area statistics, Cross section of an irrigation channel – Maintenance of irrigation channel. Regulation of channel system – Canal outlets, Requirements of a good outlet – Types of outlets; Water logging – Causes and control – Land drainage; Canal lining – methods, Design of lined canals.

Text Books:

1. Irrigation and Water Power Engineering, Punmia, B.C. and P.B.B. Lal, Laxmi Publications Pvt. Ltd.
2. Irrigation and Water Resources & Water Power, Modi, P.N., Standard Book House.
3. Irrigation and Hydraulic structures, Garg, S.K., Khanna Publishers.
4. Engineering Hydrology, Subramanya, K., Tata McGraw-Hill Education Private Limited.

Reference Books:

1. Hand book of Applied Hydrology, Chow, V.T., McGraw-Hill Book Co.
2. Impacts of climate change and climate variability on hydrological regimes, Jan C. van Dam, Cambridge University Press.
3. Hydrology: Principles, Analysis and Design, Raghunath, H.M.,New Age International.

TRANSPORTATION ENGINEERING - I

Lecture	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives: The objectives of this course are:

1. To enable students to differentiate and analyze between road pavement structures
2. To differentiate between the different types of materials used to design road pavements.
3. The introduction of the design concepts, material properties, performance criteria and vehicle loading criteria for design and construct of road pavements.
4. To distil the principles of geometric design, both vertical and horizontal.
5. To get Knowledge on Road Traffic and Traffic Management Systems
6. To get Knowledge on Characteristics and Components of Airport Engineering

Course Outcomes: Upon the successful completion of this course, the students will be able to:

1. Selecting the appropriate materials for use in different road layers
2. Perform road pavement design and analysis
3. Interpret geometric design fundamentals, in relation to safety and driver comfort, focusing on horizontal and vertical alignment
4. Design the geometric curves of a road pavement
5. Design the Traffic Management System
6. Design the Components of Airport

SYLLABUS

Highway Engineering – I : Highway development and planning, Classification of roads, Highway alignment, Highway Geometrics – Design of Cross sectional elements, Sight distance, horizontal and vertical alignment.

Highway Engineering – 2: Traffic Engineering – Traffic Characteristics, Traffic studies (Surveys), Traffic Control devices – Design of intersections. Design of pavements – Design factors, design of flexible pavements – Group Index method, CBR Methods, Design of Rigid pavements – Westergaard equations, I.R.C. recommendations for design of concrete roads.

Highway Engineering – 3: Construction of roads – Earthen roads – W.B.M. roads – Bitumen's roads – Cement concrete roads – Highway materials and their properties and tests. Maintenance of all roads – highway drainage – Arboriculture – Street lighting.

Airport Engineering: Layout of Airports – Components functions – Aircraft characteristics – Airport site selection – Airport obstructions – Runway design – Visual aids – Air traffic control.

Text Books:

1. Highway Engineering by Khanna& Justo.
2. Highway Engineering by Sharma & Sharma.
3. Airport planning and Design by Khanna&Arora.

Reference Books:

1. Highway Engineering by P H Wright and K Dixon
2. Highway Railway Airport and Harbour Engineering by Subramaniam

PROJECT PLANNING AND MANAGEMENT

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives:

1. To understand the importance of project management in today's world.
2. To understand various project time estimates and costs for project planning
3. To understand the procurement process and the risks involved in different contracts
4. To establish the importance of Construction and scientific management

Course Outcomes:

1. Define planning, controlling and scheduling to find the time estimates of a project
2. Classify the contracts and contract laws applicable to construction industry
3. Explain the Importance of project management and the role of project managers in various organizations
4. Identify and apply various networking techniques of project management

SYLLABUS

PERT and CPM : Introduction : Origin of PERT and CPM, Planning, Scheduling and controlling Bar charts, Milestone charts, weaknesses in Bar charts, PERT and CPM networks – Comparison, Event, Activity, Rules for drawing networks, Numbering the events (Fulkerson's law : Dummy activities, Time estimate-Expected time, Earliest allowable occurrence time, Latest allowable occurrence time, slack, project duration, probability of completion, Start and Finish time estimates, Floats, Project scheduling, Critical and sub-critical path.

Cost analysis / updating / resource scheduling: Cost Analysis direct and indirect costs, operation time, Normal and crash points, optimizing project cost, crash limit, free float limit, Optimization. Updating – Process of updating; when to update, Resource scheduling – Resource smoothening. Resource leveling, circle notation and arrow notation.

Contracts: Contracts – Element of contract, offer acceptance and consideration, valid contract, Department execution of works, Master Roll Form 21. Piece work Agreement form, work order; Contract system with tenders – Definitions – Contract, Contractor, Quotation, Earnest money, Security money, Tender, Tender notice, Tender form, Bidding procedure, Irregularities in Bidding, award, Types of contracts – Lump sum contract; Lump sum and schedule contract, Item rate contract, sub-contracts, joint ventures, Arbitration Disputes and claim settlement.

Management – Scope of the Construction Management, Significance of Construction management, Concept of Scientific Management, Qualities of Manager, Organization – Authority, Policy, Recruitment process and Training Development of Personnel Department.

Labour problems, Labour legislation in India, Workmen compensation Act 1923, and subsequent amendments, Minimum Wages Act 1948.

Text Books:

1. PERT and CPM – L. S. Srinath.
2. PERT and CPM – Punmia.
3. Estimating and Costing – B.N. Dutta.
4. Construction Management and Planning – Guna and Sen Gupta, B

Reference Books:

1. Construction Project management, Theory & Practice- Jha, Kumar Neeraj
2. Project Planning with PERT and CPM - Punmia, B.C., Khandelwal, K.K

COMPUTER APPLICATIONS IN CIVIL ENGINEERING LAB

Lab	: 3 Periods	Sessionals	: 50
Exam	:3 Hrs.	Ext.Marks	:50
		Credits	:2

Course Objectives: Students learn

1. To understand the purpose of computer applications in civil engineering.
2. To analyze and design the structures using various software.

Course Outcomes: Students will be able to

1. Create a program which is necessary to classify and evaluate the values.
2. Create an excel sheet for the design of slabs.
3. Model and analyze the beams and plane frames using STAAD.

SYLLABUS

Introduction

Introduction - Various software that are used in Civil Engineering based on their purpose - Drafting - AutoCAD - Coding - C Language- Excel - Analysis and design- FEM based software (Like STAAD,SAP,ETABS)

Development and Execution of Programs in C-language

- Create a program to determine the bending moment and shear force for uniformly distributed load on various beams. Also program on deflection of determinate beams
- Create a program to classify a soil sample based on the given data.
- Create a program to determine the Darcy's friction factor for a circular pipe.
- Create a program to estimate the population for the year by arithmetic increase method and geometric increase method.
- Create a program to convert a whole circle bearing to reduced bearing.

Formulation of design sheets in Excel.

- Create an excel sheet to design a one way slab for the given uniform distributed loads.
- Create an excel sheet to design a two way slab for the given uniform distributed loads.
- Create an excel sheet to design a singly reinforced beam.

Analysis and Design of RCC elements using software

- Analysis and design of a plane frame.
- Analysis and design of beam with various supports.

Reference Books:

1. Balaguruswamy. E "Object –Oriented Programming in C", Tata McGrawHill.
2. Excel: Quick start guide from beginner to expert

TRANSPORTATION ENGINEERING LAB

Lab : 3 Periods
Exam : 3 Hrs.

Sessionals : 50
Ext. Marks : 50
Credits : 2

Course Objectives: The objectives of this course are:

1. To differentiate between the different types of materials used to construct road pavements.
2. To analyze the quality of Road Aggregates
3. To analyze the quality of Bitumen.
4. To analyze the quality of Subgrade Material

Course Outcomes: Upon the successful completion of this course, the students will be able to:

1. Differentiate the Different types of materials used for Road Construction
2. Identify the quality of road aggregates
3. Identify the quality of Binder
4. Determine the strength of the Subgrade material

SYLLABUS

Testing of Aggregates: Specific gravity – Sieve Analysis – Shape test – Flakiness Index – Elongation Index – Water absorption test – Aggregate Crushing value – Impact value – Abrasion value – Stripping value & Soundness

Testing of bituminous material: Specific gravity – Penetration value – Viscosity value – Softening point – Ductility value – Flash and Fire point.

Testing on Soils : C.B.R. test (IS 2720 – Part-XVI) – N.D.C. Penetration test (IS 2720 Part-XXXII) – Group Index.

Testing on Bituminous Mixes: Bitumen Extraction Test, Marshal Stability Test (Demonstration)

Reference Books:

1. Highway material and Pavement testing by Khanna, Justo & Veeragavan

PROJECT PHASE-I**Lab : 3 Periods****Sessionals : 50****Credits : 2**

Course Outcomes:

1. Identify a current problem through literature/field/case studies and define the background objectives and methodology for solving the same.
2. Write report and present it effectively.

The phase-I of the project shall comprise of

- Problem identification in close collaboration with industry.
- Literature survey.
- Deriving work content and carry out of project requirement analysis.
- Submission of interim report.
- Presentation to an expert committee.

(Note: Sessionals 50 marks will be awarded based on
Continuous evaluation - 25 Marks
Seminar and Viva voce - 25 Marks.)

SCHEME OF INSTRUCTION & EXAMINATION
 (Regulation R16)
IV/IV B.TECH
 (With effect from **2016-2017** Admitted Batch onwards)
CIVIL ENGINEERING
II-SEMESTER

Code No.	Name of the Subject	Credits	Lecture Hrs	Tutorial Hrs	Lab Hrs	Contact Hrs/ Week	Internal Marks	External Marks	Total Marks
B16 CE 4201	Water Resource Engineering-II	4	3	1	--	4	30	70	100
B16 CE 4202	Transportation Engineering-II	4	3	1	--	4	30	70	100
B16 CE 4203	Irrigation Structures Design & Drawing	2	1	-	2	3	50	50	100
B16 CE 4204	Project Phase-II	12	-	-	9	9	50	100	150
Total		22	7	2	11	20	160	290	450

WATER RESOURCE ENGINEERING-II

Theory	: 3 Periods	Sessionals	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives: Student may know the

Gravity and earth dams, Spillways and energy dissipation works, diversion head works, river training works and the principal components of hydroelectric scheme

Course Outcomes: Student will be able to

1. Analyze gravity and earth dams (K4).
2. Design Spillways and energy dissipation works(K4).
3. Design diversion head works (K4).
4. Classify river training works(K2).
5. Use the principal components of hydroelectric scheme(K2)

SYLLABUS

Storage Works: Classification of dams, Factors governing selection of types of dam, Selection of site, Preliminary investigation.

Gravity Dams : Forces acting on a gravity dam, Stability criteria, Modes of failure – Elementary and Practical profiles, Stability analysis, Principal and shear stress – Construction joints, Openings in dams – Galleries, Foundation treatment of gravity dam.

Earth Dams: Types, Foundation for earth dams, design of earth dams, Causes for failure of earth dams, Criteria for safe design, Phreatic line, Seepage analysis – Seepage control through body and foundation.

Spillways: Essential requirements, Spillway capacity, Components, Types of spillways and their working, Design of ogee spillway, Energy dissipation below spill way, Scour protection, Use of hydraulic jump as energy dissipater – Design of stilling basins – USBR and IS standard basins; Spillway crest gates – Different types.

Diversion Head Works: Types, Location and components, effects of construction of weirs on permeable foundation, Bligh's, Lanes and Khosla's theories, Method of independent variables, Design principles of weirs and barrages, Design of weirs on permeable foundations, Design of vertical drop weir, Silt control devices.

Regulation Works: Canal falls – Definition, Necessity and location, Classification of falls, Design principles of syphon well drop, Notch fall, Sarada fall, Straight glacis fall; Offtakealignment; Cross regulator and Distributary head regulator – Design of cross regulator and Distributor head regulator.

Cross Drainage Works: Types, Factors affecting the suitability of each types, Classification of aqueducts, Design principles of different types of aqueducts.

River Training Works: River Training and its objectives, Classification of river training works, Marginal embankment, Guide banks, Groynes, cutoffs, Bank pitching, Launching aprons, Miscellaneous types of river training works.

Water Power engineering: Development of hydro power in India, Assessment of available power, Utilization factor, Load factor, Diversity factor, Storage and Pondage; Types of hydropower schemes; Components of hydel schemes – Fore bay, Intake structure, Trash racks, Surge tanks; Water hammer pressure, Substructure and Superstructure of power house.

Text Books:

1. Irrigation and Water Power Engineering, Punmia, B.C. and P.B.B. Lal, Laxmi Publications Pvt. Ltd.
2. Irrigation and Water Resources & Water Power, Modi, P.N., Standard Book House.
3. Irrigation and Hydraulic structures, Garg, S.K., Khanna Publishers.

Reference Books:

1. Hand book of Applied Hydrology, Chow, V.T., McGraw-Hill Book Co.
2. Impacts of climate change and climate variability on hydrological regimes, Jan C. van Dam, Cambridge University Press.
3. Hydrology: Principles, Analysis and Design, Raghunath, H.M., New Age International.

TRANSPORTATION ENGINEERING - II

Theory	: 3 Periods	Sessoinals	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 4

Course Objectives: The objectives of this course are:

1. Identify different Railway components and classify the different concepts of railway engineering design.
2. Analyze different geometrical elements used in design of railways.
3. Identify and understanding the importance of docks and harbours.
4. Examine the components of Tunnel Engineering

Course Outcomes: Upon the successful completion of this course, the students will be able to:

1. Explain railway track components, its importance and requirements.
2. Design elements of track geometry, points and crossings and concepts of railway signaling
3. Predict the importance and necessity of harbours and docks in transportation.
4. Evaluate the components of Tunnel Engineering.

SYLLABUS

RAILWAY ENGINEERING – 1 : Historical development of railways in India – Advantages of Railways – Classification of Indian Railways – Permanent way – Components and their functions – Rail joints – Welding of Rails – Creep of Rails – Rail fixtures & Fastenings.

RAILWAY ENGINEERING – 2 : Track Geometric design – Points & Crossings – Track drainage – Layout of Railway stations and yards – Signals – Interlocking – Track circuiting – Track Maintenance.

DOCK & HARBOUR ENGINEERING: Layout of Port components – Functions – Classification of Ports – Site selection – Natural Phenomenon – Tides, Winds, Waves, Currents – Drift – Navigational aids.

TUNNEL ENGINEERING: Alignment of tunnels – Cross-section of tunnels – Construction methods of Tunnels – Tunnel lining – Ventilation – Drainage – Muck disposal.

Text Books:

1. Railway Engineering by S.C. Saxena& S. Arora.
2. Dock&Harbour by Birdie.
3. Tunnelling by Rangwala.

Reference Books:

1. Railway Engineering by Rangwala.
2. Highway Railway Airport and Harbour Engineering by Subramaniam

IRRIGATION STRUCTURES DESIGN & DRAWING

Theory	: 1 Period	Sessionals	: 50
Lab	: 3 Periods.	Ext. Marks	: 50
Exam	: 3 Hrs.	Credits	: 2

Course Objectives: Student may know the

The paper - space environment thoroughly, components using 2D & 3D wire frame models through various editing commands, assemble of various components of compound solids and irrigation canal structures (K6)

Course Outcomes: Student will be able to

1. Understand the paper - space environment thoroughly (K2)
2. Develop the components using 2D & 3D wire frame models through various editing commands (K3)
3. Explain assemble of various components of compound solids (K5)
4. Design irrigation canal structures (K6)

SYLLABUS

Design and drawing of the following irrigation structures

1. Tank Surplus weir
2. Barrage
3. Glacis type of canal drop
4. Notch fall
5. Siphon Aqueduct- type III
6. Cross regulator and head regulator

Text books:

1. Water Resources Engineering, by C.SatyanarayanaMurthy
2. Water Resources Engineering, by SK Garg
3. Design of Irrigation Structures by RSN Murthy

Note: *ISDD External marks are allotted based on External viva-voce only

PROJECTPHASE-II**LAB : 9 Periods**

Sessionals	: 50
Ext. Marks	: 100
Credits	: 12

Course Outcomes:

1. Identify a current problem through literature/field/case studies and define the background objectives and methodology for solving the same.
2. Analyze, design and develop a technology/ process.
3. Implement and evaluate the technology at the laboratory level.
4. Write report and present it effectively.

The phase-II of the project shall consists of

Implementing, Testing and validation.
Report Writing.

Sessionals (50 Marks) will be awarded by the Project Guide based on continuous evaluation. External Evaluation (100 marks) of project report and viva voce will be conducted by a committee consisting of HOD, Guide and External Examiner.

May be carried out using in-house facilities or in an industry by specified number of students in a group.

Format for Preparation of Project Thesis for B. Tech:

1. Arrangement Of Contents: The sequence in which the project report material should be arranged and bound should be as follows:
 1. Cover Page & Title Page .
 2. Bonafide Certificate
 3. Abstract.
 4. Table of Contents
 5. List of Tables
 6. List of Figures
 7. List of Symbols, Abbreviations and Nomenclature
 8. Chapters
 9. Appendices
 10. References

*The table and figures shall be introduced in the appropriate places.